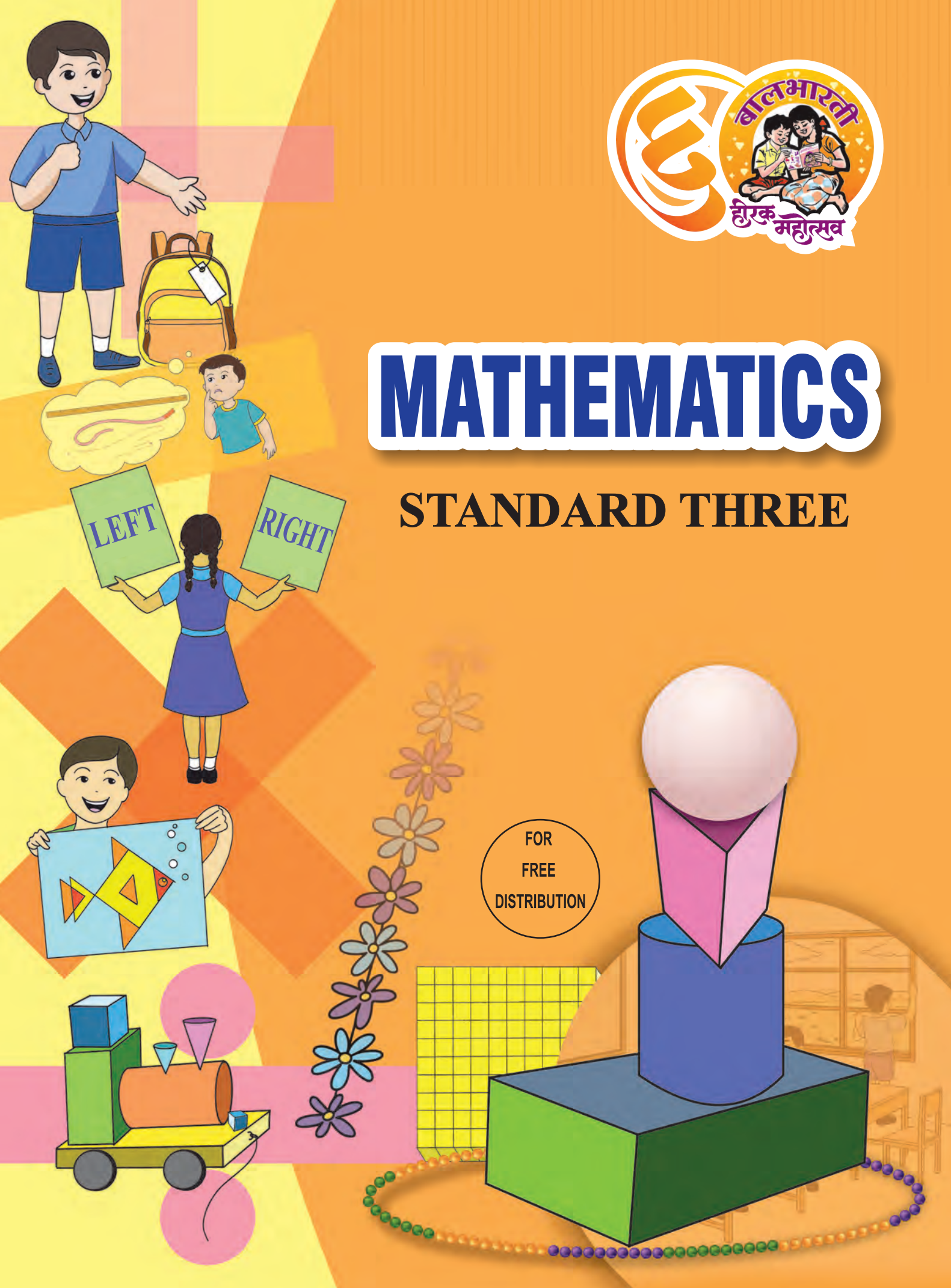




MATHEMATICS

STANDARD THREE

FOR
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Tele MANAS Program

Tele MANAS (Tele Mental Health Assistance and Networking Across States)
(14416 / 1800-89-14416)

Under the National Mental Health Program, the centrally sponsored scheme Tele MANAS is being implemented in the state of Maharashtra from October 2022. Under the Tele MANAS Program, free 24 × 7 integrated and comprehensive mental health services are provided through the helpline number 14416. This includes treatment and counselling by experts for various mental problems. The information provided by you after calling the Tele MANAS helpline remains completely confidential, so students, parents and teachers should feel free to call the helpline number and tell them about their problems and doubts and get guidance on it. Tell others to take help from this facility as well.

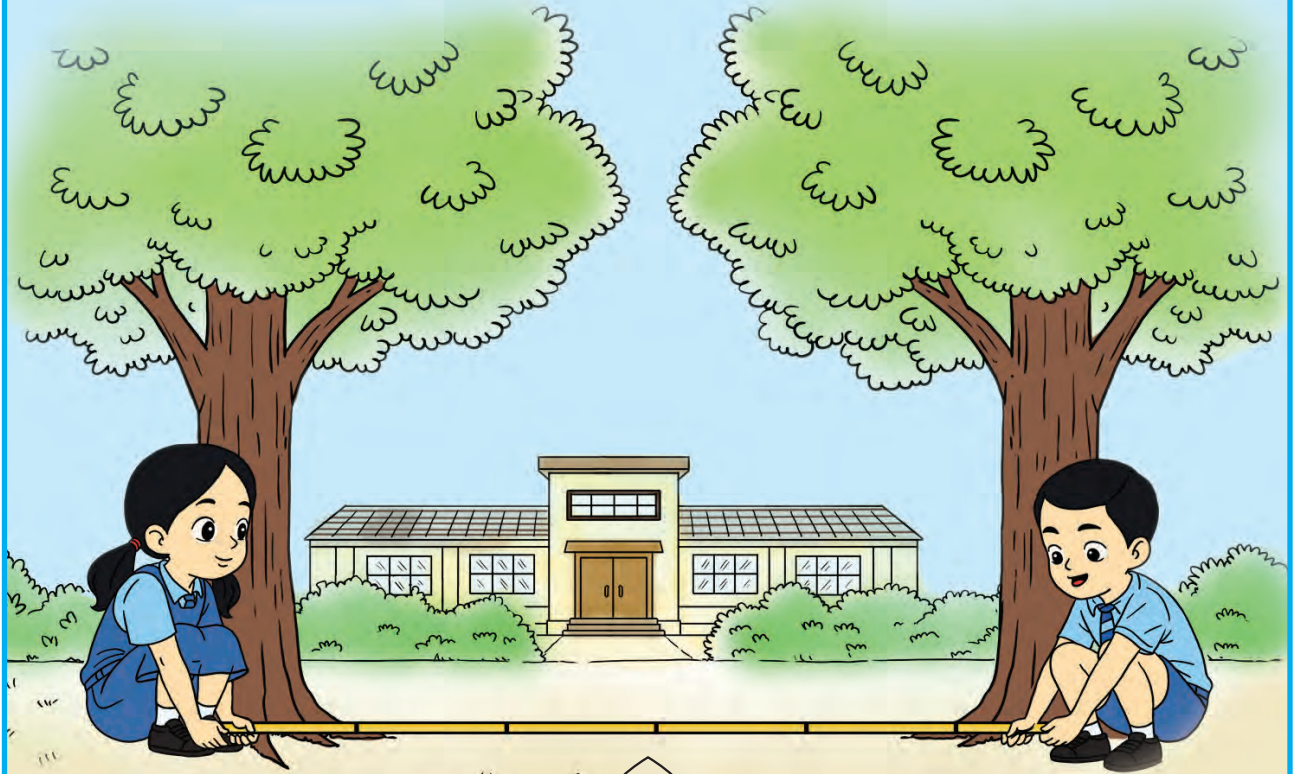
Manodarpan

The Manodarpan Helpline (8448440632) is a free telephone service that has been made available since July 2020. It is provided by the Ministry of Education through the Manodarpan Cell at the National Council of Educational Research and Training (NCERT) to offer psychological guidance and support to students, parents and teachers. This service is available every day of the week from 8:00 AM to 8:00 PM. Students, parents, and teachers can call this helpline to receive the necessary guidance and counseling.

This textbook aligns with the curriculum and syllabus developed by the State Council of Educational Research and Training (SCERT), Maharashtra, Pune-30 and has been verified by them. The Coordination Committee, established under Government Resolution by GR No.: Abhyas-2116 / (Pra.Kra. 43/16) SD-4 dated April 25, 2016, approved this textbook in its meeting held on April 1, 2026 for use from academic year 2026-27 onwards.

Mathematics

Standard Three



समग्र शिक्षा

**Maharashtra State Bureau of Textbook Production
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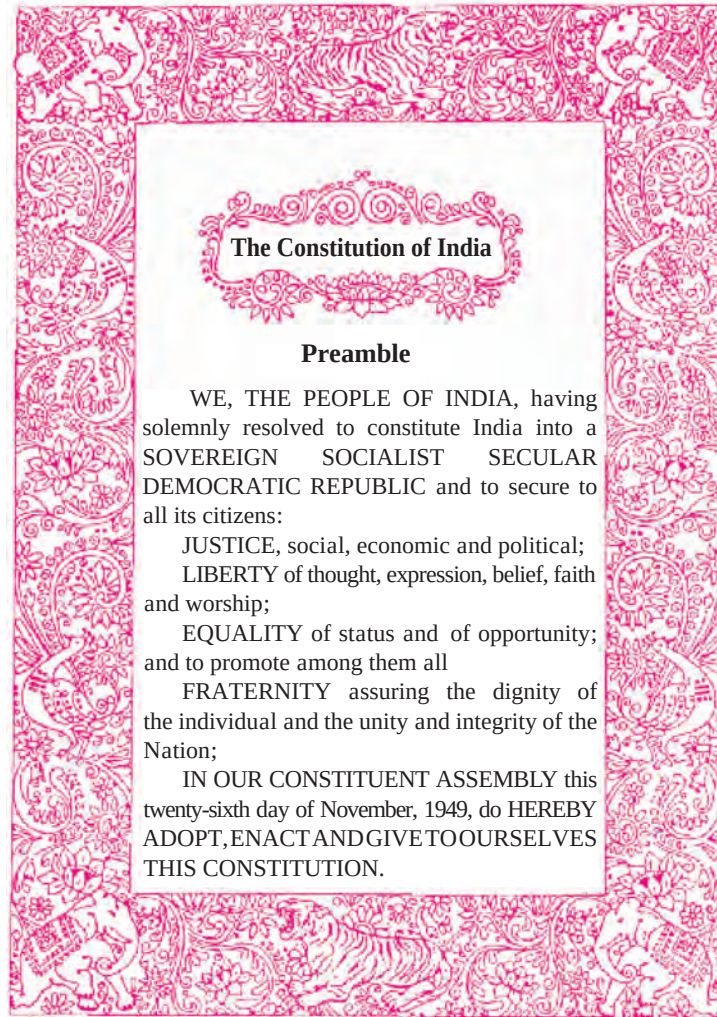
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The Constitution of India

Part IV-A*

Fundamental Duties of Citizens

(Article 51C)

Fundamental Duties - It shall be the duty of every citizen of India -

- (a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem.
- (b) to cherish and follow the noble ideals that inspired our freedom struggle.
- (c) to uphold and protect the sovereignty, unity and integrity of India.
- (d) to defend the country and render national service when called upon.
- (e) to promote harmony and the spirit of common brotherhood amongst all people, transcending religious, linguistic, regional or sectional diversities; to renounce practices derogatory to the dignity of women.
- (f) to value and preserve the rich heritage of our composite culture.
- (g) to protect and improve the natural environment including forests, lakes, rivers and wildlife, and to have compassion for living creatures.
- (h) to develop scientific temper, humanism and the spirit of inquiry and reform.
- (i) to safeguard public property and abjure violence.
- (j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement.
- (k) to provide opportunities for education to one's child or ward between the age of six and fourteen years.

NATIONAL ANTHEM

Jana-gana-mana-adhināyaka jaya hē
Bhārata-bhāgya-vidhātā,

Panjāba-Sindhu-Gujarāta-Marāthā
Drāvida-Utkala-Banga

Vindhya-Himāchala-Yamunā-Gangā
uchchala-jaladhi-taranga

Tava subha nāmē jāgē, tava subha āsisa māgē,
gāhē tava jaya-gāthā,

Jana-gana-mangala-dāyaka jaya hē
Bhārata-bhāgya-vidhātā,

Jaya hē, Jaya hē, Jaya hē,
Jaya jaya jaya, jaya hē.

PLEDGE

India is my country. All Indians
are my brothers and sisters.

I love my country, and I am proud
of its rich and varied heritage. I shall
always strive to be worthy of it.

I shall give my parents, teachers
and all elders respect, and treat
everyone with courtesy.

To my country and my people,
I pledge my devotion. In their
well-being and prosperity alone lies
my happiness.

राज्यगीत

जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ॥धृ॥

भिती न आम्हा तुझी मुळी ही गडगडणाऱ्या नभा
अस्मानाच्या सुलतानीला जबाब देती जिभा
सह्याद्रीचा सिंह गर्जतो शिव शंभू राजा
दरी दरीतून नाद गुंजला महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ॥१॥

काळ्या छातीवरी कोरली, अभिमानाची लेणी
पोलादी मनगटे खेळती, खेळ जीवघेणी
दारिद्र्याच्या उन्हात शिजला
निढळाच्या घामाने भिजला
देश गौरवासाठी झिजला

दिल्लीचे ही तख्त राखितो, महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा
जय जय महाराष्ट्र माझा ॥२॥

Introduction

Dear children, teachers, and parents, I extend my warm greetings to you all.

It gives me great pleasure to present this Mathematics textbook for Class 3, prepared in accordance with the National Education Policy 2020. As per the State Curriculum Framework 2024, it has been suggested that for the preparatory stage from Class 3 to 5, textbooks should be simple and engaging, the content should not merely be in the form of instructions, but should provide children with opportunities to work on their own.

While preparing this textbook, the emotional world and preferences of children have been fully considered. Keeping in mind the interest of children in Class 3, attractive pictures have been included. From a mathematical perspective, each unit begins with the use of various objects and situations from the child's emotional world and their surroundings to enrich the content. Various mathematical concepts have been explained clearly in simple language. To ensure children get opportunities to practice and to reinforce understanding, instructions have been provided for teachers. Efforts have been made to make the textbook more engaging through puzzles, Maths Games, and similar activities.

The pictures and examples in this textbook are child-friendly moreover, carefully chosen to maintain social balance. An effort has been made to give due priority to all sections of society in the selection of illustrations. The Panchakosha development and 21st century skills mentioned in the National Education Policy 2020 have been integrated into various units of the textbook. The examples in this textbook have been designed to maintain a progressive level of difficulty and diversity. The content of the textbook is aligned with developmental areas, curricular goals and objectives, competencies, and learning outcomes. Activities, games and stories have been incorporated to make learning enjoyable for children.

Best wishes to all for a bright future!

Pune

Date: 19 April 2026

Indian Solar Date: Vaishakh shu.2
Shake 1948, Akshaytritya.



Anuradha Oak

Director,
Maharashtra State Bureau
of Textbook Production and
Curriculum Research, Pune.

Curricular Goals, Competencies, and Learning Outcomes

Curricular Goal (CG1): Develop understanding/number sense of numbers (counting numbers and fractions). Represent whole numbers using the Indian place value system. Understand and perform the four basic operations on whole numbers. Identify and recognize patterns in number sequences.

Sr. No.	Learning Competencies	Learning Outcomes	Page Number
1.	C-1.1 : Uses place value system in the Indian Number System to read and represent numbers, identifies whole numbers, compares them, and reads large numbers.	› Understands the concept of hundred and represents it in different ways.	8, 9
		› Identifies numbers up to 999.	10, 11, 12, 13
		› Writes numbers up to 999 in figures and in words.	12, 13
		› Uses ordinal and cardinal numbers appropriately.	23, 24
		› Identifies place value and face value of digits.	14, 15, 16
		› Compares numbers up to 999.	17, 18, 19, 20
		› Form numbers based on given conditions	19, 20, 21
2.	C-1.2 : Represents commonly used parts of a whole (half, quarter, three-quarters) such as $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$ etc. in fractional form, compares them. (e.g. $\frac{1}{2}$, $\frac{1}{4}$ fractions are shown as parts of whole numbers and their position is shown on a number line)	› Identifies half, quarter, three-quarters of a shape (object) and figure.	37 to 42
		› Identifies half, quarter, three-quarters of a group of numbers.	41, 42
		› Represents fractions in numerator-denominator form.	43, 44, 45, 46
		› Identifies proper and improper fractions.	45, 46
3.	C-1.3 : Performs operations on numbers, understands and applies their inter-relationships. For example, understands the relationship between addition and multiplication (tables up to at least 10) and uses the four basic operations on numbers to solve everyday problems.--	› Performs addition and subtraction of three-digit numbers.	26 to 33
		› Forms number bonds up to 20.	34
		› Handles money transactions up to ₹999.	22, 25 to 27, 29 to 33, 76 to 79
		› Explains the relationship between addition and multiplication. Makes tables and uses them.	66, 68 to 70, 72
		› Creates and solves word problems on multiplication. (Multiplying two-digit numbers by two-digit/one-digit numbers and three-digit numbers by one-digit numbers.)	66, 73 to 79
		› Creates and solves word problems on division. (Dividing two-digit/three-digit numbers by one-digit numbers.)	80 to 89, 91
		› Explains the relationship between subtraction and division.	81
		› Identifies the relationship between multiplication and division.	86, 87
		› Solves word problems related to measurement.	51
		› Shows curiosity and awareness about the use of the zero.	7, 19, 73, 90
4.	C-1.4 : Identifies, describes, and extends patterns in numbers like odd numbers, even numbers, squares, cubes, powers of two and ten, and Virahanka Fibonacci Numbers.	› Identifies patterns in numbers.	119, 120, 122, 123

Curricular Goal (CG-2): Analyse the features and properties of 2D and 3D geometric shapes. Specify their positions and describe spatial relationships. Identify and create symmetrical shapes.

5.	C-2.1 : Identifies, compares, and analyses features of 2D and 3D shapes. Develops vocabulary to describe their shapes and properties.	› Identifies 3D shapes in the surroundings based on their features.	1, 2
		› Classifies 3D objects.	3, 6
		› Compares 3D shapes based on their properties and features.	92
		› Expresses properties and features of 3D shapes in mathematical language.	92
		› Creates 3D shapes using nets (cube, cuboid, cylinder, cone, etc.).	2, 3

		Identifies two-dimensional shapes hidden within three-dimensional objects, analyzes them, and solves puzzles.	92, 94, 95
		Identifies/finds two-dimensional shapes based on three-dimensional shapes in the surroundings.	3, 105
		Discusses the construction of specific shapes, such as triangle, square, rectangle, circle, etc.	93, 94, 95
		Identifies shapes having four or more sides (edges).	93
		Identifies/draws shapes that differ from standard geometric shapes.	93, 97
		Creates two-dimensional shapes (using sticks, threads, rangoli, dots, etc.).	97, 99, 100
		Checks the interrelationships among two-dimensional shapes.	96
		Compares the components of two-dimensional shapes (sides, edges, corners).	92, 93
		Integrates two-dimensional shapes with components of art and sports activities.	97, 99, 100
		Takes pride in Indian art forms.	99
		Identifies horizontal, vertical, and slanting lines.	100
		Identifies shapes that fit exactly or do not fit in a given area.	98
		Identifies various angles in the surroundings and angles related to objects.	101
		Identifies right angles through observation.	101
		Compares right angles with other angles.	101, 102
		Classifies angles as greater than or smaller than a right angle through observation.	101, 102
		Creates a right angle, angles larger than a right angle, and angles smaller than a right angle using tools like sticks, threads, grids, rangoli, etc.	102, 105
6.	C-2.2: Describes location and position using both spoken language and mathematical vocabulary, and understands the concept of a map.	Identifies top view, side view, and front view of simple three-dimensional household objects.	59, 60
		Identifies the view of simple 3D objects as seen from above/top.	59, 60
		Uses directional words to describe locations on maps of public places and reads maps.	61, 62, 63, 64, 65
7.	C-2.3: Identifies and creates symmetry (translational, reflectional and rotational) in familiar two-dimensional and three-dimensional shapes.	Identifies symmetry in various shapes.	103, 104
		Finds symmetry in regular shapes.	104
		Finds symmetry in irregular shapes.	104
		Draws an axis of symmetry at the correct location to make a figure symmetric.	105
		Completes a figure by considering symmetry.	105
		Solves puzzles based on properties of symmetry.	105, 106
		Completes two-dimensional figures using paper folding when an axis of symmetry is given.	105
		Predicts shapes/objects that are rotated in clockwise or anticlockwise directions.	105, 106
8.	C-2.4: Identifies patterns in two-dimensional and three-dimensional shapes, describes and explains them.	Identifies patterns in two-dimensional shapes.	120, 121, 122, 123
		Creates patterns by cutting folded paper into specific shapes.	121
		Identifies patterns in various designs in the surroundings.	120
		Compares different designs.	120, 123
		Extends a given design.	120 to 122
		Creates their own simple patterns using two-dimensional shapes and lines.	120, 121, 123, 124
		Identifies patterns in tiling and describes them.	121, 122
		Creates regular tiling using geometric shapes.	122, 124

Curricular Goal (CG-3): To understand the measurable properties of objects, to measure length, weight, area, distance, volume, and time using both non-standard and standard units, and to understand the system of unit of measurement and the process of measurement.

9.	C-3.1 : Measures using non-standard and standard units and understands the need for using standard units.	Identifies non-standard units used for measurement.	47
		Understands that measuring the same object with different non-standard tools may give different results, and recognizes the need for standard units.	47, 48
		Understands the fun in non-standard measurement practices through stories, tales, poems, etc., and discusses the need for standard units.	47, 50, 107
		Verifies that measurements remain consistent when standard tools are used repeatedly.	49, 50, 52, 53, 108, 109
		Compares and discusses the use of different tools for different types of measurements.	48, 52, 109
10.	C-3.2 : Uses appropriate units and tools to measure quantities like length, perimeter, time, weight, capacity, and volume.	Selects appropriate standard tools and units for measurement and uses the correct unit.	48, 49, 50, 51, 52, 53, 108
		Tells accurate time using a wall clock/hand watch, including fractions (e.g., quarter past, half past, quarter etc.).	111, 112, 113, 114, 115
		Identifies specific days and dates in a calendar.	116, 117, 119
		Measures weight of objects using simple balances and standard units like grams and kilograms.	52, 53
		Verifies that weight measured using standard units remains the same irrespective of who measures it.	52, 53
		Compares weights (more than 1 kilogram, less than 1 kilogram, etc.).	53
		Discusses and gathers information about old and new tools used for weighing.	52
		Identifies appropriate units and tools for measuring volume.	107, 108, 109
11.	C-3.5 : Estimates and verifies length, distance, time, perimeter, area (regular and irregular shapes), weight, and volume using appropriate standard units.	Identifies standard units and tools for measuring the capacity of given liquids.	108, 109, 110
		Estimates and verifies length by comparing measured and unmeasured objects.	47
		Estimates length and distance using centimetres and metres as units and verifies by actual measurement.	48, 49, 50
		Estimates the weight of objects and verifies by actual measurement.	50, 51, 53
		Compares capacities/volume of different containers using non-standard units.	107, 110
12.	C-3.7 : Checks for conservation of the measurable attribute related to length and applies it in daily life.	Expresses time intervals in hours for daily events and verifies by actual measurement.	114, 115
		Measures length of objects in daily life (horizontally/vertically/diagonally) using non-standard units and verifies consistency.	49
		Measures the length of a stick using non-standard units. Verifies that the total length of the stick is the same as the sum of lengths of its pieces when cut and measured with non-standard units.	49
		Fills two identical glasses with water, measures water from one glass using a small cup, combines it, and refills the glass to verify the result.	107

Curricular Goal (CG-4): Develop problem-solving skills to solve mathematical puzzles and real-life problems, and thereby initiate the development of computational thinking.

13.	C-4.1 : Solves puzzles involving whole numbers with one or more operations, and examples based on daily life. This includes word problems, recreational puzzles such as creating a magic square, etc.	Analyzes information from daily life examples, games, and activities.	4, 5, 9, 20, 24, 35, 36, 60, 61, 63, 79, 91, 95, 97, 103, 105, 120, 123
		Finds step-by-step solutions for various real-life problems, mathematical games, and activities.	
		Uses logical thinking to select one or more appropriate options from those available to reach the answer.	
		Solves recreational puzzles based on this process.	

		▶ Discusses with other students to understand various approaches and other options to reach the answer to a problem. ▶ Solves and creates magic squares, word puzzles, problems.	
14.	C-4.2 : Lists all possible permutations and combinations by systematically counting as per the given condition in a simple situation. (e.g., forming a committee of two people from a group of five people.)	▶ Using dots, lines, shapes draws, creates and identifies the maximum possible patterns, arrangements and also analyze it. ▶ Systematically counts all possible pairs of given numbers/letters/colors, objects/shapes, etc. ▶ Creates patterns/designs using numbers/letters/colors, etc., as per the given condition.	123, 124 124, 125 125, 126
15.	C-4.3 : Selects appropriate methods and tools as per context for operations on whole numbers. e.g., mental computation, estimation, or using paper and pencil.	▶ Selects appropriate operation on numbers, methods, tools to solve real-life problems and draws conclusions. (Market transactions, measurement of objects)	8, 22, 25, 31, 30, 41, 66

Curricular Goal (CG-5): Understand the development of the Indian decimal place value system, which is now used worldwide, and develop pride in it.

16.	C-5.1 : Understands the development of zero in India and the history of the Indian place value system used for writing numbers, its spread across the world, and comprehends its modern impact on our lives and all sciences.	▶ Imagines how counting would have been done if zero did not exist. ▶ Recognizes zero as a digit and understands its role in counting. ▶ Knows how zero developed in India.	7, 90, 73 67, 72, 90 7
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Curricular Goal (CG-6): Develop skills to collect information from daily life experiences, organize it visually, estimate, and use it in daily life.

17.	C-6.1 : Collects information, describes it and infers meaning, uses pictures/symbols for representing information. Understands scale, prepares pictographs according to scale.	▶ Collects various types of information from the surrounding environment. ▶ Uses tally marks while classifying the collected information. ▶ Uses symbols/pictures while representing information. ▶ Observes pictorial information, estimates, compares, and interprets it.	54 54 56, 57, 58 57, 58
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While Handling the Textbook...

Teachers, Parents, and Facilitators should -

- Use materials from the surroundings and from the Maths Kit to clarify all mathematical concepts, activities, projects, practicals and Maths Fun given in the textbook.
- Reinforce concepts by having students do some textbook activities individually, in pairs, or in groups.
- The book includes some new concepts. Expand on them at your level. Study additional contexts based on these concepts. The expansion of specific concepts should be done by teachers at their level as per their teaching plan. For that, guidelines have been provided in the box titled 'For Teachers'.
- Provide students opportunities to apply acquired knowledge and encourage them to think about it.
- Each unit includes learning outcomes. Provide learning experiences to develop outcomes accordingly.
- Instructions in the textbook are guidelines. Make appropriate changes to your teaching-learning method considering local conditions, dialects and local context.
- Instead of just solving examples or questions, students should get higher-level opportunities to create examples or questions themselves. This has been considered at some places in the textbook.
- Try to integrate other subjects while giving learning experiences based on math concepts and principles.
- Emojis have been used in the textbook to prominently highlight the interaction between the teacher and students. Examples: Let's Recall, Let's Understand, I Understood, Do the Activity, Fun with Maths.
- In the process of teaching and learning for special children use sign language as well as complimentary method. Provide opportunities to these students to express themselves in any possible way.



* Index *

Sr.No.	Lesson Name	Page No.
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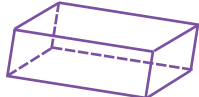
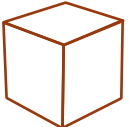
Let's Recall.

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☉ Observe the picture and identify the shapes of the objects in it.



☉ Observe and complete the table.

Object	Geometric shape similar to the object	Name of the geometric shapes
		Cuboid.....
		
		
		
		

Identify who am I.

1. I have two curved edges and one curved surface, but no corners.
2. I have one corner, one curved surface and one flat surface.
3. I have no corners and no edges. I have only one curved surface.



I have got so many lovely gifts!



Hey! Look at this box.



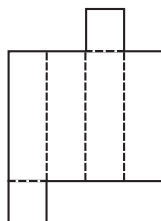
Oh, this is fun! This cuboid-shaped box has 12 edges, 8 corners, and 6 faces.



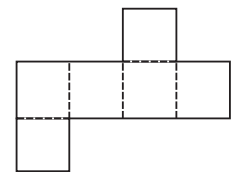
Shall we open this box? Then we'll understand how this box is made.



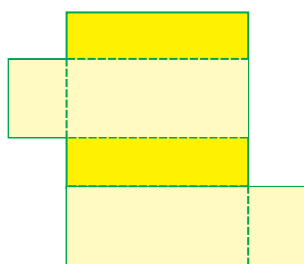
The shape from which a box is formed is called the net of that shape. Look at the net of this cuboid.



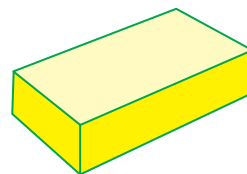
Shall we open (unfold) the boxes of other shapes and see their nets too?



Three Dimensional Shapes and Nets



Using this net, a cuboid is formed.



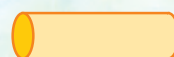
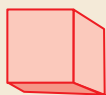
● Observe the table below and match the objects with their shapes based on their nets.

Sr. No.	Net of the shape	Shape
(1)		Cuboid
(2)		Cone
(3)		Cube
(4)		Cylinder



For Teachers : Using pictures, videos, the internet, or other media, show students 3D shapes found in the surroundings. For example: planets in the solar system, dewdrops, tree trunks, bamboo, minerals, crystals, etc. Point out the shapes present in the surroundings. Also, unfold different boxes of 3D shapes and show how their nets look.

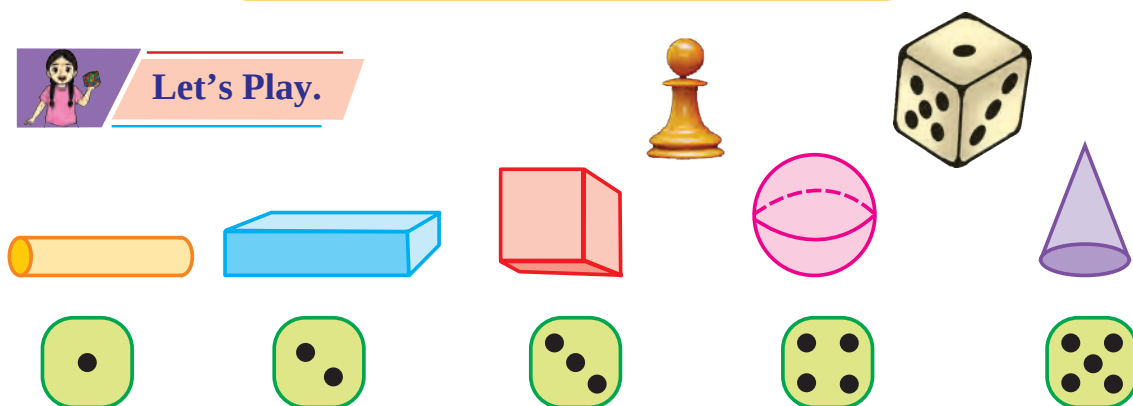
Start



Snakes and Ladders with Shapes



Let's Play.



Move forward two spaces and get one more chance to roll the dice.

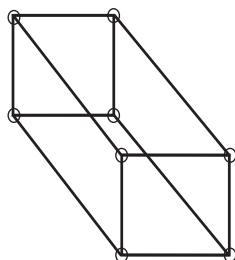


Instructions: This game is to be played using a dice and tokens. After rolling the dice, move your token forward according to the number you get. Place your token on the corresponding shape in order. The player who reaches the final shape (●) first wins the game.



Do the Activity.

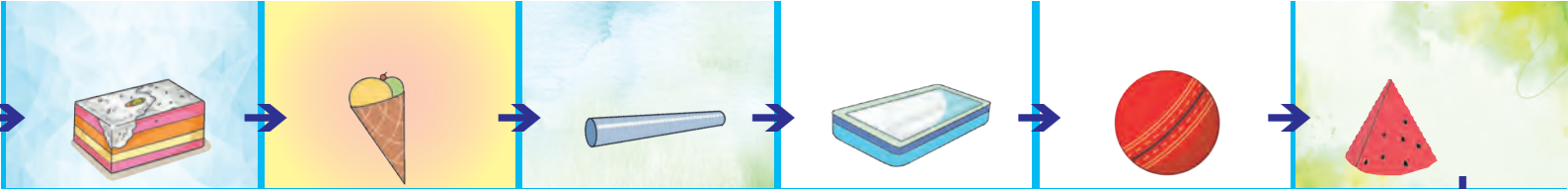
Fatima has made a cuboid using sticks and clay / jodo Blocks and connectors. In the same way, make other 3D shapes.



For Teachers :

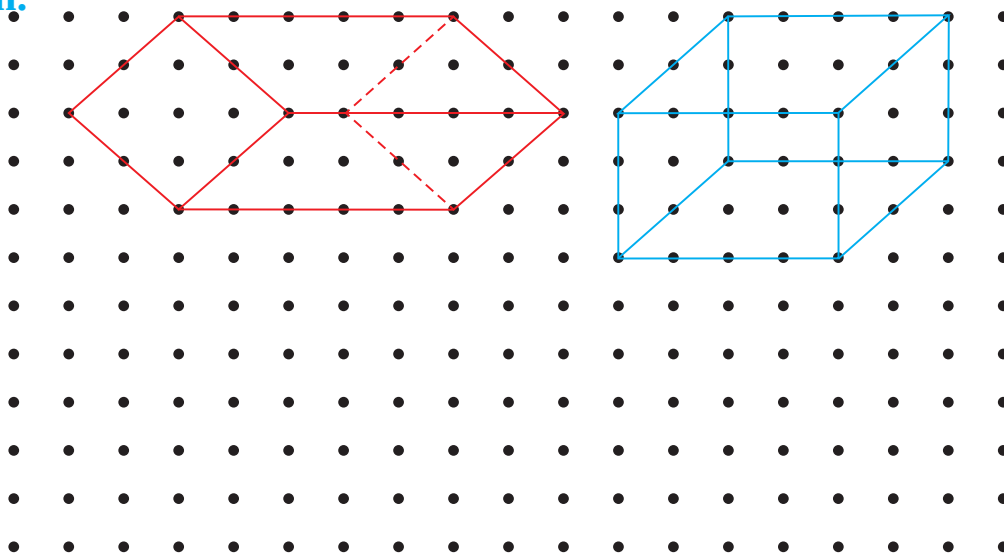
1. Replace the dots on the dice with pictures of related shapes, and let the students play the game.
2. Use available materials or materials from the mathematics kit to make different shapes.



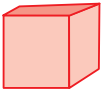


Do the Activity.

- ◎ Observe the shapes below and draw the same shapes beneath them.

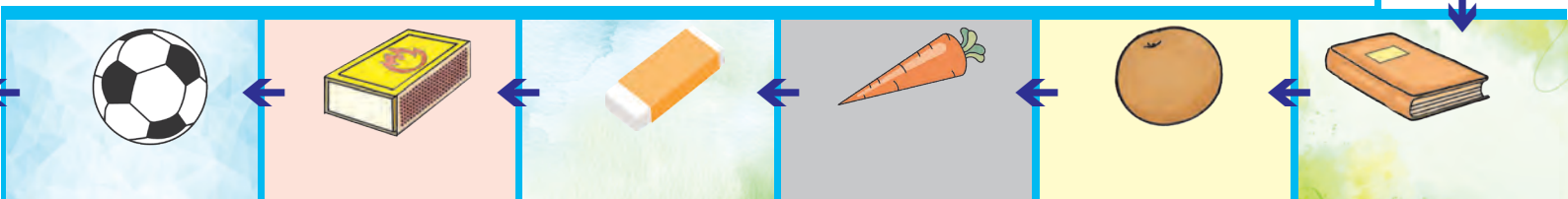
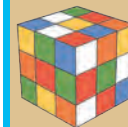


- ◎ Observe the shapes in the table below. Identify objects from your surroundings that correspond to these shapes and write their names.

				
Cylinder	Sphere	Cube	Cuboid	Cone
Box	Ball	Dice	Brick	Radish

1. Which shape did you observe the most? Write its name.
2. Which shape did you observe the least? Write its name.
3. Write the names of two objects made from more than one shape.

.....,



Exercise

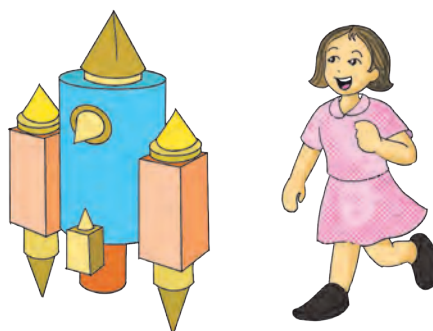
- (1) Poonam has made a rocket using some three-dimensional shapes. For that, how many blocks of each shape has she used? Observe and write.

Cube

Cuboid

Cone

Cylinder



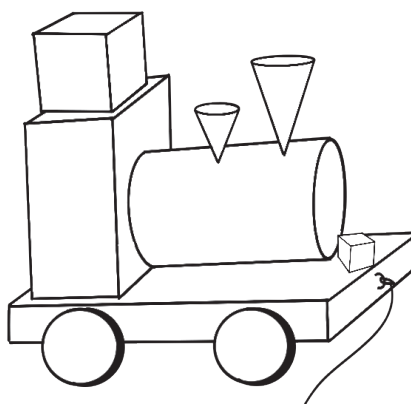
- (2) Kartik has drawn a picture of an engine using various three-dimensional shapes. Colour the picture according to the shapes.

Cone

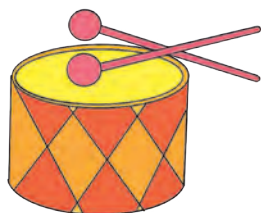
Cylinder

Cube

Cuboid



- (3) Some toys from a toy shop are shown here. These toys are made from different shapes. Write the names of the shapes used in each toy below the pictures.

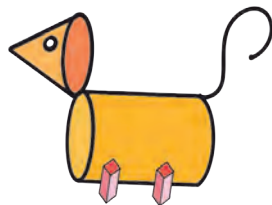


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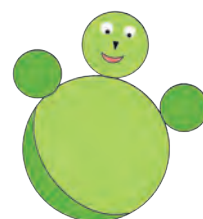


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How did people count things in ancient times? Did number exists at that time?



In ancient times, humans used to count by making marks on walls of cave and tree trunks. Afterwards they began using symbols to count groups and started counting in collection.



When and how were numbers discovered?



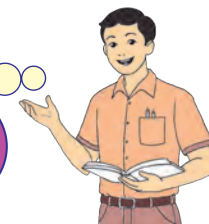
For a single object, the name “one” and the symbol “1” was assigned. When one more object was added to a single object, the group formed was called “two” and was represented by the symbol “2.” In this way, for counting objects, the number names one, two, three,..., nine were given, along with the corresponding digits 1, 2, 3, 4, 5, 6, 7, 8, 9.



The ‘zero’ (0) is used to indicate the absence of any quantity. When 1 is added to 9, The number formed is called ‘Ten’ . We write ten and it is denoted by the numeral ‘10’.



The invention of zero by Indian mathematicians is the greatest contribution to the world.

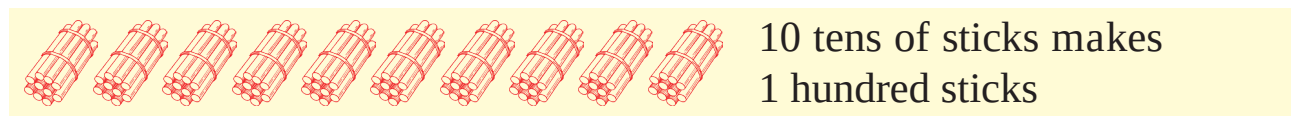
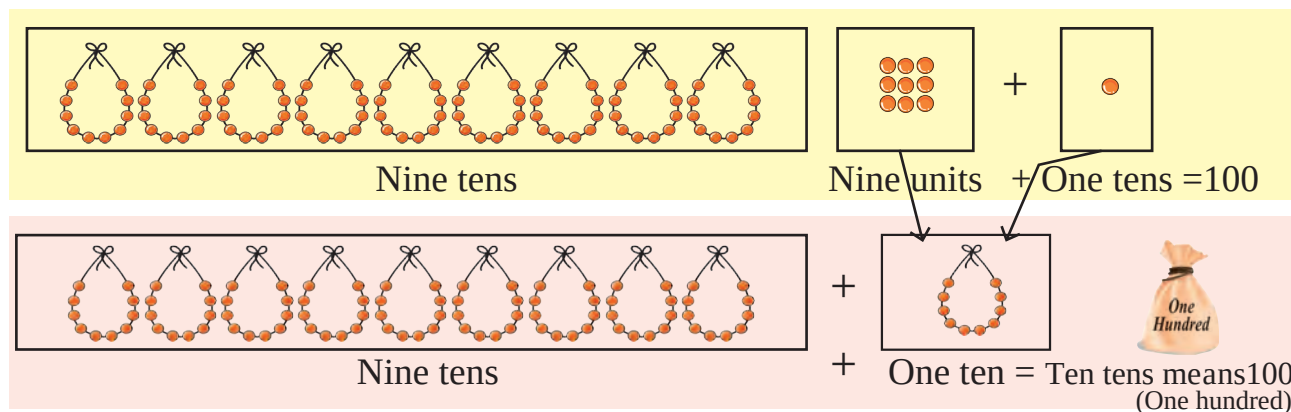
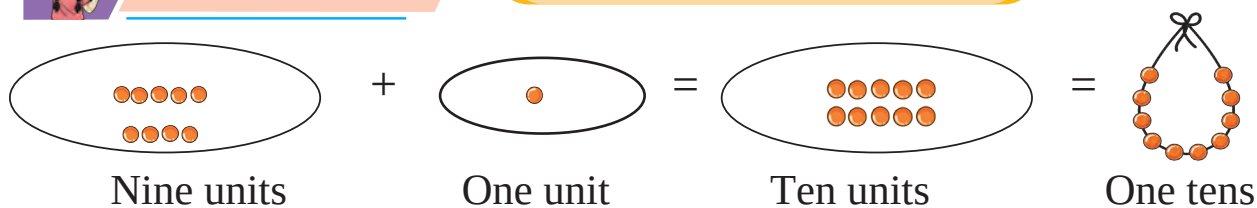


For Teachers: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 are the ten digits. Using these, one-digit, two-digit... such numbers are formed. No matter how big the number is, it becomes easy to identify, read, and write it. This became possible because of the discovery of zero. Give students experience of using numerals for groups of objects.



Let's Understand.

Introducing of a hundred

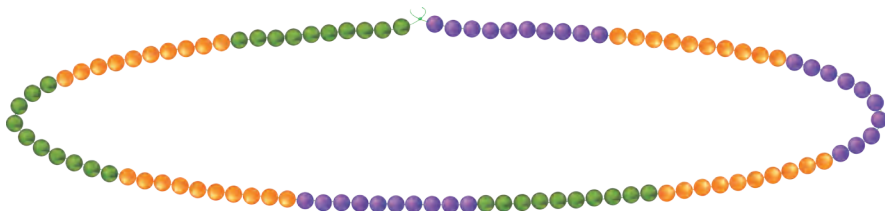
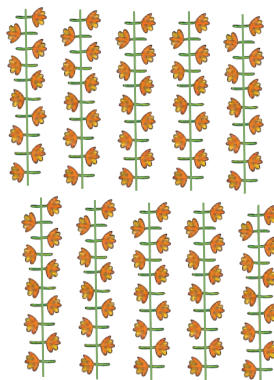
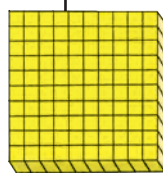


I scored 100 runs
I scored a century

1 hundred



100
Hundred



A string of one hundred beads

Let's Remember: A century means 100 units or simply hundred.

Ten tens means hundred. One hundred means hundred.

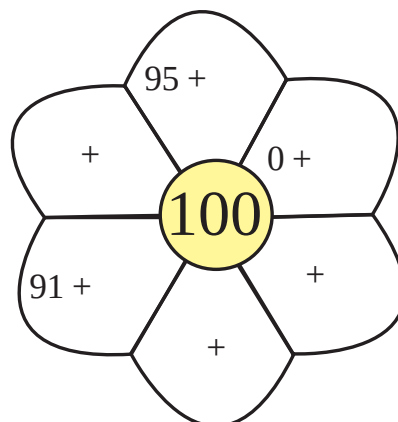
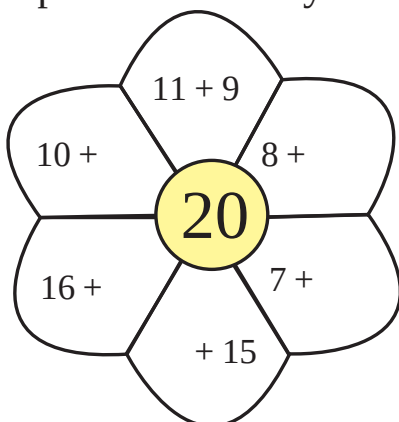
100 is the smallest three-digit number.

H	T	U
1	0	0



Fun with Maths.

Complete the activity to obtain the number in the circle.

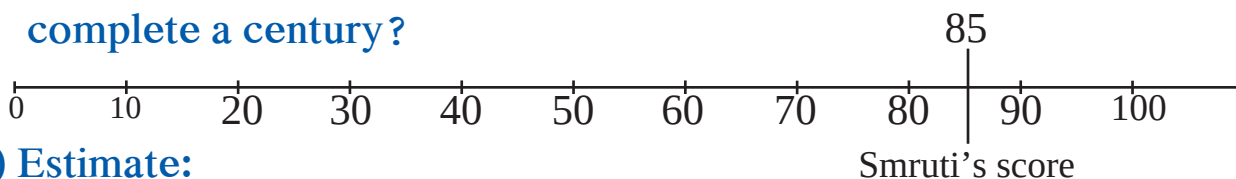


Let's do it.

(1) Count the blocks of tens and write the correct number in the blank space.

	+		70 and 30 together make 100
	+		 and 10 together make 100
	+		80 and together make 100

(2) Smruti scored 85 runs. How many more runs will she need to complete a century?



(3) Estimate:

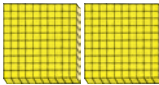
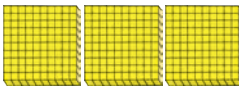
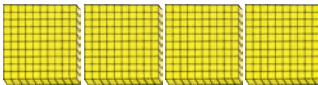
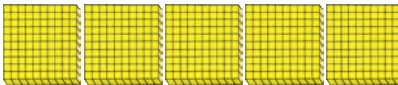
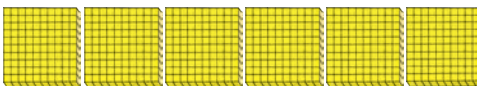
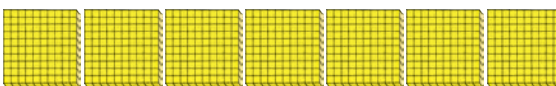
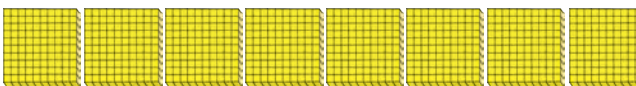
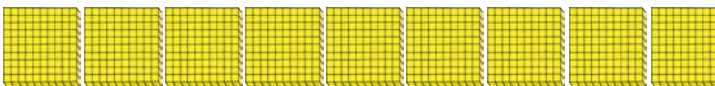
Take some seeds in your hands.

- ⦿ How many seeds are there in your hands? Estimate.
- ⦿ The number of seeds you estimated is less/more than 100.
- ⦿ Count the seeds in your hands.
- ⦿ The number of seeds you counted is less/more than 100.



For Teachers: Through the videos based on the life of Shrinivasa Ramanujan, various mathematical situations and incidents should be shown. For example, a scene like estimating the number of chickpeas (as shown in the movie Ramanujan) can be presented. Students should be given an opportunity to enact such scene through dramatization, focusing on developing the skill of estimation.

Counting in Hundreds

Symbols	Meaning of the symbol	Number in figures	Number in words
	One hundred	100	One hundred
	Two hundred	200	Two hundred
	Three hundred	300	Three hundred
	Four hundred		Four hundred
	Five hundred	500	
		600	Six hundred
	Seven hundred		Seven hundred
		800	
		900	



100200300500600800900



For Teachers : With the help of students, arrange 10 blocks of tens means 100 blocks and give practice of recognizing numbers.

Numbers after 100



Let's Understand.

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Symbols	Meaning of the symbol	Number in figures	Number in words
	100 and 1	101	One hundred and one
	100 and 2	102	One hundred and two
	100 and 3	103	One hundred and three
	100 and 4		One hundred and four
	100 and 5	105	
		106	One hundred and six
	100 and 7		
			One hundred and eight
		109	
			



111

112

120

Three-digit Numbers



Let's Understand.

Symbols	Hundreds	Tens	Units	Number in figures	Number in words
	2	2	0	220	Two hundred and twenty
	4	3	6	436	Four hundred and thirty six
	7	4	4	744	Seven hundred and forty four
	3	6	9	369	Three hundred and sixty nine
	5	8	5	585	
	8	0	1		
	6	6	9		
					Nine hundred and ninety

- ◉ Viraj, Dhavan and Sharayu are using bead strings to form numbers, observe and write the numbers that are formed.



4 hundreds

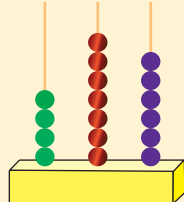


7 tens



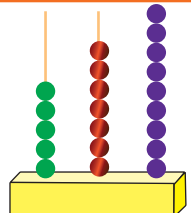
6 units

H T U



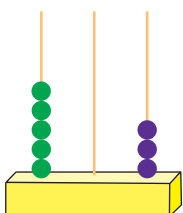
476
Four hundred and seventy six

H T U



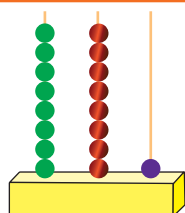
5 7 9
Five hundred and seventy nine

H T U



5 0 3
.....

H T U

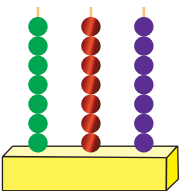


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Exercise

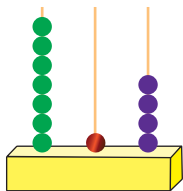
- (1) Write the number in figures and in words.

H T U



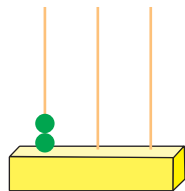
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.....

H T U



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.....

H T U



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.....

- (2) Write the given numbers in words. (3) Write the given numbers in figures.

In Figures	In Words	In Words	In Figures
435		Eight hundred and sixty seven	
279		Four hundred and forty nine	
508		Six hundred and one	
777		Three hundred and forty	

Face value and Place value of digits in a number



Let's Understand.

Face value of digits

Face value of 5 is 5
Face value of 2 is 2
Face value of 4 is 4

524		
H	T	U
5	2	4
5 hundreds 500	2 tens 20	4 units 4

Place value of digits

Place value of 5 is 500
Place value of 2 is 20
Place value of 4 is 4

Face value of 8 is
Face value of 0 is 0
Face value of 9 is

809		
H	T	U
8	0	9
8 hundreds	0 tens	9 units

Place value of 8 is
Place value of 0 is 0
Place value of 9 is



What I Understand

- ⦿ The face value of a digit in a number is same as the digit.
- ⦿ The place value of a digit in a number depends on its position.
- ⦿ If zero is at any place in a number, its face value and place value both are zero.

Exercise

⦿ Fill in the blanks with correct answers.

(1) In the number 460

Face value of 4 is
Face value of 6 is
Face value of 0 is

460		
H	T	U
4	6	0
4 hundreds 400	6 tens 60	0 units 0

In the number 460

Place value of 4 is
Place value of 6 is
Place value of 0 is

- (2) ⦿ In the number 519, the face value of 5 is ___ and the place value is ___
- ⦿ In the number 915, the face value of 5 is ___ and the place value is ___
- ⦿ In the number 951, the face value of 5 is ___ and the place value is ___
- ⦿ In the number 905, the face value of 0 is ___ and the place value is ___

(3) Write the place value of the underlined digit in the number.

123
708

910
189

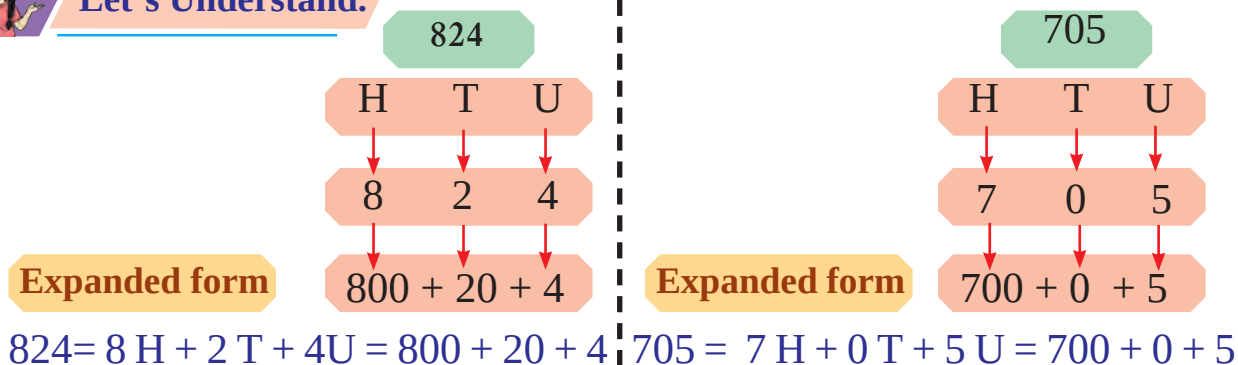
657
807

680
189

Expanded form of number



Let's Understand.



I Understood.

The expanded form of a number is the sum of the place values of the digits in that number.

Exercise

(1) Write the number from expanded form.

⊙ 5 H + 6 T + 7 U =

⊙ 3 H + 3 T + 0 U =

⊙ 2 H + 0 T + 5 U =

⊙ 9 H + 9 T + 9 U =

⊙ 4 H + 6 T + 7 U =

⊙ 8 H + 0 T + 0 U =

(2) Write the number from expanded form.

⊙ 500 + 30 + 8 =

⊙ 600 + 0 + 5 =

⊙ 100 + 30 + 9 =

⊙ 400 + 70 + 0 =

(3) Write the following numbers in expanded form.

⊙ 596 = 500 + +

⊙ 469 = + +

⊙ 600 = + +

⊙ 901 = + +

⊙ 999 = + +

(4) Write the given numbers in expanded form.

⊙ 287 = 2 H + 8 T + 7 U

⊙ 410 =

⊙ 549 =

⊙ 790 =

⊙ 666 =



Do The Activity.

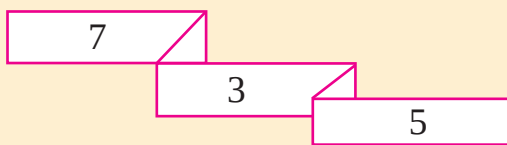
Expanded form of numbers using a folding strip

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Take a paper strip as shown in the picture and divide it into seven equal parts. Fold the paper along the lines of the strip. Write the expanded form of the number 735 on the strip as shown below.

7	00	+	3	0	+	5
---	----	---	---	---	---	---

Then fold the strip on the lines step by step as shown in the picture.

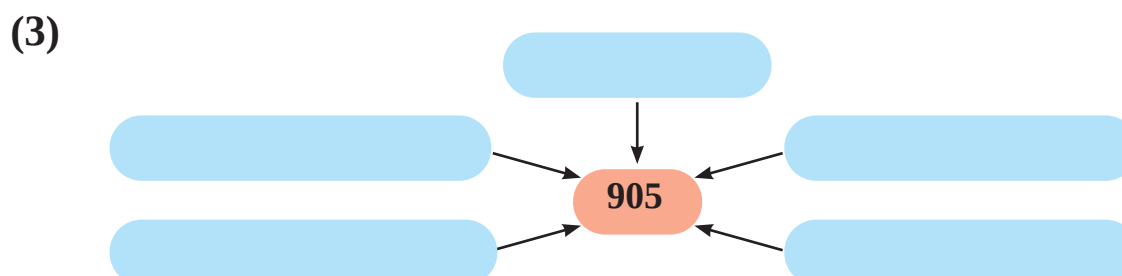
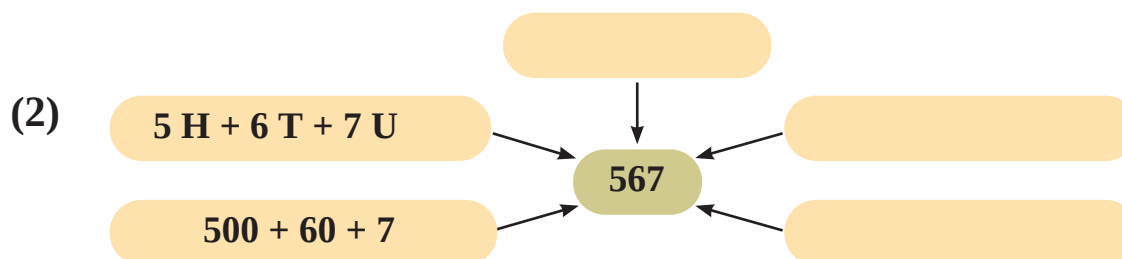
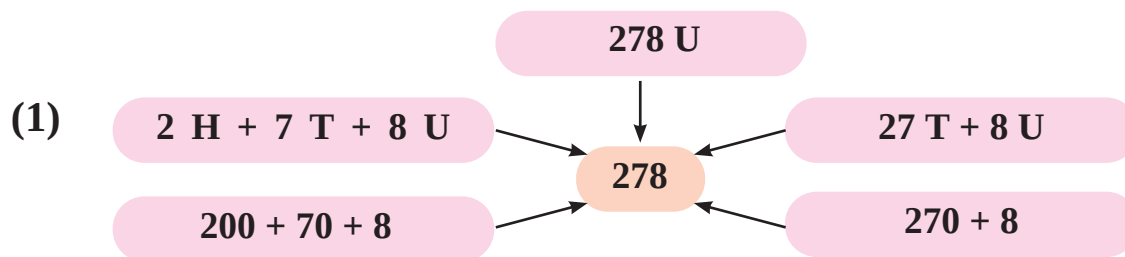


After folding, '00 +' and '0 +' get hidden. Now the number 735 is seen.

7	3	5
---	---	---

On the folded strip, the number is seen, and on the opened strip, its expanded form is seen.

◎ Write different forms of the same number. (The first example is solved.)



Comparing numbers (greater–smaller)



Let's Understand.



Which is the greater number between 6 and 9?

$9 > 6$, I know that 9 is greater than 6.



Which is the greater number between 36 and 29?

$36 > 29$ because I compared the digits at the tens place.



Which is the greater number between 178 and 99?

178 is a three-digit number and 99 is a two-digit number. Therefore, $178 > 99$. Hence, 178 is greater than 99.



Identify the smaller number between 495 and 812 and write the correct sign.

Comparing the digits at the hundreds place, $495 < 812$.



How to determine the smaller number between 314 and 319?

Here the hundreds are equal, so look at the digit at tens and units place. Digits at the tens are also equal. So compare the digits at units place. Since $4 < 9$, therefore $314 < 319$.



I Understood.

While determining the greater or smaller number among three-digit numbers, first compare the digits at the hundreds place. If the hundreds place digits are equal, then compare the digits at the tens place. If the hundreds and tens place digits are equal, then compare the digits at the units place.

Exercise

(1) Use the correct sign $>$, $<$ in the empty box.

721 589

423 723

600 497

901 919

813 99

300 624

150 501

549 559

459 452

(2) Match the correct pairs with a line

527 is greater than 272,
because

239 is smaller than 392,
because

$110 > 101$ because

475 is greater than 457,
because

$153 < 157$, because

475 and 457 have equal hundreds. In 475 there are 7 tens and in 457 there are 5 tens. $7\text{tens} > 5\text{ tens}$

153 and 157 have equal hundreds and tens, but the units are different. $3 < 7$

In 239 there are 2 hundreds. In 392 there are 3 hundreds. And $2\text{ hundreds} < 3\text{ hundreds}$

In 527 there are 5 hundreds. In 272 there are 2 hundreds. And $5\text{ hundreds} > 2\text{ hundreds}$

Both numbers have equal hundreds. $1\text{ tens} > 0\text{ tens}$

(3) According to the instruction, interchange the digits and write the resulting number in numerals and in words.

⦿ The number formed by interchanging the digits at tens and units place in 523.

⦿ The number formed by interchanging the digits at hundreds and tens place in 716.

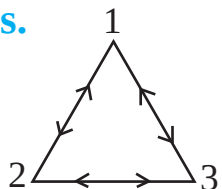
⦿ The number formed by interchanging the digits at hundreds and units place in 804.

Forming three-digit numbers



Let's Understand.

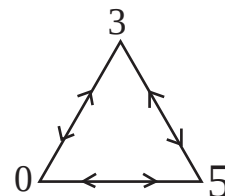
- Using each digit only once from the given digits, form three-digit numbers.



Formed three-digit numbers:

123, 132, 213, 231, 312, 321

- The greatest three-digit number from the above:
- The smallest three-digit number from the above:



Formed three-digit numbers:

305, 350, 530, 503

- The greatest three-digit number from the above:
- The smallest three-digit number from the above:
- Are 035, 053 three-digit numbers? Think over it.

Exercise

- From the given digits, using each digit only once, form the greatest and the smallest three-digit number.

Given digits	Greatest number	Smallest number
3, 4, 5		
6, 7, 8		
5, 0, 1		
7, 2, 0		

- Circle the smallest number.

- 239, 132, 321, 456
- 574, 573, 547, 537

- Circle the greatest number.

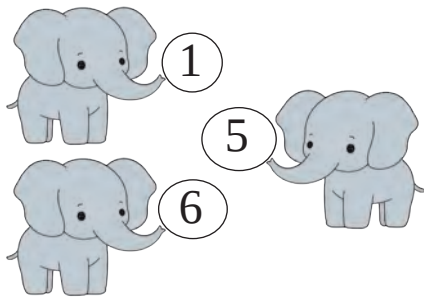
- 863, 384, 468, 348, 268
- 666, 637, 639, 647, 683

- Using digits 5, 2, 6, form all possible three-digit numbers using each digit only once and circle the greatest number.



For teachers: Three-digit numbers should not have zero in the hundreds place. Conduct discussion accordingly in class. Give practice using such examples.

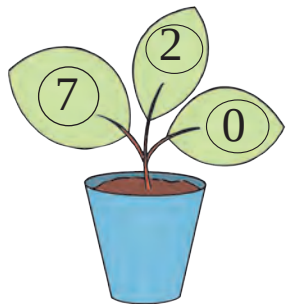
(5) Using each digit only once from the given digits, form three-digit numbers and write them in words.



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Guess, who am I?



Fun with Maths.

- (1) I am a three-digit number. I have only the digits 4 and 0. The digits in the units and tens place are the same. Who am I?
- (2) I am a number greater than 300 but less than 400. I do not have a tens digit, but the units and hundreds digits are the same. Who am I?
- (3) I am a number between 200 and 300. The digit in my units place is the greatest one-digit number and the digit in the tens place is the smallest one-digit number. Who am I?



For teachers: Using digits 8, 3, 4, form all possible three-digit numbers with repetition of digits. Tell how many numbers are formed.

Ascending and descending order of numbers

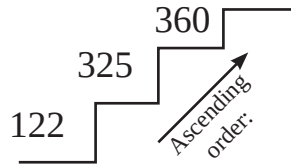


Let's Understand.

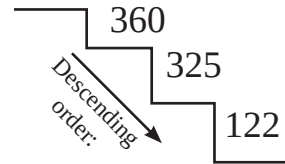
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- Arrange the numbers 122, 360, 325 in ascending and descending order.

Smallest number 122



Greatest number 360



Ascending order: 122, 325, 360

Descending order: 360, 325, 122

From the digits 7, 5, 8, the three-digit numbers formed using each digit only once:

587

578

758

785

857

875

When these numbers are written from smallest to greatest. This order is called ascending order.

578

587

758

785

857

875

When these numbers are written from greatest to smallest. This order is called descending order.

875

857

785

758

587

578

Exercise

- Write the given numbers in ascending and descending order.

Numbers	Ascending order	Descending order
117, 69, 50, 8		
285, 407, 589, 360		
217, 271, 270, 207		
99, 909, 990, 999, 90		

Coins and Currency Notes



Let's Understand.

① Write the value of these currency notes .



₹



₹

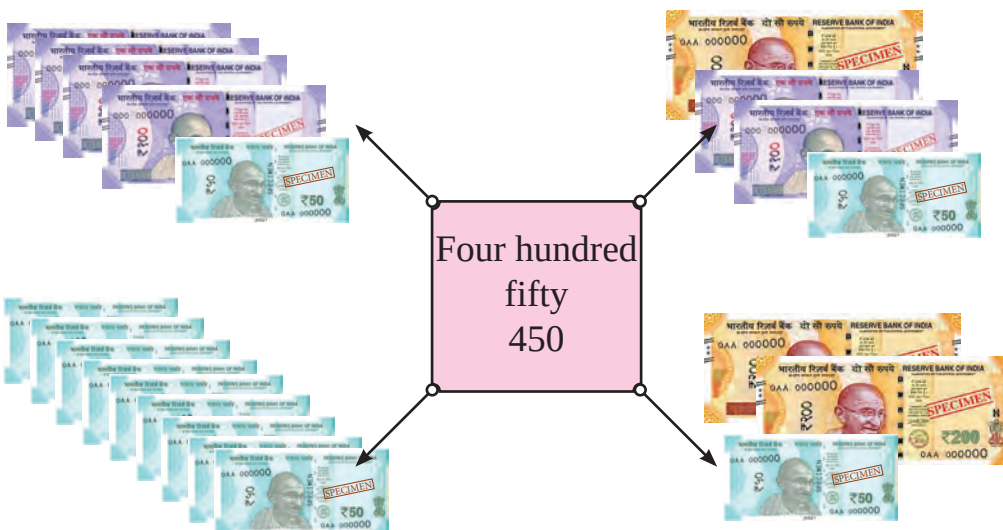


₹

② Count the currency notes and coins. Write the total amount in rupees.

Pictures of coins and currency notes	Amount in figures	Amount in words
	₹101	One hundred and one rupees
	₹102	
		One hundred and fifty rupees
		
		

③ Raviraj, brought a school bag worth ₹450. In how many different ways can this amount be given?



Cardinal and Ordinal Numbers



Let's Understand.



Names of children	Hasan	Sumit	Rohan	Mary	Gauravi	Arnav	Aditi
Position	First	Second	Third	Fourth	Fifth	Sixth	Seventh

- There are total 7 children in the bus line. Here **7** is a cardinal number.
- What is Gauravi's position in the line? Fifth
- At what position is Mary? Fourth
- What is Aditi's position in the line? Seventh

Here fifth, fourth, seventh are ordinal numbers.

1, 2, 3, 4, 5, ... are cardinal numbers.

First, second, third, ..., fiftieth, ..., two hundredth, ..., five hundred first are ordinal numbers.

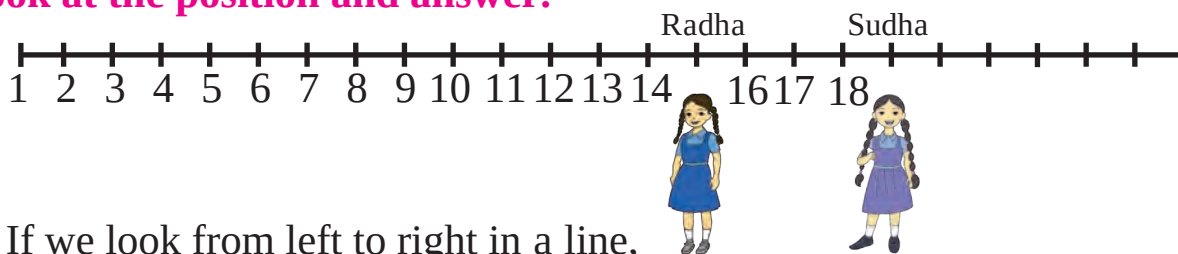
Exercise

- (1) Look at the picture. Count the compartments of the train after the engine. Write the correct answer in the box.



- How many compartments are there in total?
Here ____ is a cardinal number.
- In which numbered compartment the horse is?
- In which numbered compartment the rabbit is?
- Which animal is in the third compartment?
- What is the number of the lion's compartment?

(2) Look at the position and answer.



- ⦿ If we look from left to right in a line,
Radha is at the 15th position. Then Sudha is at the 19th position.
- ⦿ What is Sudha's position from Radha?

Math Puzzles

- There was a king who could count only up to 9. The king had many horses but he could not count them. A stable was built for the horses. The caretaker of the horses was Ramu. Ramu use to tie the horses in such a way in the stable that when the king looked from every side, he saw only 9 horses. How many horses did the king have in total?
- One day a traveller came to that village. He had 4 horses. It became dark. The traveller went to meet Ramu. He asked, "Can my 4 horses stay in your stable today?" Ramu said, "No, because our king counts the horses every night. There is no space in my stable to tie your horses. The king will not allow it." Ramu explained the reason to him. The traveller said, "I will arrange my horses in your stable in such a way that the king will not know that new horses have come here." The traveller arranged the horses in such a way that when the king counted from each side, he saw only 9 horses. Now how many horses are there in total?
- The next day that clever traveller used another trick. He took his 4 horses along with the king's 4 horses and arranged the remaining horses in such a way that when counted from each side, it appeared as 9. Because of this arrangement, the king did not realize that the traveller had taken the king's 4 horses. Now how many horses are there in total?
- Try to make one more such arrangement yourself.

3	3	3
3		3
3	3	3

2	5	2
5		5
2	5	2

4	1	4
1		1
4	1	4



For teachers : Encourage students to solve various types of puzzles based on numbers.

3.

Addition and Subtraction



Let's Recall.

Write the correct number in the empty boxes.

Add 2

57 → → →

Add 5

28 → → → →

Subtract 6

→ → → → 79

Subtract 3 and add 3

→ → 42 → →

Addition

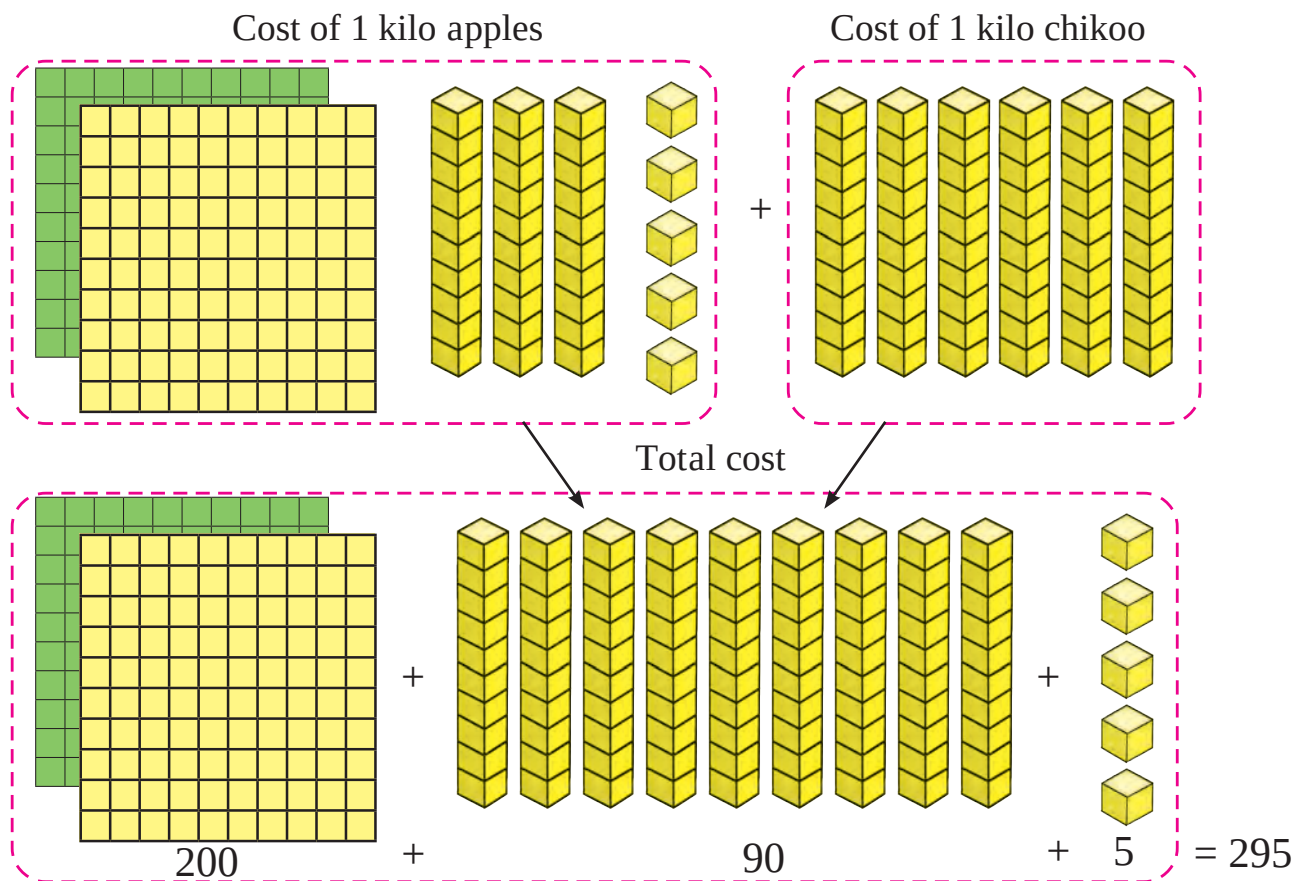


Let's Understand.



- ⦿ Gauri bought 1 kilo apples and 1 kilo chikoo. Then what is the total cost of fruits purchased? $\boxed{235} + \boxed{60} = \boxed{?}$

To do this addition, use blocks of hundreds, tens rods and unit cubes.

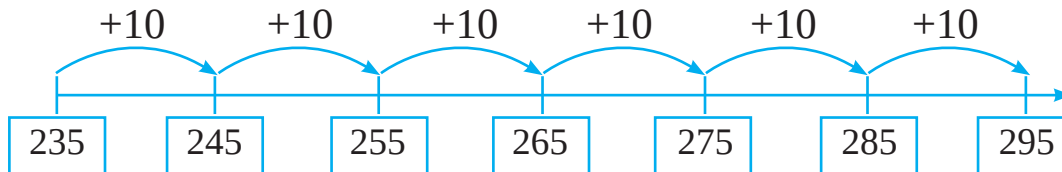


- ⦿ Addition by horizontal arrangement ⦿ Addition by vertical arrangement

$$\begin{array}{ccccccc} 2 & 3 & 5 & + & 0 & 6 & 0 \\ \hline 2 & 3 & 5 & + & 0 & 6 & 0 \\ \hline 2 & 9 & 5 \end{array}$$

H	T	U
2	3	5
+ 0	6	0
2	9	5

- ⦿ Addition using number line



Gauri bought fruits worth Rs. 295 from the shopkeeper.



Let's Think .

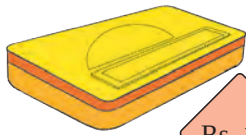
For the above fruit purchase, Gauri gave the shopkeeper 3 notes of 100 rupees, then how much money will the shopkeeper return to her?



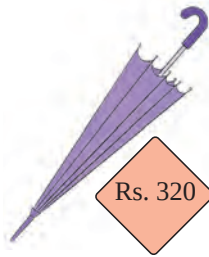
Let's Understand.

Meena bought a compass box and an umbrella from a shop. What is the total amount she paid to the shopkeeper?

Given information:



Rs. 145



Rs. 320

Cost of
compass box

145

Cost of
Umbrella

320

Total
cost

= 465

Required Information: Total amount

H	T	U
1	4	5
+ 3	2	0
4	6	5

Cost of compass box

Cost of umbrella

Amount paid to the
shopkeeper

Total amount paid by Meena to the
shopkeeper is 465 rupees.



Do The Activity

Add 1	Add 2	Add 5	Add 10	Add 100
476	476	476	476	476
477				
		491		

Exercise

(1) Add by vertical arrangement.

321 + 4

H	T	U

267 + 21

H	T	U

407 + 312

H	T	U

$457 + 210$

H	T	U

$354 + 513$

H	T	U

$675 + 324$

H	T	U

(1) Add by horizontal arrangement.

$147 + 521$

$209 + 580$

$210 + 79$

(3) Using the given digits only once, form the greatest three-digit number and the smallest three-digit number and find their sum.

$2, 4, 3$

$4, 5, 1$

$7, 2, 0$

$6, 4, 3$

Addition with carrying over



Let's Understand.

- Perform the addition: $628 + 357$
- First add the unit digits 8 and 7. The sum of the digits at the units place is 15.
15 units means 1 tens and 5 units. Write 1 as carry in the tens place.
Write 5 under the line in the units place.
The sum of the digits in the tens place is 8.
So write 8 under the line in the tens place.
The sum of the digits in the hundreds place is 9.
 $628 + 357 = 985$

H	T	U
	1	
6	2	8
+ 3	5	7
9	8	①5

- A trader sold rice 212, 386 and 259 kilograms in three days respectively. Then how many kilograms of rice did he sell in total?
The trader sold a total of kilograms of rice in three days.

H	T	U
2	1	2
+ 3	8	6
+ 2	5	9

Exercise

(1) Add

H	T	U
4	7	8
+ 3	4	5

H	T	U
2	7	6
+ 5	0	8

H	T	U
4	5	5
+ 2	6	7

H	T	U
6	3	2
+	5	7
+		6

H	T	U
3	2	7
+ 2	6	4
+	9	5

H	T	U
4	7	8
+ 2	0	9
+ 1	6	5

(2) Solve the following examples.

- ⦿ There were 381 books in a library. Subash donated 51 books to the library, then how many books are there in the library now?
- ⦿ In a farm there are 478 mango trees and 256 chikoo trees, then how many trees are there in total in that farm?
- ⦿ **Based on the given information, frame an addition word problem and solve it.**

Information: Workers 352, new workers 187

Example: In a factory there are 352 workers. If 187 new workers are recruited, then what will be the total number of workers in the factory?

Total workers will be in the factory.

H	T	U	
3	5	2	Workers
+ 1	8	7	New Workers
			Total Workers

Exercise

(3) From the given information, frame examples and solve.

- ⦿ In a basket – mangoes 35, guava 48 and chikoo 24.
- ⦿ Clothes – price of shirt 399, price of pants 499.
- ⦿ In a school – Third standard has 134 students, Fourth standard has 143 students, Fifth standard has 139 students.

Subtraction



Let's Understand.

GENERAL STORES

Things	Price
Drum	₹340
Football	₹330
Pant	₹499
Carrom board	₹240
Tshirt	₹470
Chess	₹350
Shoes	₹588



⊙ How much more is the price of the T-shirt than the price of the football?

Cost of the T-shirt = ₹470

Cost of the Foot ball = ₹330

To find the difference between the prices of the T-shirt and football, let's do subtraction.

Cost of the T-shirt

₹100	₹100	₹100
₹100	₹10	₹10
₹10	₹10	₹10
₹10	₹10	

H	T	U
4	7	0
-	3	0
1	4	0

Cost of T shirt

Cost of the football

Difference in the cost

The price of the T-shirt is 140 rupees more than the price of the football.

Exercise

⊙ Solve:

(1) $287 - 5$

H	T	U
-		

(2) $592 - 71$

H	T	U
-		

(3) $385 - 263$

H	T	U
-		

(4) $676 - 300$

H	T	U
-		

(5) $764 - 341$

H	T	U
-		

(6) $972 - 432$

H	T	U
-		

Subtraction (with borrowing)



Let's Understand.

FOR FREE DISTRIBUTION

- Yash purchased a carrom board and a chess set. Then how much money did Yash spend on purchase?

$$\boxed{240 \text{ rupees}} + \boxed{350 \text{ rupees}} = \boxed{\dots\dots \text{ rupees}}$$

Yash spent rupees on purchase.

Yash realized that his purchase crossed ₹500, so he gave one note of ₹ 500 and one note of ₹ 200 to the shopkeeper, i.e., ₹ 700. Then how much amount will the shopkeeper return to him?

$$700 - 590 = \text{how much?}$$

H	T	U
6	10	
7	0	0
- 5	9	0
1	1	0

First subtract units from units.

$$0 - 0 = 0$$

Now 9 tens cannot be subtracted from 0 tens, so take 1 hundred from 7 hundred, by untying.

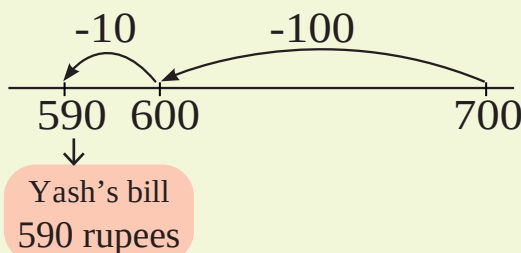
That becomes 10 tens. From 10 tens subtract 9 tens.

From the remaining 6 hundreds subtract 5 hundreds.

The shopkeeper returned 110 rupees to Yash.

- Showing subtraction using number line.

You can also find how much money the shopkeeper should return to Yash like this.



$10 + 100 = 110$ rupees
The shopkeeper returned ₹110 to Yash



For teachers : Take Practice of addition and subtraction examples. Use currency notes from mathematics kit to bring into effect real-life transactions.

◎ 507 – 268

H	T	U
4	⁹ 10	17
5	0	7
- 2	6	8
2	3	9

- 8 units cannot be subtracted from 7 units, so let's untie 1 tens.
 - But in tens place there is nothing, so untie 1 hundred from 5 hundred, so we get 10 tens. From that, 10 tens untied. 1 ten and 9 tens remain.
 - The untied 1 ten becomes 10 units; adding the earlier 7 units makes 17 units. From these 17 units subtract 8 units. We get 9 units.
 - From tens place subtract 6 tens from 9 tens, we get 3 tens.
 - From hundreds place subtract 2 hundreds from 4 hundreds. We get 2 hundreds.
- $507 - 268 = 239$

Exercise

(1) Solve:

◎

H	T	U
3	5	4
-	5	8

◎

H	T	U
7	8	5
- 3	5	8

◎

H	T	U
8	6	4
- 1	9	6

(2) Sayali has 500 rupees. She bought books worth 375 rupees. Then how much money is left with her?

(3) There are 832 students in a school. Out of them, 749 students went on a field trip. Then how many students did not go for trip?

(4) John's house electricity bill is 469 rupees. To pay it, he gave a 500 rupee note in the electricity bill payment center. Then how much money will he get back?

(5) In Adarsh library there are a total of 820 Marathi, Hindi and English books. Out of these, 252 are Hindi books and 279 are English books. Then how many Marathi books are there?

H	T	U
5	0	0
- 3	7	5

Number of Marathi books + Number of Hindi books + Number of English = TOTAL NUMBER OF BOOKS

?	+	252	+	279	=	820
---	---	-----	---	-----	---	-----

H	T	U
2	5	2
+ 2	7	9

Hindi Books
English Books
Hindi and
English Books

H	T	U
8	2	0
-		

Total number of books
Hindi and English
books
Marathi books

(6) Mother had 700 rupees. She bought sugar for 245 rupees and tea powder for 128 rupees. After the purchase, how much money was left with her? Fill in the blanks and complete the activity.

Money paid for sugar: rupees

Money paid for tea powder: rupees

Total expense: rupees

Mother had: rupees

Money left with mother

$$\begin{array}{r} 245 \\ - 128 \\ \hline \end{array}$$

$$\begin{array}{r} 700 \\ - \\ \hline \end{array}$$

(7) In a milk collection center, initially 624 litres of milk was stored. Later, 347 litres of milk was collected from farmers. Out of this, 512 liters of milk was given to restaurant owners and 346 liters of milk was given to other customers. How many litres of milk remained in the collection center?

☉ Frame the word problem for subtraction using the given information and solve it.

Given information:

Total pages in a book = 150

Pages read = 126

Example framed from the information:

Sayali read 126 pages out of a 150 page book. How many pages are left for her to read?

H	T	U
	4	10
1	5	0
- 1	2	6
0	2	4

Total pages
Pages read
Pages not read

☉ Book exhibition: Total books = 775, out of which 283 books were sold.

☉ Mango trees = 217, Chikoo (sapota) trees = 476



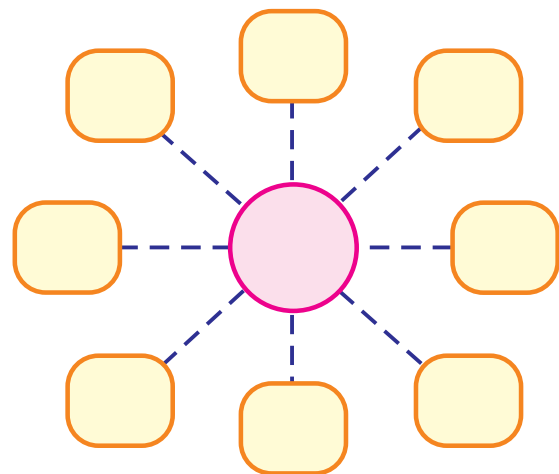
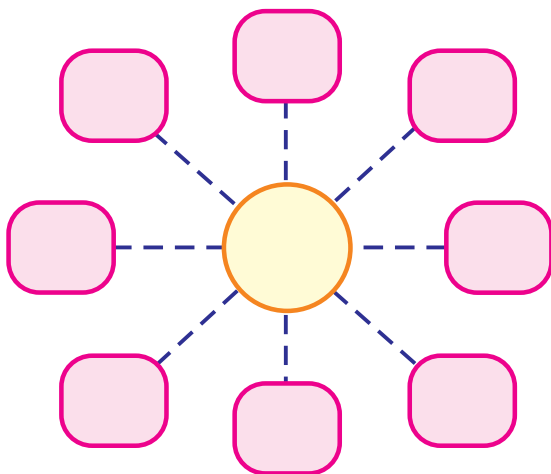
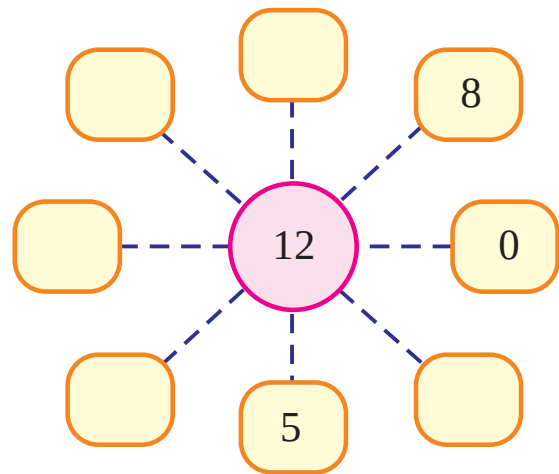
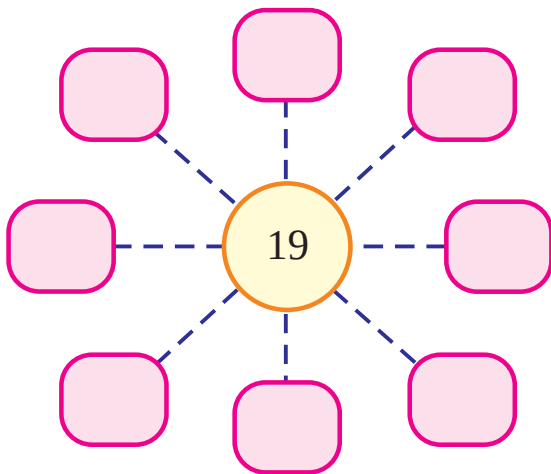
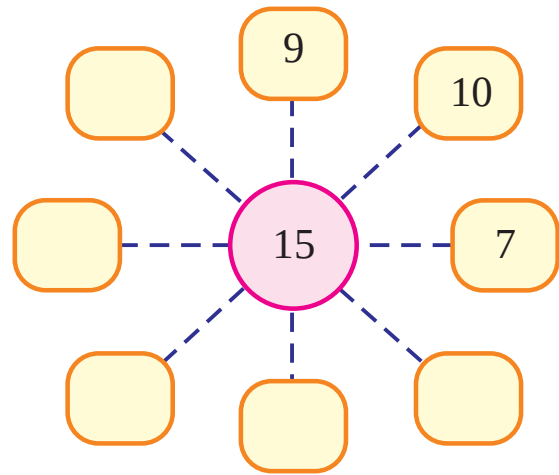
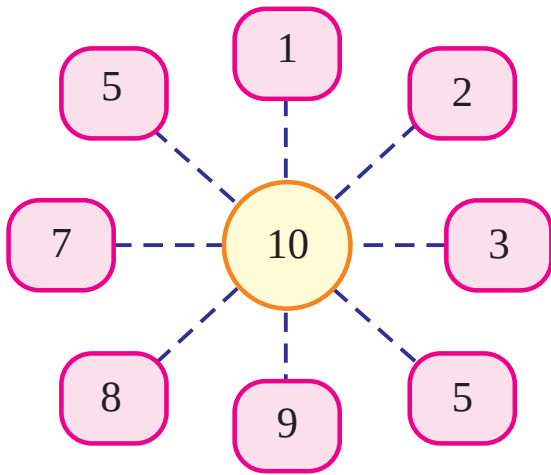
For Teacher : Ask the students to prepare word problems for addition and subtraction, revision and solve them in class.



Let's Understand.

Number Bonds

☉ Create number bonds up to 20.



For Teacher : Create number bonds for numbers up to 20 and take sufficient practice. This should be used while teaching addition and subtraction.

Co-relation between Addition and Subtraction

① Write the correct numbers in the empty boxes.

5	+	4	=	9
4	+		=	9
9	-	4	=	5
	-		=	

$$5 + 4 = 9$$

From the sum of the two numbers if one of the numbers is subtracted then we get the other numbers.

Exercise

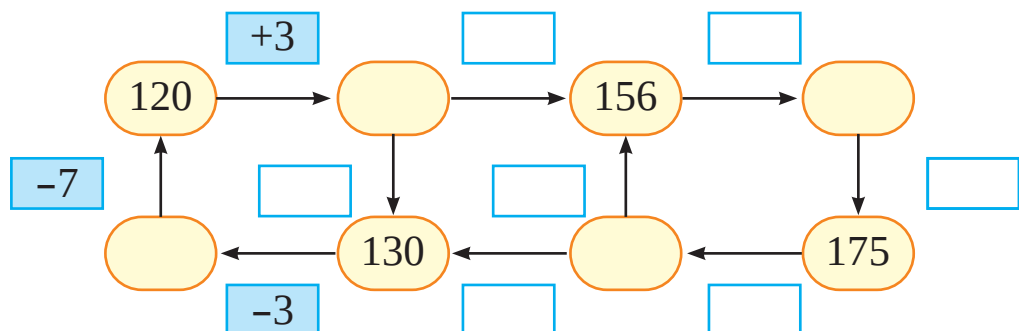
(1) Solve.

T	U		T	U		T	U		T	U		
2	6	→			→			→	6	7		
+			-	4		1	+		2	6	-	
6	7		2	6		6	7		4	1		

H	T	U	H	T	U	H	T	U
2	3	7	3	5	8			
+			+			-	4	3
5	5	7	7	4	3	1	5	1

H	T	U	H	T	U
9	6	7	3		2
-			-		
5	3	5	2	1	3

(2) Solve the following puzzle.



(3) Write the nearest hundred.

Number	Nearest hundred
$178 + 24$	200
$134 - 56$	
$457 + 67$	
$863 - 156$	

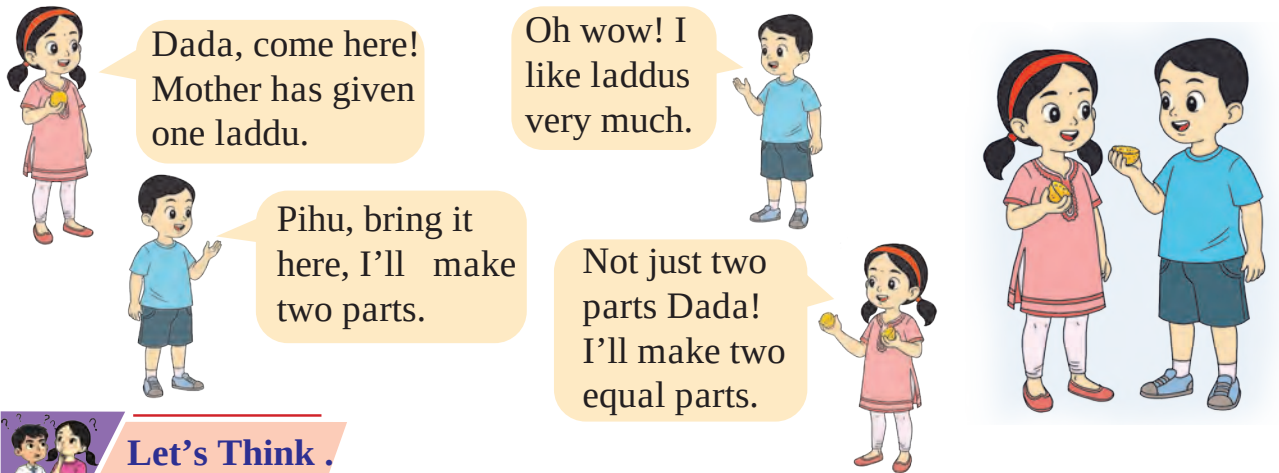
(4) From the following pairs, circle the one whose answer is greater.

- ☒ $352 + 14$ $352 - 14$
- ☐ $456 + 20$ $436 + 50$
- ☐ $654 + 27$ $598 + 67$
- ☐ $967 - 12$ $897 + 66$

(5) Find the pairs that have the same answers.

$256 + 292$	$813 - 237$	$462 + 106$
$259 + 289$	$719 - 143$	$916 - 348$





Let's Think .

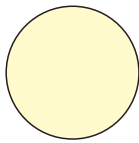
- (1) Pihu divided one laddu into two equal parts.
- (2) Dividing a bhakari into two equal parts and dividing a laddu into two equal parts, which one is easier to you? Why?

Half

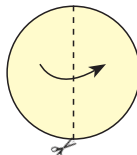
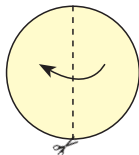


Let's Understand.

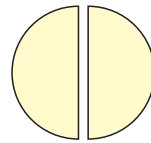
⦿ Divide a circular paper into two equal parts.



Take a circular Paper. Fold it.



Open the fold.
Cut on the fold.



Two equal parts are formed. Each part is half of the whole circle.

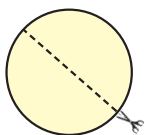
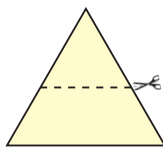
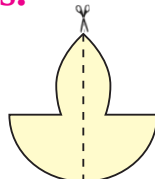
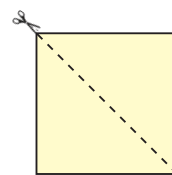
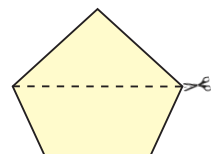


I Understood.

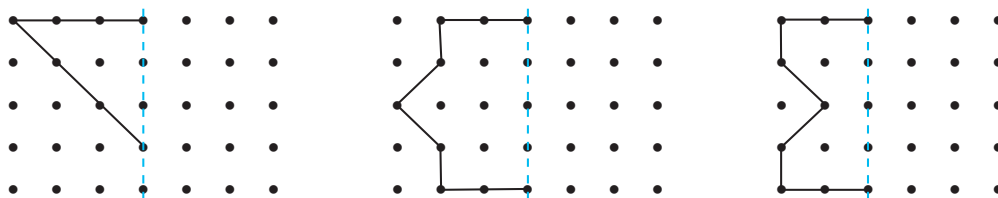
If any one shape, object or group is divided into two equal parts, then each part is called half of that shape, object or group.

Exercise

- (1) From the following shapes, tick (✓) for those in which the line divides the shape into two equal parts.


☐

☐

☐

☐

☐

(2) Draw the remaining half part and complete the figure.



Do The Activity.

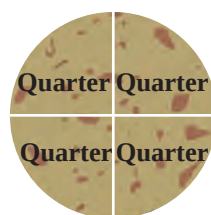
- (1) From the English alphabet A to Z, which letters can be divided into two equal parts?
- (2) Take paper cut-outs of the shapes of a circle, a square and a triangle, fold each shape and find in how many different ways it can be divided into two equal parts.



Let's Understand.

Quarter

- Meghana has one bhakari She divided that bhakari into four equal parts.



Half of a half is a quarter.

The bhakari is divided into four equal parts. Each part is a quarter of the whole.

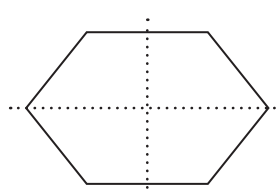
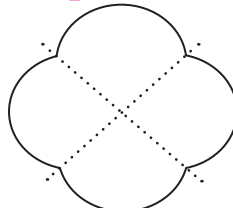
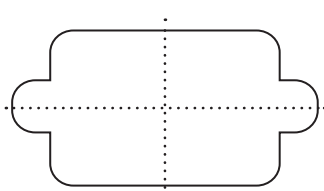
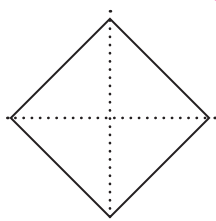


I Understood.

If any object or shape is divided into four equal parts, each part is called a quarter. Two quarters together make a half.

Exercise

● Colour one quarter of the following shapes.



For Teachers : Give students papers of different shapes and conduct the activity of dividing them into four equal parts. Provide opportunities for students to understand fractions through hands-on activity.



Let's Understand.

Three quarters

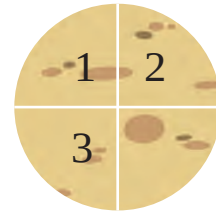


It was time for lunch,

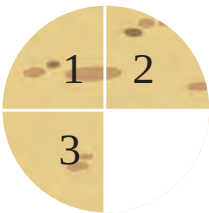
Puranpoli was made to munch.

Pihu was very hungry for lunch,

She divided the poli into four equal parts to munch.



Four equal parts were made to much



Pihu ate three quarters,

Remained only one quarter.



I Understood.

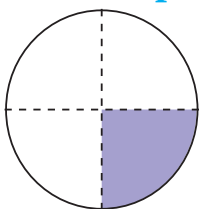
If an object or shape is divided into four equal parts and three parts are taken, then it is called three quarters.

Half part and one quarter part together make three quarter parts.

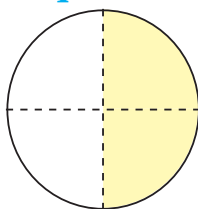
If one quarter is removal from a whole, then the remaining part is three quarters.

The part formed by combining three quarter parts is called three quarters.

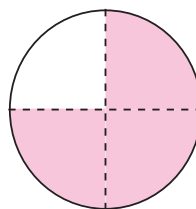
☉ Shaded part: From quarter to whole



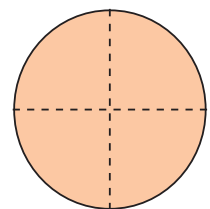
1 quarter
= Quarter



2 quarters
= Half



3 quarters =
Three quarters



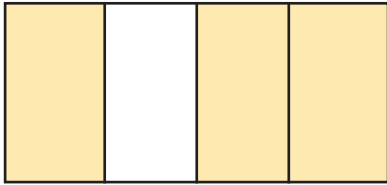
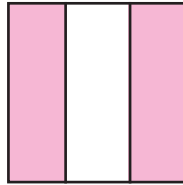
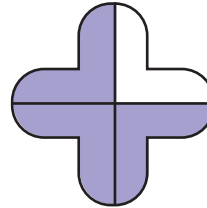
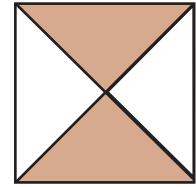
4 quarters
= One whole



For Teachers : Prepare songs based on mathematical concepts and present them rhythmically with actions.

Exercise

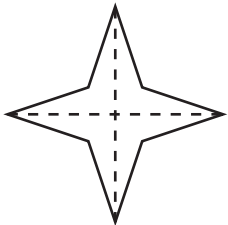
(1) Tick (✓) the box below the figure in which three quarters are shaded. Otherwise, mark (×).


☐

☐

☐

☐

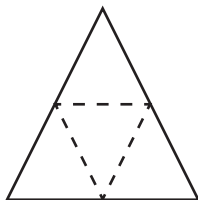
(2) Complete the following table.

Figure					
Shaded part	Quarter				Half
Unshaded part		Half		Three quarters	

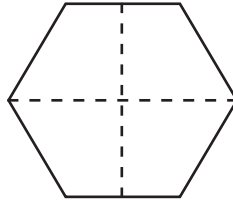
(3) Colour the indicated parts of the figures as given below:



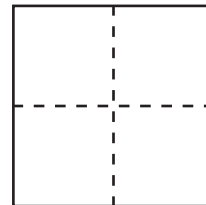
Three quarters



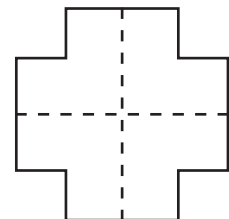
Quarter



Half



Half

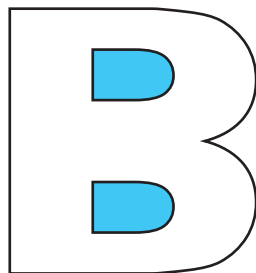


3 quarters

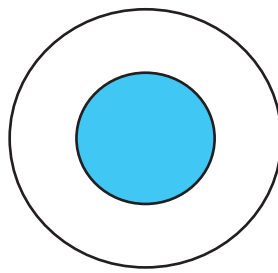
(4) Draw a line to show the part of letters as give below the letters. Colour that part.



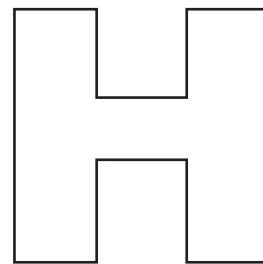
Half



Half



Three quarters



Quarter

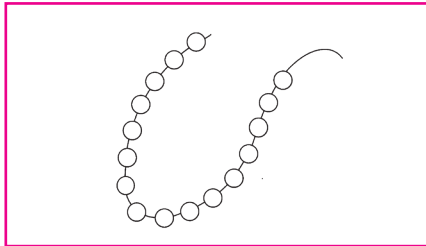
Half, quarter and three quarters of collection



Do The Activity.

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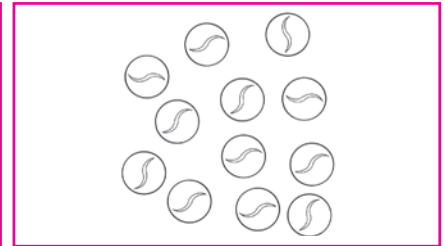
Consider the following collection and colour the suggested part:



(1) In the given picture, colour half of the beads of the necklace red and the remaining half black.



(2) In the given picture, colour one quarter of the balloons yellow.



(3) In the given picture, colour three quarters of the marbles blue.

Exercise

(1) Angel brought 1 dozen bananas. If she gave half a dozen bananas to her friend, how many bananas are left with her?

12 bananas

Gave 6 bananas to her friend

bananas remained



Write down the count :

1 Dozen = 12 bananas

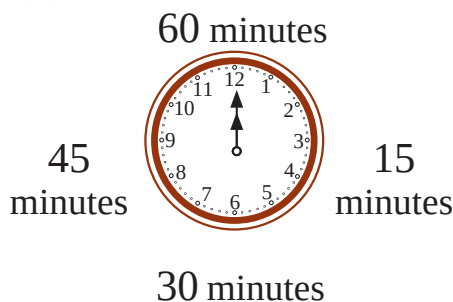
Half dozen = bananas

Quarter dozen = bananas

Three quarters of a dozen = bananas

(2) There are 12 months in a year. Then how many months are there in half a year? How many months are there in a quarter of a year?

(3) Solve:



Write down the count:

1 hour = 60 minutes

Half hour = minutes

Quarter hour = minutes

Three quarters of an hour = minutes

Write down the count

Half of 60 =

Quarter of 60 =

Three quarters of 60 =



For Teachers : Discuss and ask students to colour half, quarter and three quarters of the objects, based on the parts given in the collection. Explain that dividing the collection into 2 equal parts gives half, into 4 equal parts gives quarter, and combining half and a quarter gives three quarters. Take practice using different examples.

(4) Solve

Fill in the box:

Half of 100 =
Quarter of 100 =
Three quarters
of 100 =

Match the pairs:

Half of 12 25
Quarter of 100 60
Three quarters
of 80 6

Fill in the blanks:

15 minutes = hours
3 bananas = dozen bananas
100 runs = century

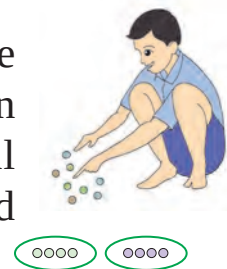
Half and Double

Let's Understand.

- I have 4 marbles.
I have half of your marbles.



- I have 8 marbles.
So, I have double
the marbles than
you have. Two equal
groups are formed
from 8 marbles.



I Understood.

- 'Half' means half of the collection.
- Double means multiplying a number by 2.

Exercise

(1) Fill in the blanks using the words double and half:

Raju and Meena were skipping. Raju did 10 skips. Meena did 20 skips.

- 10 skips are of 20 skips.
- 20 skips are of 10 skips.

(2) Write the correct number in the empty box:

- In a class of third standard, there are 30 students. Half of them participated in a running race, how many students participated in the running race? students.
- Father brought wheat and rice. The weight of wheat is double the weight of rice. If the weight of rice is 5 kilograms, what is the weight of wheat? kilograms.
- Sachin's father brought half a dozen notebooks. Half of them were given to his sister. The remaining notebooks were given to Sachin. How many notebooks did Sachin get? notebooks.
- A quarter of the amount Ajay has is Rs. 10. So how many rupees does he have in total? rupees.

Reading and Writing Fractions



Let's Understand.

Shape	Part	Written as number
	4 whole	4
	2 whole	2
	1 whole	1
	Half of one whole	$\frac{1}{2}$

Dada : Meghna, how many parts of a whole shape does represent the last figure in this table ?

Meghna: It is a half part.

Dada : Meghna, “half” is a fraction. Similarly, quarter and three-fourths are also fractions.

Meghna: Dada, how do we write “half” in numbers? And how do we read it?

Dada :

$$\frac{\text{Parts considered (numerator)}}{\text{Total equal parts of the whole (denominator)}} = \frac{1}{2}$$

$\frac{1}{2}$ is read as “one upon two” or “half”

Meghna: It's so easy! I understood.

(1) How to write fraction in numbers? (2) How to write ‘quarter’ in numbers?

$$\frac{\text{Shaded Parts}}{\text{Total equal Parts}} = \frac{\text{Numerator}}{\text{Denominator}} = \frac{4}{5}$$

$\frac{4}{5}$ is read as “four upon five.”

$$\frac{\text{Shaded Parts (Numerator)}}{\text{Total equal Parts of the whole (Denominator)}} = \frac{1}{4}$$

$\frac{1}{4}$ is read as “one upon four.”

(3) How to write ‘three-fourths’ in numbers?

$$\frac{\text{Shaded Parts (Numerator)}}{\text{Total equal Parts of the whole (Denominator)}} = \frac{3}{4}$$

$\frac{3}{4}$ is read as “three upon four” or “three parts out of four.”

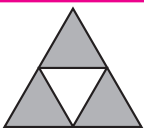
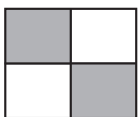
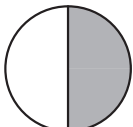
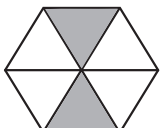
(4) Fill in the blanks and write the fraction:

$$\frac{\text{Shaded Parts (Numerator)}}{\text{Total equal Parts of the whole (Denominator)}} = \frac{\text{Numerator}}{\text{Denominator}} = \frac{\quad}{\quad}$$

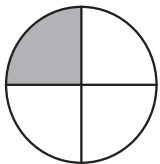
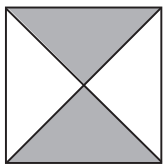
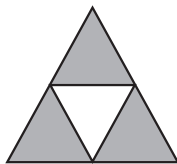
$\frac{3}{8}$ is read as

Exercise

(1) Complete the table

Shape	Shaded Parts	Total equal Parts	Numer-ator	Denomi-nator	Fraction form	Reading
	3	4	3	4	$\frac{3}{4}$	Three upon four
	2	4	2	4	$\frac{2}{4}$	Two upon four
		2			$\frac{\boxed{}}{2}$	One upon two
	3			5	$\frac{3}{5}$	upon.....
	4	6			$\frac{\boxed{}}{}$	
	2				$\frac{\boxed{}}{}$	

🕒 **Observe and compare the numerator and denominator.**

			
1 quarter	2 quarters	3 quarters	4 quarter
‘Quarter’ ($\frac{1}{4}$)	‘Half’ ($\frac{2}{4}$)	($\frac{3}{4}$)	‘One whole’ ($\frac{4}{4}$)



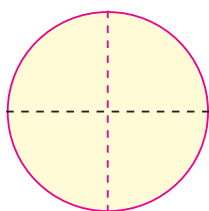
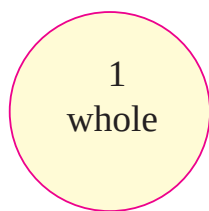
I Understood.

- 🕒 In fractions like ‘half’, ‘quarter’, ‘three quarters’, numerator is always smaller than denominator.
- 🕒 When numerator is equal to denominator, it becomes one whole.

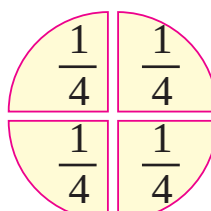


Let's Understand.

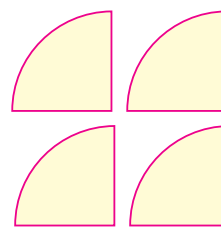
Unit Fraction



Divided into
4 equal parts



4 quarters



Each part = $\frac{1}{4}$

If you take fraction $\frac{1}{4}$ four times, it becomes 1 whole

Therefore, $\frac{1}{4}$ is called a unit fraction.



A flower has 5 petals out of them, 2 petals are coloured.

Coloured part is denoted by a fraction $\frac{2}{5}$.

A whole is divided into 5 equal parts. Each part is $\frac{1}{5}$.

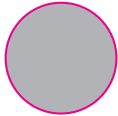
Here $\frac{1}{5}$ is a unit fraction. $\frac{2}{5}$ means $\frac{1}{5}$ taken 2 times.

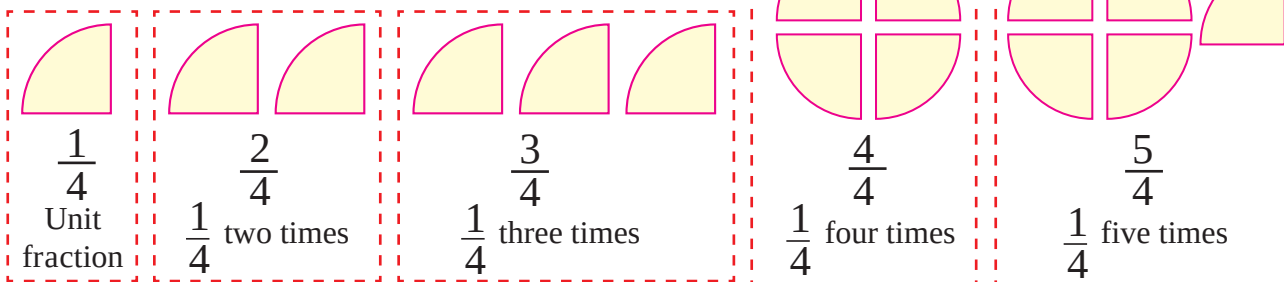
Improper and Proper Fraction



Let's Discuss.

- In fractions, is the denominator always bigger than numerator?
- If numerator becomes double of denominator, will it still be a fraction or become a whole?

A full  this  is a quarter part. Observe when such quarter parts are taken many times.



For Teachers : Using pictures, dots, lines and groups of items give students opportunities to solve examples related to half, quarter and three quarters. Encourage students to create practical examples on their own related to half, quarter, three quarters, one quarter and double. As required, suggest taking the help of friends and parents.



I Understood.

We can take a **unit fraction many times**, e.g. 3 times $\frac{1}{4}$ is $\frac{3}{4}$,
7 times $\frac{1}{4}$ is $\frac{7}{4}$

- $\frac{1}{4}, \frac{2}{4}, \frac{3}{4}$ in such fractions, the denominator is greater than the numerator, so they are Proper fractions.
- $\frac{5}{4}, \frac{6}{4}, \frac{7}{4}$ in such fractions, numerator is greater than the denominator, so they are Improper Fractions.

Fractions in which the denominator is greater than the numerator are proper fractions.

for example $\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{5}{9}, \frac{11}{14}$

Fractions in which the numerator is greater than the denominator are improper fractions.

for example $\frac{5}{4}, \frac{7}{2}, \frac{6}{5}, \frac{13}{9}, \frac{19}{11}$

Exercise

(1) Classify the given fraction into proper fraction and improper fraction

$$\frac{3}{5}, \quad \frac{2}{6}, \quad \frac{6}{5}, \quad \frac{3}{8}, \quad \frac{5}{4}, \quad \frac{7}{9}, \quad \frac{4}{3}$$

Proper
fractions

Improper
fractions

(2) Write the number in numerator to make improper fractions

$$\frac{\boxed{}}{4}, \quad \frac{\boxed{}}{5}, \quad \frac{\boxed{}}{3}, \quad \frac{\boxed{}}{7}, \quad \frac{\boxed{}}{2}, \quad \frac{\boxed{}}{4}$$

(3) Write the numbers denominator to make proper fractions

$$\frac{1}{\boxed{}}, \quad \frac{4}{\boxed{}}, \quad \frac{2}{\boxed{}}, \quad \frac{5}{\boxed{}}, \quad \frac{3}{\boxed{}}, \quad \frac{6}{\boxed{}}$$

(4) Write the numbers in the boxes to make improper fractions.

$$\frac{\boxed{}}{17}, \quad \frac{37}{\boxed{}}, \quad \frac{59}{\boxed{}}, \quad \frac{3}{\boxed{}}, \quad \frac{\boxed{}}{\boxed{}}$$



For teachers : Using the materials from the mathematics kit, give students practice in forming improper fractions and proper fractions.

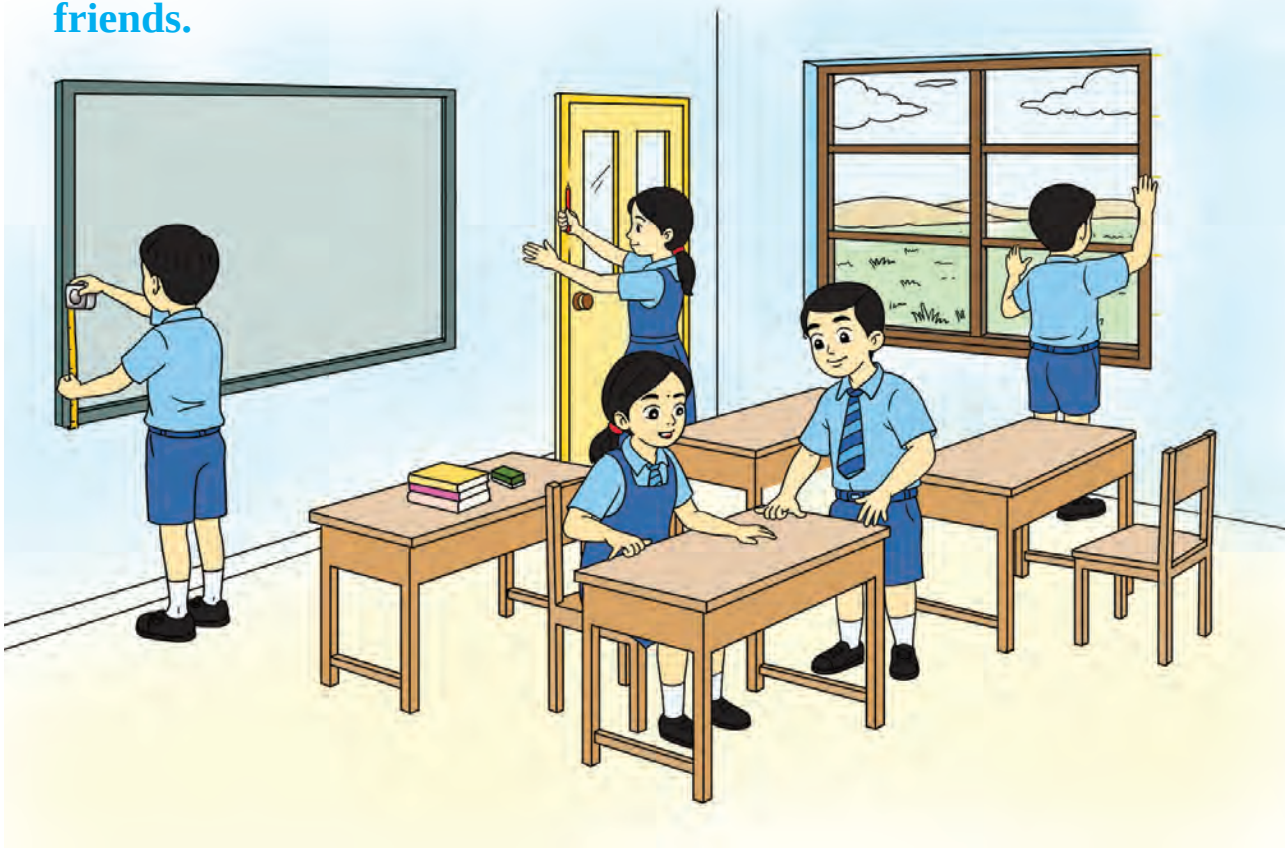


Measuring Length



Let's Discuss.

- Observe the picture given below and discuss the questions with your friends.



- (1) What is Shyam using to measure the length of the side of the window?
- (2) What are Anuj and Rina using to measure the length of the table?
- (3) Will the length of the table measured by Anuj and Rina be the same?
If not, why not?
- (4) What is Tanushree using to measure the length of the door?
- (5) What is Arnav using to measure the length of the board?
- (6) Which of the following will give the same measurement of length, regardless of who measures it - hands-span, foot-span, pencil, sticks, measuring tape or measuring rope.



For Teachers : Motivate students to perform activities of measuring length using different non-standard tools such as hand-span, pencil, rope, hand, footsteps, etc. Discuss in class the need for standard units over non-standard units.

Measurement using a metre scale

A sheet of cloth must measure the same no matter who measures it. That is why a long metal scale is used to measure cloth in a cloth shop.

If we need 3 metres of cloth, then no matter from which shop we buy it, its length will be 3 metres.



The length of a metre scale is one metre. **A metre is a standard unit for measuring length.** If the length is less than one metre, then it is measured in centimetres.

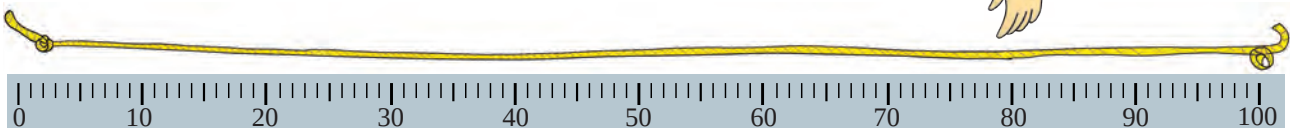
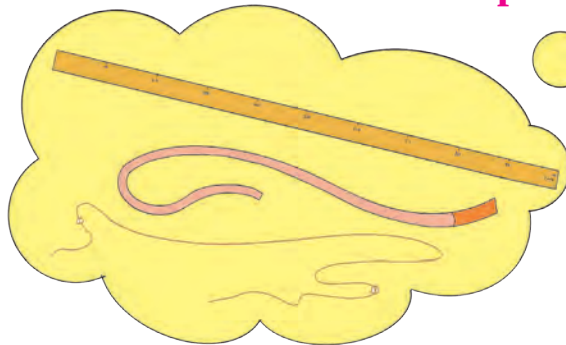
If 1 metre is divided into 100 equal parts, each part is 1 centimetre.

1 metre = 100 centimetres



Do the Activity.

Let us make a rope of length 1 metre.



For Teachers : Ask students to prepare a rope of length 1 metre. Using this rope, take practice of measuring the lengths of different objects in metres, for objects shorter than a metre and in centimetres.

- ◎ Measure the length of the following objects using the rope you have made and complete the table.

Object	Estimated length (less than / more than / equal to one metre)	Verify by actual measurement
Table		
Window		
Chair		

Quarter, Half, Three-fourth of One metre



Let's Understand.



This is a rope of one metre length.



Join both ends together. Now a rope of half metre length is ready. Mark half metre.



Join both ends of the half-metre rope again. Now a rope of quarter metre length is ready. Mark quarter metre.



Open the rope again. Now you can see 3 marks.



Do the Activity.

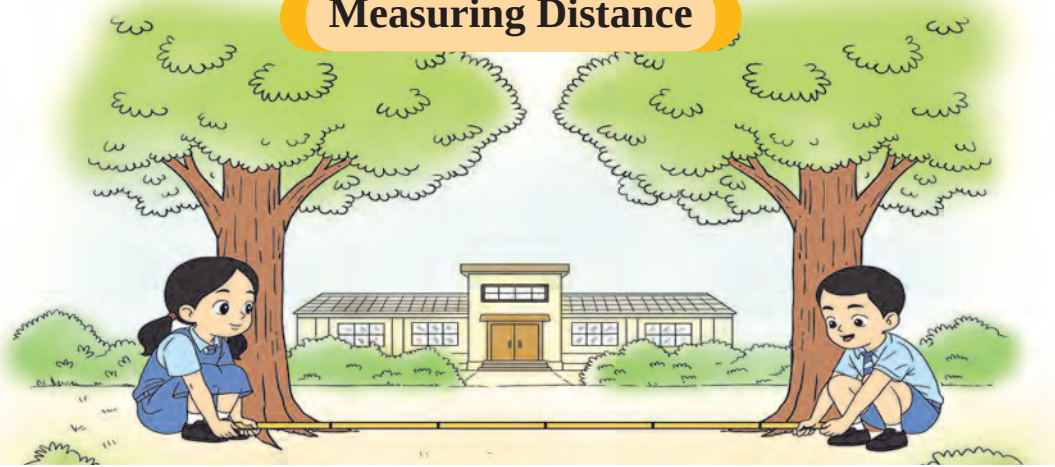
- (1) Using ropes marked as one metre, half metre, and quarter metre, determine whether the length of different objects is less than or more than quarter metre, half metre, and one metre. Put a tick ☒ in the correct box.

Name of objects	Compared to quarter metre		Compared to half metre		Compared to one metre	
	Less	More	Less	More	Less	More



For Teachers : Take a rope of one metre length. Hold it vertically, horizontally and diagonally and measure its length. Divide the one-metre rope into two equal parts. Then divide those equal parts again into two equal parts. Then measure the length of the four parts together.

Measuring Distance



The distance between two trees measured using a metre rope is more than 5 metres. Now let's measure that distance again using a metre scale in metres and centimetres. The distance between the two trees is more than 5 metres, so let's measure the remaining distance using a measuring scale. The distance between the two trees is 5 metres 40 centimetres.

To measure longer distances, metre is used, and for shorter lengths, centimetre is used.



Do the Activity.

- (1) Write whether the following lengths/distances will be measured in centimetres or metres:

Length of a pencil		Length of a notebook	
Distance between two houses		Distance from home to school	
Width of a road		Distance from classroom to playground	

- (2) Go to a ground/open space. From a fixed place, throw a ring far away and measure the distance the ring was thrown using a metre tape.
- (3) Measure the distance between the classroom door and the board using a metre scale.
- (4) Measure the length of the classroom using a metre scale.

Measuring Weight

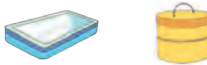
Heavy or Light?

Bhavana is holding a textbook in one hand and a pen in the other hand.

Put a tick ☒ below the heavier object in Bhavana's hand.



- ☉ For each of the statements given below, write your opinion and your friend's opinion by marking ○ around the correct word.

Object for comparison	My Experience	My friend's experience
	Lunch box is heavier/lighter compared to compass box.	Lunch box is heavier/lighter compared to compass box
	Lunchbox is heavier/lighter compared to school bag.	Lunchbox is heavier/lighter compared to school bag.
	Cricket bat is heavier/ lighter compared to a cricket ball.	Cricket bat is heavier/ lighter compared to a cricket ball.
	Notebook is heavier/lighter compared to a bat.	Notebook is heavier/lighter compared to a bat.



Do the Activity.

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- ☉ **Weigh the notebook.**

- (1) Everyone should take 15 to 20 different marbles.
- (2) Place the notebook on one pan of the weighing balance and start placing marbles on the other pan. Do this until both pans become equal.
- (3) How many marbles were needed to weigh the notebook?
Also note how many marbles your friend used to weigh the same notebook.



- ☉ **Complete the following table.**

Names of friends	Number of marbles required to weigh one notebook
Anuja	15 marbles
Meena	20 marbles
.....	marbles



Let's Discuss.

- ☉ Anuja found the notebook's weight to be 15 marbles, while Meena found it to be 20 marbles. Do you think why this happened?
- ☉ The number of marbles used by each person to weigh the notebook depends on the size of the marbles and the material they are made of (glass, lead, etc.). Discuss this.
- ☉ If someone else uses the same marbles as Anuja to weigh the same notebook, will there be any difference in weight?



For Teachers : Students should understand that heavier objects tilt the balance. Ask students to prepare a balance and take their practice in weighing different objects using it.



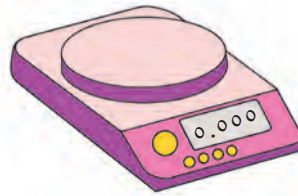
I Understood.

The weight of an object remains the same no matter who measures it using the same weights. For measuring weight, standard units and standard weights are used.

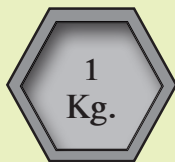
Kilogram is a standard unit of measuring weight.

1 kilogram = 1000 grams

Different instruments for measuring weight



Where have you seen such instruments for measuring weight?



In daily life, these measures are used to measure weight of objects.

Observe the pictures.



For Teachers : Using a weighing scale and 1 kg bag of salt, give students practical experience of 1 kilogram, half kilogram and quarter kilogram.



Do the Activity.

- (1) Find objects whose weight is 1 kilogram with the help of your parents and experience lifting them by hand. Estimate which other objects having weight of 1 kilogram.
- (2) From your daily life, write the name of objects that have '1 kilogram' written on them.

.....

.....

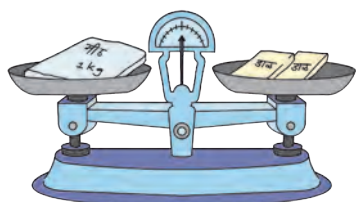
In ancient times units like *Man*, *Sher*, *Tola*, *Gunj* were used to measure weights.

Exercise

- (1) Estimate whether the given objects weigh less than or more than 1 kilogram. Then measure and verify. Mark a tick ✓ in the correct place in the boxes given below.

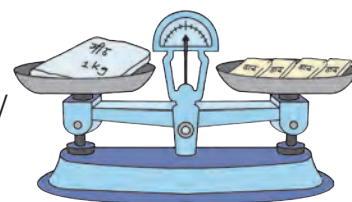
	Bucket filled with water	Book	A handful of rice	Objects
Objects				
Estimated: less/more than 1 kilogram				
After actual measurement: less/more than 1 kilogram				

- (2) Observe the weighing balance as shown below and circle the correct word in the sentences.



- ⦿ The weight of a bag of dal is 1 kilogram / half kilogram.

- ⦿ The weight of the bag of dal is 1 kilogram / half kilogram / quarter kilogram.





Do the Activity.

For the Fun Fair, children came to school in coloured dress. At that time, Manisha, Neeraj, and Salma recorded the colours of their dresses.



Everybody looks very nice in their different coloured dresses.



By looking at the pictures, can we find out who likes which colour? To do this, we will draw a vertical line for each colour.

In this way, we can record the colours of their dresses in a table.



(1) Tally marks have been used while classifying the information collected by Manisha, Salma, and Neeraj. Count those marks and write the number of children.

Colours	Red	Green	Yellow	Blue	Total number of children
Tally Marks					
Number of Students	6				

Manisha: In all there are 28 children in total. Among them, the number of children wearing blue-coloured dresses is the highest, which is 10.

Salma : The number of children wearing yellow-coloured dresses is the least, which is 4.

Manisha: The number of children wearing green-coloured dresses is 2 more than those wearing red-coloured dresses.

Teacher : How did you collect this information?

Salma : We prepared this table by observing the colours of dresses of 28 children. For each child, we used one vertical line.

Neeraj : Yes, we counted those lines and wrote them in the form of numbers.

Teacher : These lines are called tally marks.

The school had organized a field trip for the students. It was decided that each students should bring a fruit of their choice along with their lunchbox for the trip.



(1) Observe the above picture. Discuss the information shown in it. Write the answers to the following questions with the help of the given table.

Names of fruits	Chikoo	Guava	Banana	Orange
Number of children who brought their favourite fruit	5	2	3	4

- ⦿ How many students came for the field trip in total?
- ⦿ Number of children who brought chikoo
- ⦿ The fruit brought by most of the students is
- ⦿ The fruit brought the least by the students is

(2) Fill in the blanks with the correct word: more / less / equal.

- ⦿ Number of students who likes guava is, than the number of students who likes chikoo.
- ⦿ Number of students who likes chikoo is,than the number of students who likes banana .
- ⦿ Number of students who likes oranges is,than the number of students who likes guava.



For Teachers : While classifying the collected information, the lines used are called tally marks. While drawing lines, instruct to leave a little space after every group of five lines, and explain how the concept of tally marks was used in early mathematics for counting in story-format.

Exercise

(1) Observe the table prepared from the picture and write the answers of the following questions.



Modes of transports used by students of a class to go to school	Number of students
	6
	15
	5
	2
	4

- Most the students come by
- Least number of students come by
- The number of students coming by school bus is than the number of students coming by rickshaw.
- The number of students coming by bicycle is than the number of students come walking.
- The number of students coming to school by bicycle is less than the number of students coming by
- The number of students coming to school without using fuel is



For Teachers : Observe the picture and discuss its co-relation with the prepared table. Guide students how to read and understand the table.

(2) Ahmad prepared a table showing the information of students who participated in the interschool sports competition. Based on that, answer the questions.

Type of Game	Students participating in the sports competition	Number of Students
Long jump	   	4
Running	     	6
Skipping	  	3
High jump	    	5

- ⦿ In which game did the maximum number of students participate?.....
- ⦿ How many students participated in long jump and high jump together?
.....
- ⦿ How many students participated in skipping?
- ⦿ What is the total number of students who participated in sports?.....

(3) Information about students participating in different art forms in your school's cultural program is given. Complete the table from it and answer the questions.

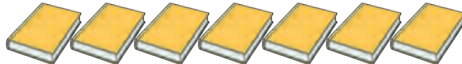
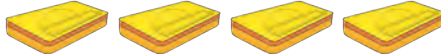
Art Form	Participating Students	Number of Students
Dance	   	
Monologue	    	
Instrumental	  	
Singing	 	
Costume	     	

- ⦿ How many types of art forms were organized for the cultural program?
- ⦿ How many students participated in singing?
- ⦿ In which art form did the maximum number of students participate?
- ⦿ In which art form did the minimum number of students participate?



For Teachers : Prepare a table using tally marks showing how many students in the class come to school by different modes of transport. Conduct a discussion based on it.

(4) The information about the items sold in one day from the school cooperative store is given in the table below.

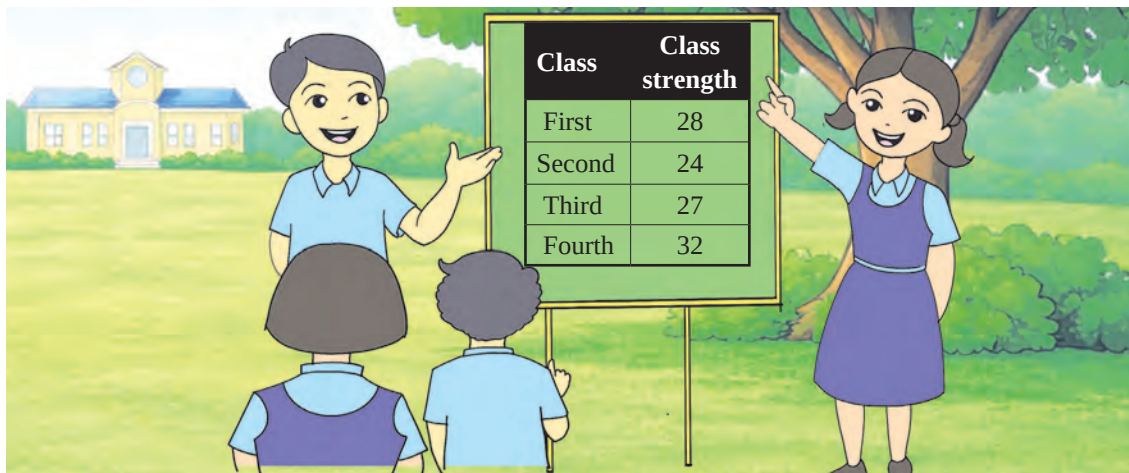
Name of Item	Items Sold
Notebook	
Pen	
Pencil	
Compass Box	

Frame questions based on the above table.

Example: Which item had the highest sales in the cooperative store?

- -----
- -----
- -----

(5) Observe the following picture. Write what kind of information is given.



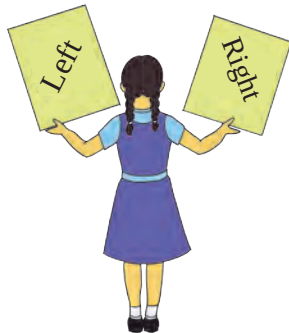
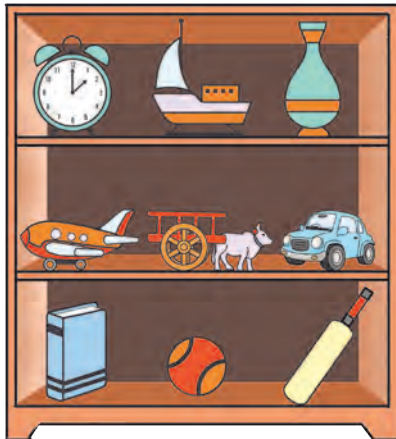
For Teachers : While presenting information, explain how to use tables, pictures/diagrams, and tally marks with examples. Give students newspapers, observe the given information, and help them read, compare, and interpret meaning. Explain how information is managed based on this.



Let's Understand.

Position and Location

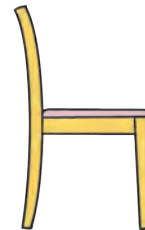
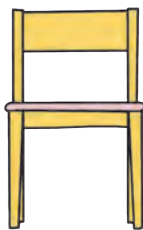
Observe the cupboard and write the answers.



- In the upper most compartment of the cupboard, on the left side of the ship, there is a **clock**.
- In the middle compartment of the cupboard, between the aeroplane and the car, there is a
- In the compartment of the cupboard, the aeroplane is the book.
- In the cupboard, exactly above the car, there is a
- To the right of the aeroplane, there is a

Types of Views: Front, Top, Side

Different views of a chair are shown below:



Front view of the chair

Top view of the chair

Side view of the chair

Objects have different views such as top view, side view and front view.



For Teachers : Take a game of hide-and-seek in the school and conduct a discussion among the students about its information, rules, and scenes. Also, facilitate a discussion on spatial concepts such as up-down, left-right, near-far, etc. Further, discuss how different objects in the surroundings appear when viewed from the front, from above and from the side.

◎ Razia, Yash, and Meena have drawn pictures of the same bus from different views, shown below.



Bus view



Bus view



Bus view

Game (1):

Various 3D objects are placed on the table. From these, draw in your notebook the front view, side view and top view of each object.

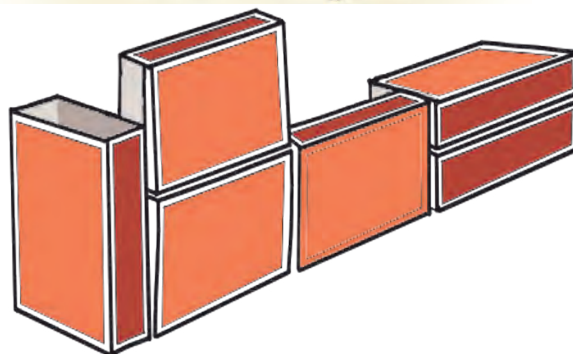


Game (2):

Saniya and Michael are playing a game of building a structure using empty match boxes.

They are drawing different views of that structure. You also draw the same.

You may arrange the boxes in any way and create different structures, and then draw their front view, side view, and top view.



Exercise

(1) Look at the following pictures. Write the name of the object. Also write whether it is the front view, top view or side view of that object.



Name of the object:

.....

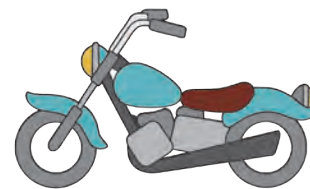
View:.....



Name of the object:

.....

View:.....



Name of the object:

.....

View:.....



For Teachers : Give opportunities to the students to play game with their friends that help them identify front view, side view, and top view using objects like eraser, water bottle, pencil, notebook etc.

(2) Using the available material, make a model of a house and draw its diagram. Draw the views of how the house model from the front, side, and top views.

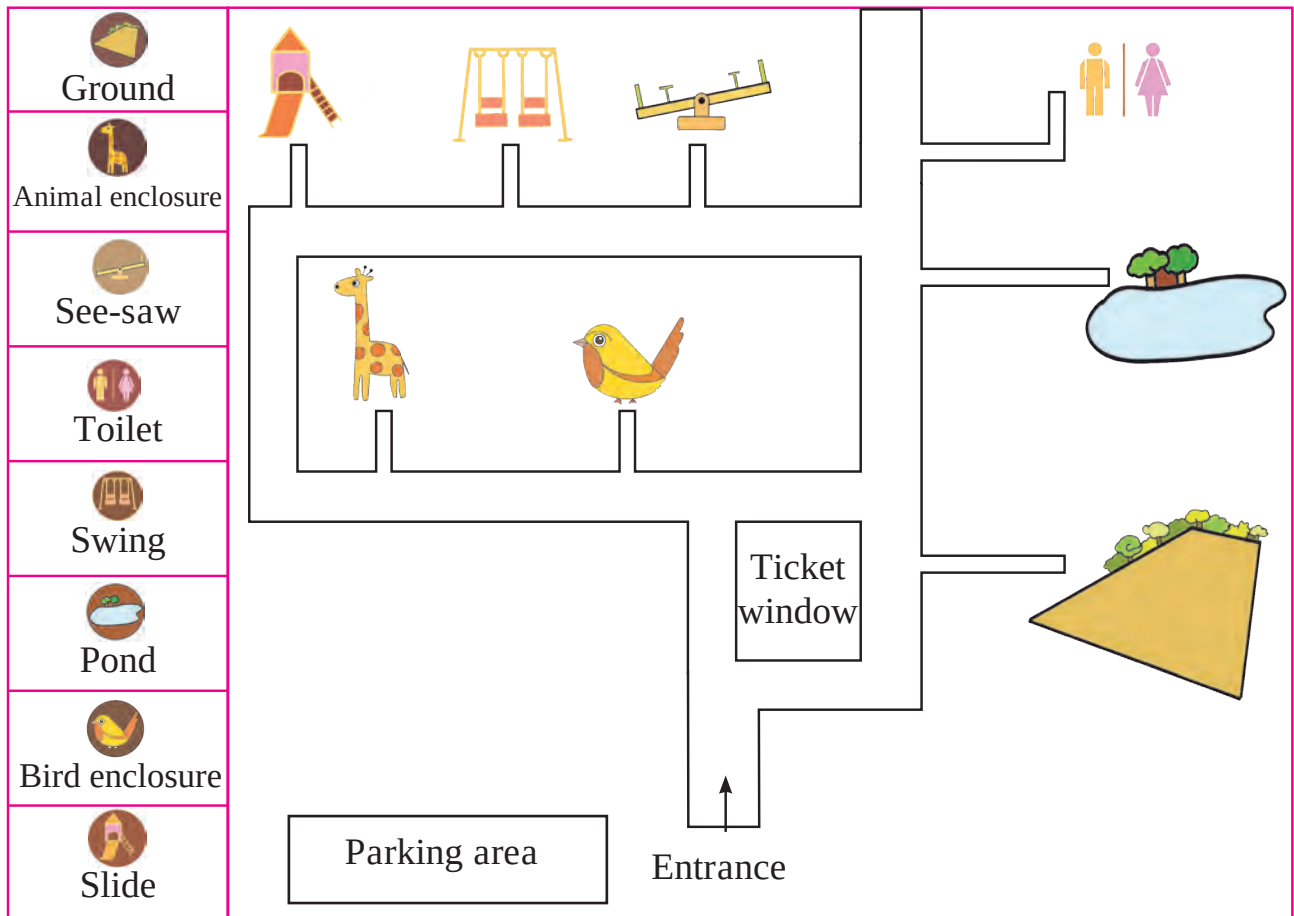
Pictorial Map of Garden

FOR FREE DISTRIBUTION



Let's Understand

Symbol and Name



Solve

Observe the given instructions of the garden map and answer the following questions:

- (1) What does this symbol represent?
- (2) From the entrance of the garden, colour the shortest path leading to the slide.
- (3) On which side of the swing is the slide located?
- (4) What is on the left side of the bird enclosure?

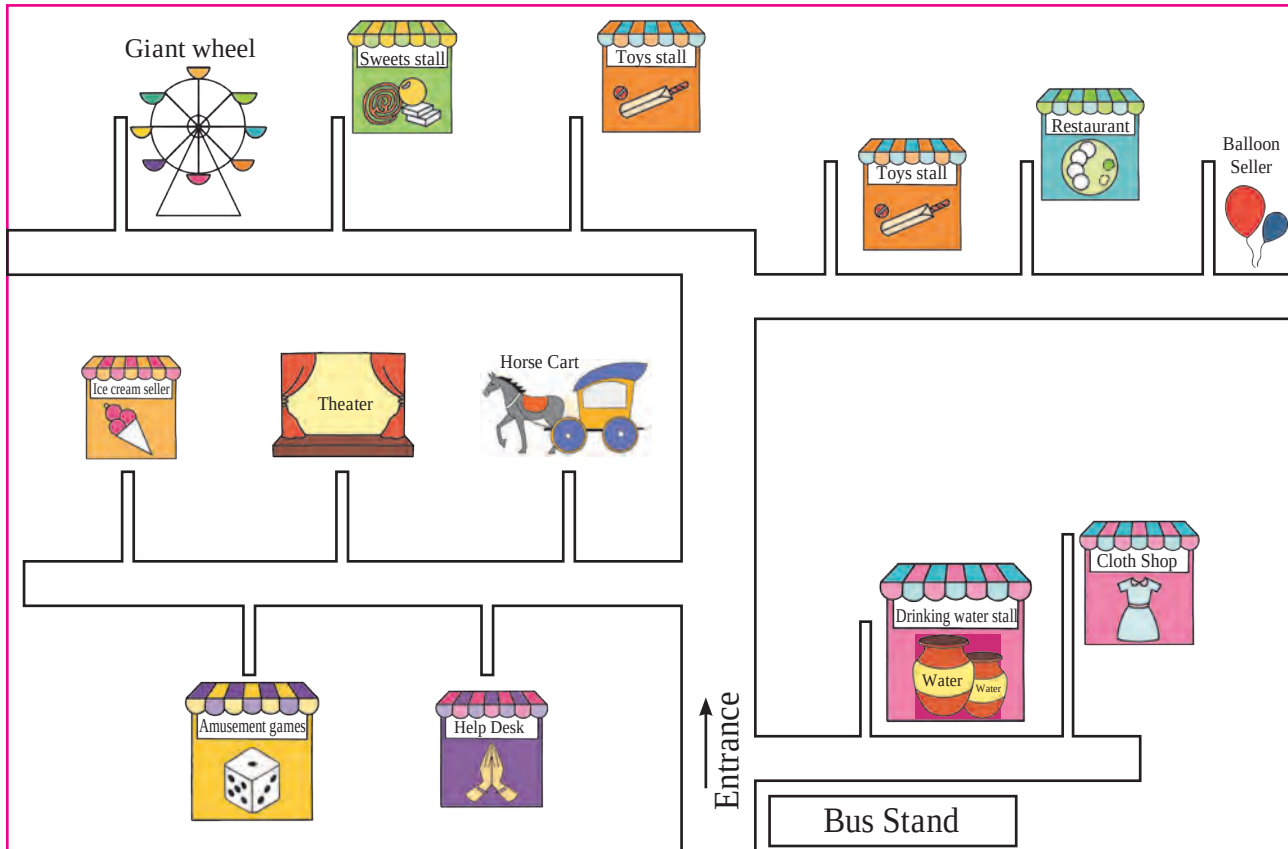


For Teachers : Encourage students to discuss and classify the different views they have drawn.

Fair pictorial Map

Neha and Sadhana went with their families to see the village fair. The pictorial map of that fair is given below.

Symbols and Names



Exercise

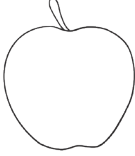
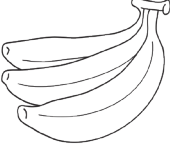
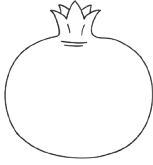
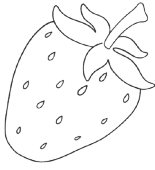
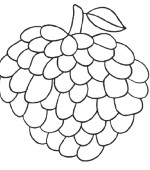
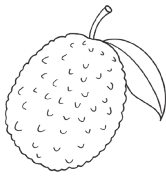
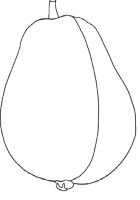
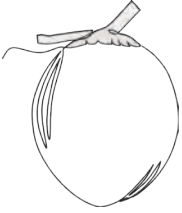
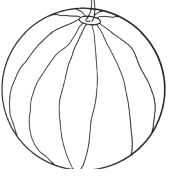
◎ Observe the above pictorial map and answer the following questions.

- (1) How many toy shops are there?
- (2) On which side of the restaurant is the balloon seller?
- (3) What is on the left side of the horse cart?
- (4) From the bus stop, circle ○ the place that is the farthest from the following?
 - 1) Drinking water stall 2) Giant wheel
 - 3) Horse cart 4) Restaurant
- (5) In your notebook write the directions you would give Neha to reach the Help Desk from the Giant wheel?



For Teachers : Teachers should ask various questions related to direction and location, and conduct related games.

☉ Using the grid, find the exact position of the objects.

4				
3				
2				
1				
	A	B	C	D

In the above grid, there are 4 columns (vertical lines) and 4 rows (horizontal lines). For the columns, the letters A, B, C, D are given below. For the rows, the numbers 1, 2, 3, 4 are given in order. Make use of the gride to write the position of objects. For example, the position of the banana is C–4.

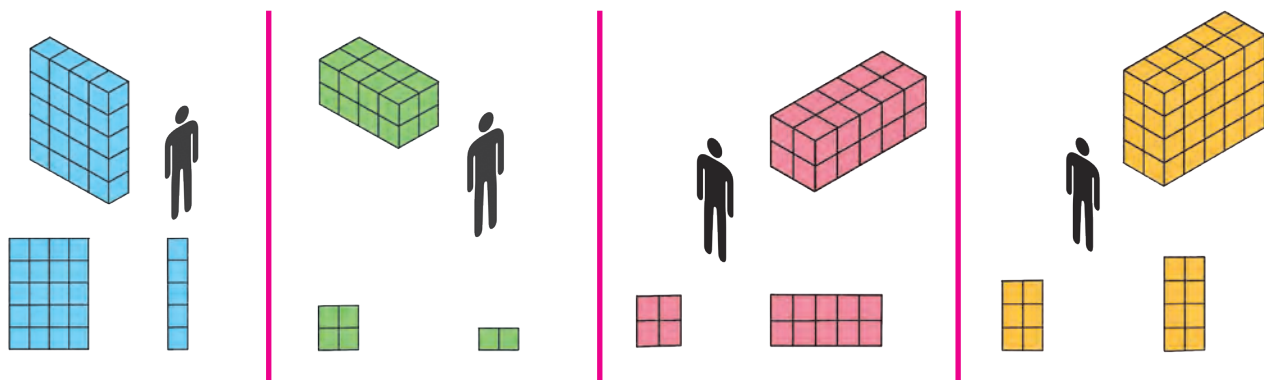
☉ Do the following activities according to the instructions given below:

- (1) Colour the fruit to the right of the grapes with orange colour and its stem with green colour.
- (2) Colour the picture in A–2 with yellow colour.
- (3) Write the name of the fruit that is above D–3.
- (4) Write the name of the fruit in B–2.
- (5) State the position of the pineapple.

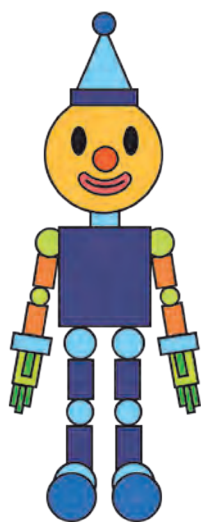


For Teachers : Using a computer, explain the concept of a grid in Excel. Give practice to explain the exact position of objects in a 3×3 or 4×4 grid.

Put a tick (✓) on the correct figure of blocks visible to the observer. 



Riya and Neel, while playing with different 3D shapes, made a clown.



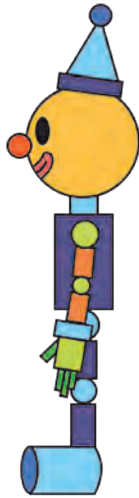
Clown's picture

Wow! What a beautiful clown has been made. I liked it.



This is its front view.





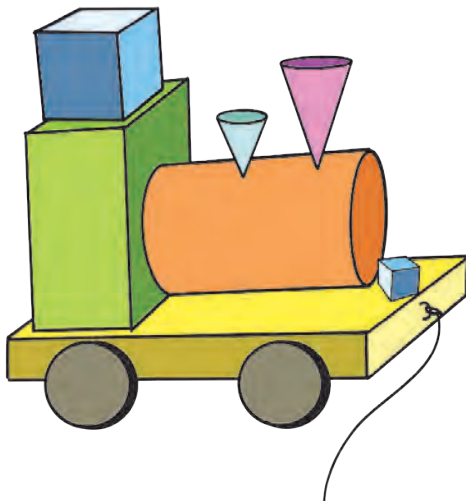
This is the side view of the clown. It looks very nice.



This top view of the clown looks nice.



Let's Discuss.



In the adjacent picture, observe the engine made using three-dimensional shapes like cylinder, cube, cone, cuboid etc. Discuss with your friends about its side view, front view and top view.

- ◎ Using three – dimensional shapes from the mathematical kit such as cuboid, cube, cylinder, cone, etc., make objects like a bus, house, table, truck, bridge, etc. Draw pictures showing their side views, top views and front views.



For Teachers : Provide students with three-dimensional shapes like cube, cuboid, sphere, etc., give them opportunities to create different objects and encourage them to develop their creativity.



- During the holiday, Ramesh and Surekha helped their parents to fill the boxes with oranges. If Surekha filled 5 boxes with 6 oranges in each box, how many oranges did she fill in total?

$6 + 6 + 6 + 6 + 6 = 30$, which means, five times six. This is written as $6 \times 5 = 30$ and is read as 'six five times equals thirty'. Surekha filled total 30 oranges in the boxes.



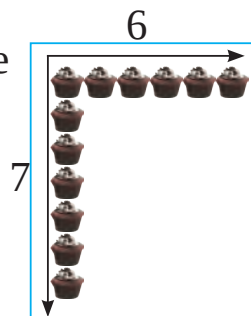
Let's do it.

- (1) Complete the picture and write how many cupcakes are there in total.

Here 6 cakes..... times or 7 cakes..... times = ... cakes

We can also say that seven times... or six times 7 means that there are a total of cupcakes.

$$6 \times 7 = 7 \times 6 = \boxed{}$$



- (2) The price of an ice cream cone is Rs.5. If 9 people buy one ice cream cone each, how much will they have to pay in total?

5 rupees will have to be taken 9 times

That is, $5 \times \dots$ or $9 \times \dots = \dots$

Let's say 9 times of $\dots = \dots = \dots$

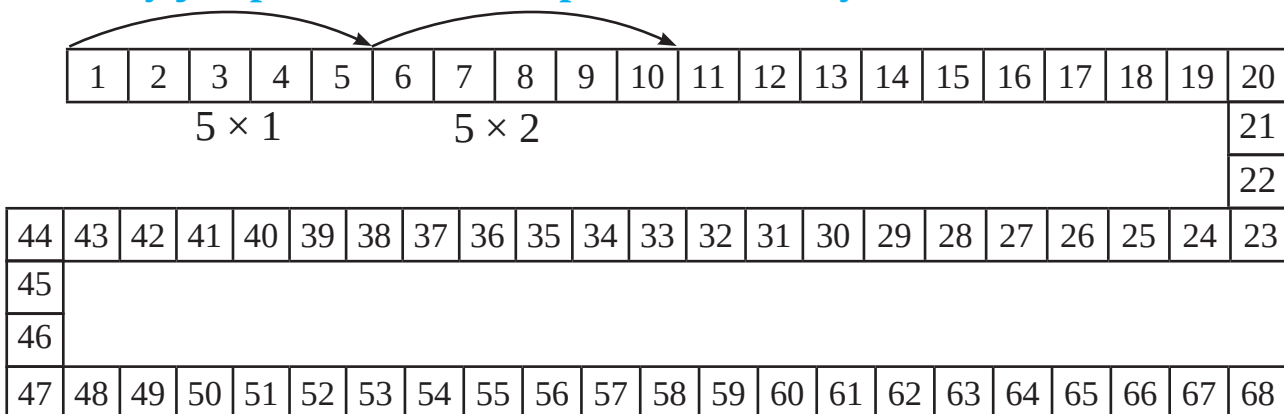
That is, 9 people took one ice cream each. So, Rs. will have to be paid in total.



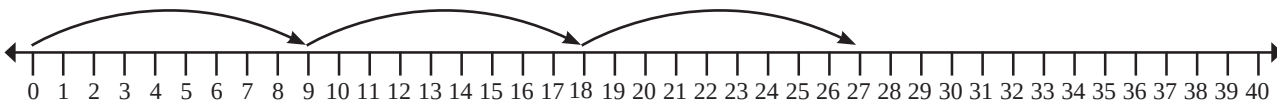


Tell me.

- ☉ Lets play the game of jumping. How many in each step and, how many jumps? For that complete the activity.



- ☉ Sadu started from zero



On which numbers does Sadu reach? 9, 18, 27, ..., ..., ..., ..., ..., ..., ..., ..., ..., ..., ...

1 jump of 9 means $9 \times 1 = 9$

$9 \times 1 = 9$ $9 \times 4 = \dots\dots$

$9 \times 2 = 18$ $9 \times 5 = \dots\dots$

$9 \times 3 = 27$ $9 \times 6 = \dots\dots$

- ☉ Pooja also started from zero and jumped, Gopal recorded her numbers. Some initial numbers were not written. So, what are those numbers?
,,, 32, 40, 48, 56

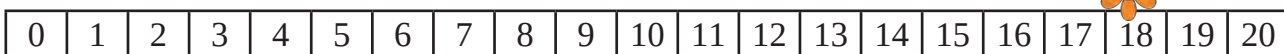
- ☉ Identify the steps and write the next three numbers.

- (1) 10, 20, 30,,, (2) 7, 14, 21,,,
 (3) 20, 40, 60,,, (4) 6, 12, 18,,,



Let's do it.

I placed a flower on the number 18. To reach this flower, you have to jump in equal steps. You don't have to jump directly to the number on the flower. The one who reaches the flower in the least number of jumps wins on which numbers will you jump to?

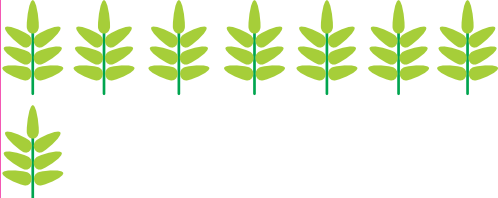
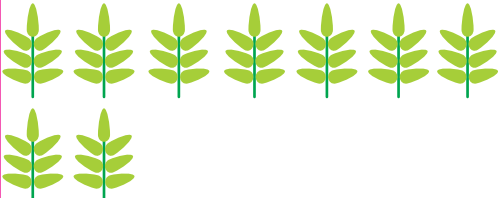
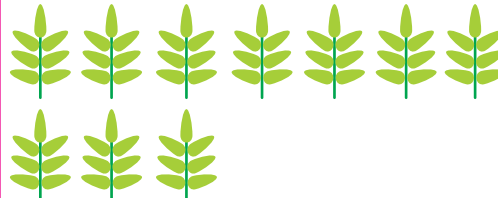


In daily practice, we multiply with the help of multiplication tables. Multiplication tables are simply multiplication done in order.



For teachers : Students should be given hands on learning experiences by allowing them to handle materials directly. Students should be given a number strip or similar material to reinforce the concept of multiplication Table. Encourage the use of various strategies.

Table of 7

Picture	Addition form	Multipli- cation	Table of 7	Reading Multiplication Table
	7	7×1	7	Seven ones are seven
	$7 + 7$	7×2	14	Seven twos are fourteen
	$7 + 7 + 7$	7×3	21	Seven threes are twenty one
	$7 + 7 + 7 + 7$	7×4	28	Seven fours are twenty eight
	$7 + 7 + 7 + 7 + 7$	7×5	35	Seven fives are thirty five
	$7 + 7 + 7 + 7 + 7 + 7$	7×6	42	Seven six are forty two
	$7 + 7 + 7 + 7 + 7 + 7 + 7$	7×7	49	Seven sevens are forty nine
	$7 + 7 + 7 + 7 + 7 + 7 + 7 + 7$	7×8	56	Seven eights are fifty six
	$7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7$	7×9	63	Seven nines are sixty three
	$7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7$	7×10	70	Seven tens are seventy

☉ Lets read Tables.

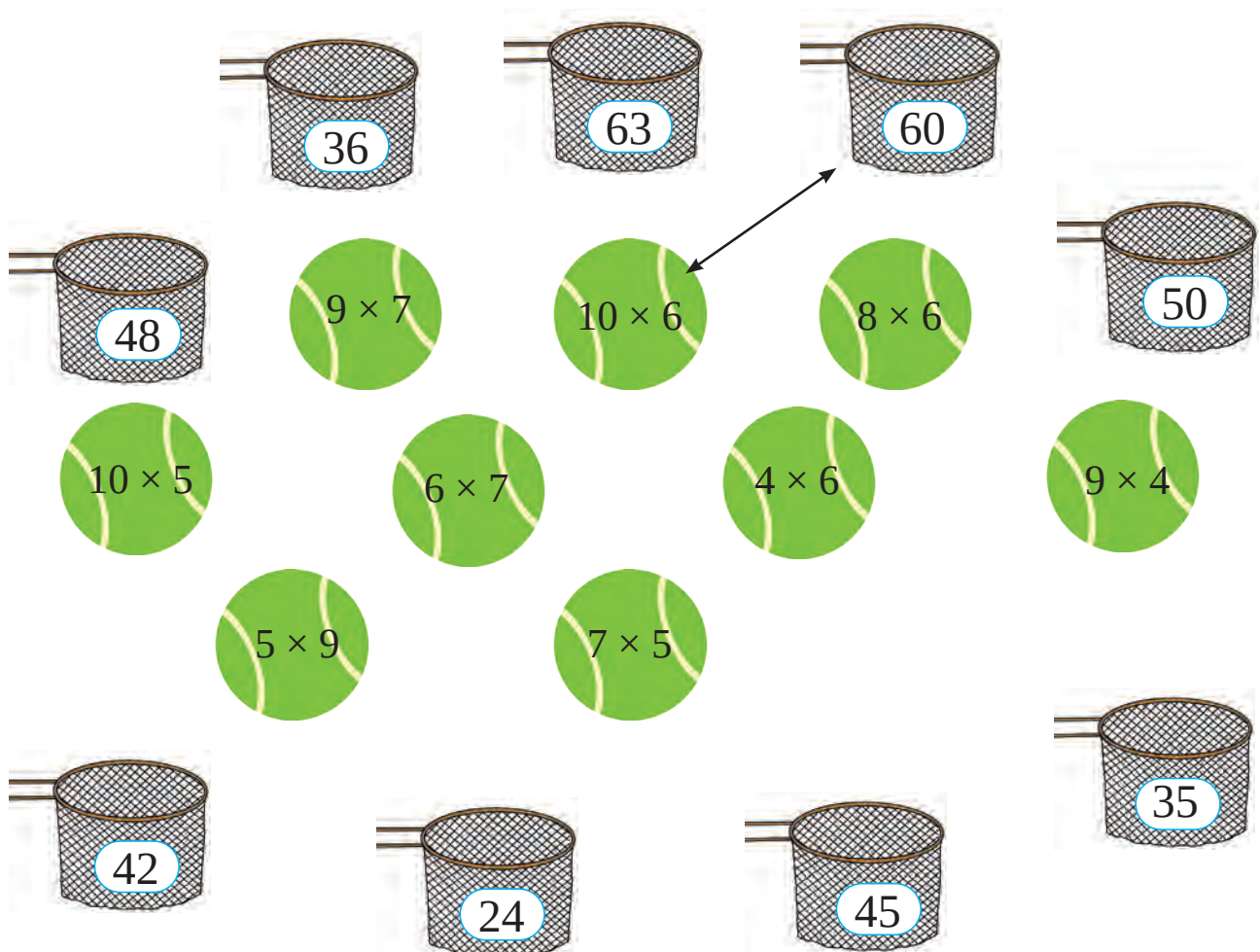
Table of 6
$6 \times 1 = 6$
$6 \times 2 = 12$
$6 \times 3 = 18$
$6 \times 4 = 24$
$6 \times 5 = 30$
$6 \times 6 = 36$
$6 \times 7 = 42$
$6 \times 8 = 48$
$6 \times 9 = 54$
$6 \times 10 = 60$

Table of 8
$8 \times 1 = 8$
$8 \times 2 = 16$
$8 \times 3 = 24$
$8 \times 4 = 32$
$8 \times 5 = 40$
$8 \times 6 = 48$
$8 \times 7 = 56$
$8 \times 8 = 64$
$8 \times 9 = 72$
$8 \times 10 = 80$

Table of 9
$9 \times 1 = 9$
$9 \times 2 = 18$
$9 \times 3 = 27$
$9 \times 4 = 36$
$9 \times 5 = 45$
$9 \times 6 = 54$
$9 \times 7 = 63$
$9 \times 8 = 72$
$9 \times 9 = 81$
$9 \times 10 = 90$

Table of 10
$10 \times 1 = 10$
$10 \times 2 = 20$
$10 \times 3 = 30$
$10 \times 4 = 40$
$10 \times 5 = 50$
$10 \times 6 = 60$
$10 \times 7 = 70$
$10 \times 8 = 80$
$10 \times 9 = 90$
$10 \times 10 = 100$

☉ Show by drawing line which net the ball will go.



☉ Observe the given tables of 3, 4 and 7. What do you see?

$3 \times 1 = \textcircled{3}$	→	$4 \times 1 = \textcircled{4}$	→	$7 \times 1 = \boxed{7}$
$3 \times 2 = \textcircled{6}$		$4 \times 2 = \textcircled{8}$		$7 \times 2 = \boxed{14}$
$3 \times 3 = \textcircled{9}$		$4 \times 3 = \textcircled{12}$		$7 \times 3 = \boxed{21}$
$3 \times 4 = \textcircled{12}$		$4 \times 4 = \textcircled{16}$		$7 \times 4 = \boxed{28}$

Rupa: What correlation is seen between the numbers in the circles and in the box?

Sadu : After adding the numbers in the circle, the number in the next box is obtained.

Rupa : Oh yes, this is so much fun! Now let's make tables..

I will make a table of 6.

Table of 2:	2	4	6	8	10	12	14	16	18	20
Table of 4: +	4	8	12	16	20	24	28	32	36	40
Table of 6:	6	12	18	24	30	36	42	48	54	60

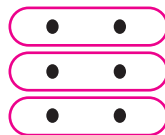
Add the tables of 3 and 5 to make the table of 8.

Table of 3:	3	6			15	18		24		30
Table of 5: +	5		15			30	35			
Table of 8:	8									80

Fun with Multiplication

(1) Group of 2, three times

$$2 \times 3 = 6$$

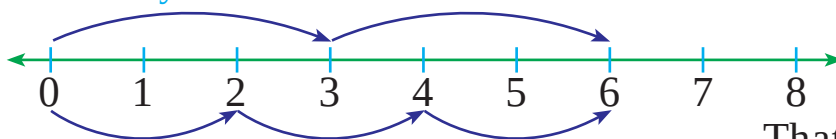


Group of 3, two times

$$3 \times 2 = 6$$



☉ Let's try this on a number line.



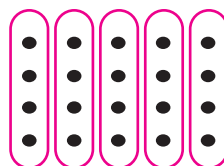
$$3 \times 2 = 6$$

$$2 \times 3 = 6$$

$$\text{That is, } 3 \times 2 = 2 \times 3 = 6$$

(2) Group of 4, five times

$$4 \times 5 = 20$$



Group of 5, four times

$$5 \times 4 = 20$$



that is $4 \times 5 = 5 \times 4$



For multiplication even if the order of the number is changed the product remains the same.

☉ Write the correct number in the blank

(1) $8 \times 4 = \dots \times 8 = \boxed{32}$

(2) $7 \times 3 = 3 \times \dots = \boxed{}$

(3) $6 \times \dots = 9 \times \dots = \boxed{54}$

(4) $9 \times 5 = 5 \times \dots = \boxed{}$

☉ Multiply

(1) $12 \times 1 = 12$

(2) $17 \times 1 = \dots$ (3) $55 \times 1 = \dots$

(4) $\dots \times 1 = 32$

(5) $\dots \times 1 = 25$ (6) $84 \times \dots = 84$

If you multiply any number by 1, the product is the original number.

Exercise

(1) Complete the multiplication chart.

×	1	2	3	4	5	6	7	8	9	10
1	1				5					
2										
3	3					18				
4										
5										
6										60
7										
8						48	56			
9									81	
10						60	70			

$6 \times 3 = 18$

$3 \times 6 = 18$

$10 \times 6 = 60$

$6 \times 10 = 60$

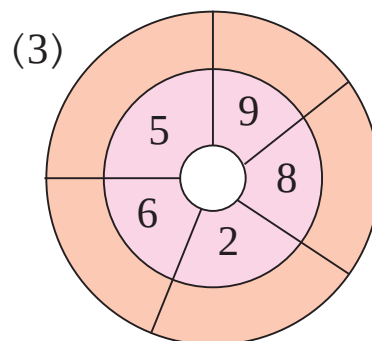
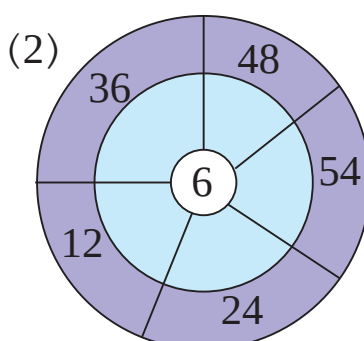
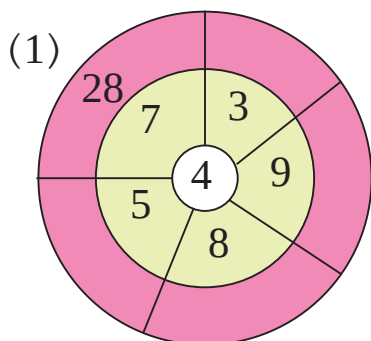
Look at the multiplication table above. Answer the following questions

- ☉ The price of a pen is Rs.7 If you buy 8 pens, how much will you have to pay? Rs.....
- ☉ A box hold 9 laddus. How many laddus will be there in 9 such boxes? Laddus.
- ☉ There are 10 pencils in a box. How many pencils are there in 6 such boxes? Pencils.



For teachers : Teachers should ask students to complete the multiplication chart and ask them various types of questions related to daily life based on the table. They should also give students the opportunity to create and solve their own questions for example, shapes, numbers, etc

(2) Observe the following figures. Write the correct number in the blanks places.

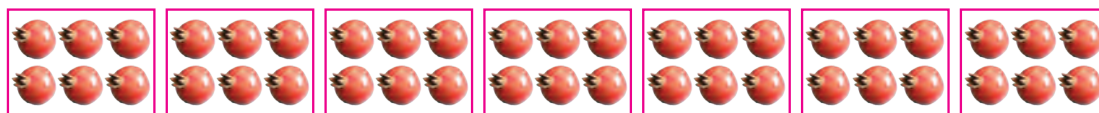


Multiplicand, Multiplier, Product



Let's Understand.

- There are 6 pomegranates in one box. So how many pomegranates in 7 such boxes?



6 pomegranates in each box $\times$ 7 boxes

Total 42 pomegranates in 7 boxes

here, 6 Multiplicand
 $\times$ 7 Multiplier
 42 Product

In multiplication, the number by which we multiply is the '**Multiplier**'. The number that is multiplied is the '**Multiplicand**', and the answer is '**Product**'

- (2) How many children?

Multiplicand $\times$ Multiplier = Product

8 children in each row, 6 such rows

8 children $\times$ 6 rows =

All together children are there

..... $\times$ 6 =

Multiplicand =

Multiplier =

Product =

- Observe the mangoes in the basket one by one and write the correct number in the empty box.

Observe picture	No. of mangoes by Addition	No. of mangoes by Addition
	$4+4+4 = 12$	$4 \times 3 = 12$
	$3+3+3 = 9$	$3 \times 3 = 9$
	$2+2+2 = 6$	$2 \times 3 = 6$
	$1+1+1 = 3$	$1 \times 3 = 3$
	$0+0+0 = 0$	$0 \times 3 = $

$$0 \times 3 = 0 \text{ and } 3 \times 0 = 0$$

$$8 \times 0 = 0$$

$$0 \times 9 = 0$$

$$0 \times 100 = 0$$

$$15 \times 0 = 0$$

$$0 \times 10 = 0$$

$$999 \times 0 = 0$$



I Understood.

If you multiply any number by 'zero' or multiply zero by any number, the product is 'zero'

Exercise

Solve.

$$\begin{array}{r} 6 \\ \times 3 \\ \hline \end{array}$$

$$\begin{array}{r} 8 \\ \times 4 \\ \hline \end{array}$$

$$\begin{array}{r} 7 \\ \times 5 \\ \hline \end{array}$$

$$\begin{array}{r} 9 \\ \times 0 \\ \hline \end{array}$$

$$\begin{array}{r} 5 \\ \times 1 \\ \hline \end{array}$$

$$\begin{array}{r} 4 \\ \times 7 \\ \hline \end{array}$$

$$\begin{array}{r} 0 \\ \times 6 \\ \hline \end{array}$$

$$\begin{array}{r} 1 \\ \times 9 \\ \hline \end{array}$$

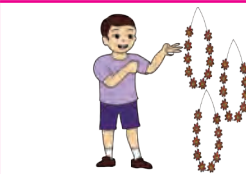
Multiplying by tens



Let's Understand.

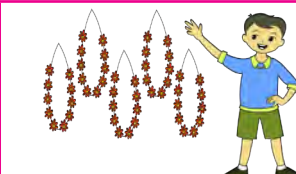
- (1) Raju and Tony brought a basket full of flowers. They made garlands out of them as follows. How many flowers did each person need to make the garland?

Raju: I made 3 garlands of 10 flowers each



$3 \times 10 = 30$ Flowers

Tony: I made 5 garlands of 10 flowers each



$\dots \times \dots = \dots$ Flowers

- Multiplying a number by 10 means taking ten times that number.
- To multiply a number by 10 means in the product we just need to write zero after that number.

- $3 \times 20 = \square$ In this example, we want to multiply 3 by 20 that is, 3 by 2 tens. $3 \times 20 = 3 \times 2 \text{ tens} = 6 \text{ tens} = 60$

Exercise

- (1) Write the correct number in the box

$$6 \times 10 = \square$$

$$7 \times 10 = \square$$

$$9 \times 10 = \square$$

$$10 \times 10 = \square$$

- (2) Write the correct number in the box

$$9 \times 20 = \square$$

$$7 \times 40 = \square$$

$$5 \times 60 = \square$$

$$3 \times 80 = \square$$

$$8 \times 30 = \square$$

$$6 \times 50 = \square$$

$$4 \times 70 = \square$$

$$2 \times 90 = \square$$

To multiply a two-digit number by a one-digit number.



Let's Understand.

⦿ Multiplication by lattice method.

- (1) Vishal on his birthday gifted 3 coloured pencils each to 36 students in his class. How many pencils did he gift altogether?

Total number of students in the class is 36. Each student has given 3 pencils. So total pencils gifted = $36 \times 3 = ?$

Now let's first guess what will be the product of 36×3 . Since 36 has 3 tens, $3 \text{ T} \times 3 = 9 \text{ T} = 90$. That is, the product of 36×3 is greater than 90.

How do you do 36×3 ?
36 means 30 and 6

×	30	6
3	30×3	6×3
	90	18

$$36 \times 3 = 108$$

So altogether Vishal has gifted 108 pencils.

×	30	6
3		

30 times 3 is 90 and 6 times 3 is 18.

90 and 18 are added to, then

We get the answer 108.

90
+ 18
108



Let's Think .

How much difference was there between the multiplication you guessed and the actual product? Why?

- (2) If 43 students participated in the Vanabhojan. Each student was given 5 fruits, then in all how many fruits were given?

$$43 = 40 + 3$$

×	40	3
5		

×	40	3
5	40×5	3×5
	200	15

200
+ 15
215

That is, $43 \times 5 = 215$, so in all 215 fruits were given.

Exercise

- (1) Solve the following multiplication problem in your notebook using the lattice method in your notebook. Write the answer in the box provided.

⦿ $22 \times 4 = \square$ ⦿ $32 \times 7 = \square$ ⦿ $38 \times 9 = \square$ ⦿ $49 \times 8 = \square$



For teachers : Take practice of multiplying a number by tens. Help students to understand the pattern in it. Conduct a discussion on this in the classroom

(2) First guess what the answer will be, then multiply by the lattice method. Find out by multiplication with lattice method.

- There are 6 flowers in a bunch. If 12 such bunches are made then, how many flowers will be required?
- The price of one book is Rs. 48. What will be the total price of 3 such books?
- If 21 students are standing in a row, what is the total number of students in 8 such rows?
- Varad deposited Rs. 20 every day in his school savings bank, so how much did he deposit in 9 days?

Multiplication: By Vertical Arrangement



Let's Understand.

Multiply 21×4 .

- First multiply 1 in the units place by 4. Four ones are four. So, we write 4 in the units place in the answer.
- Now multiply the 2 in the tens place by 4.
- Four twos are eight. Let's write 8 in tens place in the answer
- That is, $21 \times 4 = 84$

T	U
2	1
$\times$	4
8	4

Multiply 34×6 .

First, multiply 4 in the units place by 6, six fours are twenty four.

In the answer, only 4 is written in the units place and write carried 2 above 3 in the tens place as shown. Now multiply 3 in the tens place by 6, which is 6 threes are eighteen.

Add carried 2 and 18, we get 20.

Write 2 in the hundredth place and zero in the tens place. The product is 204. That is, $34 \times 6 = 204$

H	T	U
	2	
	3	4
	$\times$	6
2	0	24

Exercise

Multiply.

(1)

H	T	U
	4	3
	$\times$	5

(2)

H	T	U
	5	3
	$\times$	3

(3)

H	T	U
	6	5
	$\times$	4

(4) If there are 25 laddus in one box, how many laddus are there in total in 7 boxes?

H	T	U
	3	
	2	5
	$\times$	7

Laddu in one box

boxes

Total laddus

(5) The price of one kilogram of sugar is Rs. 43, then what is the price of 8 kilograms of sugar?

43

$\times$ 8

Price of sugar per kilogram

Kilograms of sugar

The cost of 8 kilograms of sugar is rupees.



Let's Understand.

Table of Ten multiplied by tens:

$$20 \times 30 = ?$$

$$2 \text{ T} \times 30 = 60 \text{ T} = 600$$

In the multiplication of tens, two zeros are written in front of the number obtained by multiplying the digit in the tens place.

Write the correct number in the box

(1) $20 \times 40 =$

(2) $30 \times 30 =$

(3) $10 \times 50 =$

Multiply a two-digit number by a two-digit number.



Let's Understand.

Multiplication by the lattice method.

Sonu brought 43 dozen bananas, how many bananas did he bring in total?

Vishal: We have to multiply 43×12 , because 1 dozen bananas = 12 bananas.

$$43 = 40 + 3$$

$$12 = 10 + 2$$

Both numbers are written in expanded form

$\times$	40	3
10	$40 \times 10 = 400$	$3 \times 10 = 30$
2	$40 \times 2 = 80$	$3 \times 2 = 6$

$$\begin{array}{r} 400 \\ + 30 \\ + 80 \\ + 6 \\ \hline 516 \end{array}$$

$43 \times 12 = 516$ Sonu brought 43 dozen bananas means 516 bananas.

Exercise

(1) Multiply by the lattice method.

⦿ 45×14

⦿ 37×19

⦿ 28×23

(2) If 74 passengers can sit in one compartment of a train, how many passengers can sit in 12 such compartments?

(3) There are 43 chairs arranged in a row in the auditorium, then how many chairs are there in 23 such rows?

(4) If the cost of one hat is Rs.65, then what is the total cost of 11 such hats?

Multiplication : By Vertical Arrangement

⦿ $43 \times 12 = ?$

H	T	U
	4	3
×	1	2
	8	6

H	T	U
1		
×	4	3
	1	2
	8	6
+4	3	0
5	1	6

- First, let's do the vertical arrangement. Multiply the number 3 of the units place by 2. Two threes are 6
- Let's write 6 in the unit place below the line. Then multiply 2 by 4 at tens place.
- Two fours are eight. Write eight in tens place.
- Now 1 in the tens place means 1 tens
- Now, multiplying by one tens means multiplying by 10.
- $1T \times 3U = 3T$, $3T = 30$
The zero in 30 is written in the units place and 3 in the tens place.
- 1 tens multiplied by 4 tens is $1T \times 4T = 10 \times 40 = 400 = 4 \text{ Hundreds}$, Write 4 in the hundreds place.
- Now, after multiplying both the numbers, we need to add the products 86 and 430. This sum is the answer of the given multiplication.
- Therefore $43 \times 12 = 516$

⦿ $35 \times 26 = ?$

H	T	U
1	3	
×	3	5
	2	6
2	1	0
+7	0	0
9	1	0

When multiplied by 6 units

Multiplied by 2 tens

Sum of both.

$35 \times 26 = 910$

Exercise

(1) Multiply by vertical arrangement

$\odot 21 \times 11$ $\odot 32 \times 23$ $\odot 29 \times 25$ $\odot 18 \times 12$ $\odot 21 \times 19$

(2) There are 13 desks in each classroom in the school. How many total desks are there in 18 such classrooms?

(3) One bag contains 15 kilograms of total rice, how many kilograms of rice are there in total in 22 such bags?

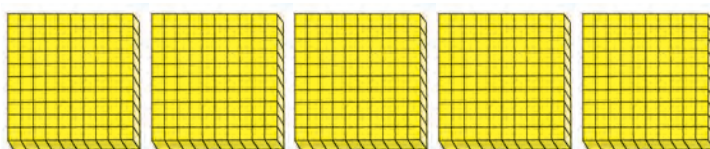
(4) If the price of one book is Rs. 45, then what is the price of 12 such books?

Multiply a three-digit number by a single-digit number



Let's Understand.

$100 \times 5 = 500$ $1 \text{ H} \times 5 = 5 \text{ H} = 500$



$\odot$ Write the correct number in the blank.

(1) If 100 chikoos are placed in a box, how many chikoos can be placed in 6 such boxes?

$100 \times 6 = \dots\dots\dots$ 600 chikoos can be stored in 6 boxes.

(2) If there are 100 *pedhas* in a box, how many *pedhas* will there be in 7 such boxes?

$100 \times 7 = \dots\dots\dots$ 7 boxes have $\dots\dots\dots$ *pedhas*.

$\odot$ The price of a book is Rs.125. If Meena buys 5 such books, then find the amount to be paid.

Let's do this multiplication using the lattice method.

$\times$	100	20	5
5	100×5	20×5	5×5
	500	100	25



$$\begin{array}{r}
 + 500 \\
 + 100 \\
 + 25 \\
 \hline
 625
 \end{array}$$

$125 \times 5 = 625$

To buy 5 books, Rs. 625 is to be paid.

$\odot$ I would like to give gifts to my friends, each worth Rs. 215.

I bought 4 gifts. So how much did I spend?

Let's do 215×4 . Now let's do this multiplication using vertical arrangement.

	H	T	U
		2	
$\times$	2	1	5
			4
	8	6	0

Spent Rs. 860 to buy 4 gifts



For teachers: Discuss in class the similarities and differences between multiplication using the lattice method and vertical arrangement.

Exercise

(1) Multiply, by lattice method and vertical arrangement.

- ⦿ 121×4
⦿ 323×3
⦿ 247×2
⦿ 138×3

(2) Solve the examples

- ⦿ If a factory produces 215 toys every day, how many toys will be produced in 4 days?
- ⦿ If 415 bricks are loaded in one cargo rickshaw, then how many bricks can be loaded in 2 such cargo rickshaws?
- ⦿ If 313 beads are placed in a box then how many beads will be placed in 3 such boxes?
- ⦿ If a book has 128 pages, then what is the number of pages in 7 such books?
- ⦿ 65 ropes, each of 4 meters long, were purchased for the girls to play with, at school. If all the ropes are placed one after the other in a straight line, then how many metre long will be the total length of one long rope?
- ⦿ A shopkeeper sold 132 bags of jaggery, each weighing 5 kilograms, how many kilogram of jaggery did he sell in total?
- ⦿ If 35 cans of 20 litres of water are needed per day for drinking water stall, how many litre of water do they need in total per day?

(3) Solve the examples.

- ⦿ 227×3
⦿ 348×2
⦿ 260×3
⦿ 189×4
⦿ 897×1

(4) Form and solve multiplication examples using the given information.

- ⦿ 64 pages in a book, 15 books
- ⦿ 9 boxes, 24 mangoes in each box
- ⦿ 42 passengers in one bus, 4 buses
- ⦿ Each Rs. 165, 6 toys
- ⦿ 35 kilograms of onions, Rs.25 per kilograms
- ⦿ 66 per litre, 12 litres
- ⦿ 3 m, Rs. 15 per metre

(5) Solve the puzzle.

 $\times$  $\times$  $= 27$

 $\times$  $\times$  $\times$  $= 24$

 $\times$  $\times$  $\times$  $= 96$

 $\times$  $\times$  $= ?$

So, what numbers should be substituted for apple, mango and watermelon.

 $=$

 $=$

 $=$





Division by distribution equally

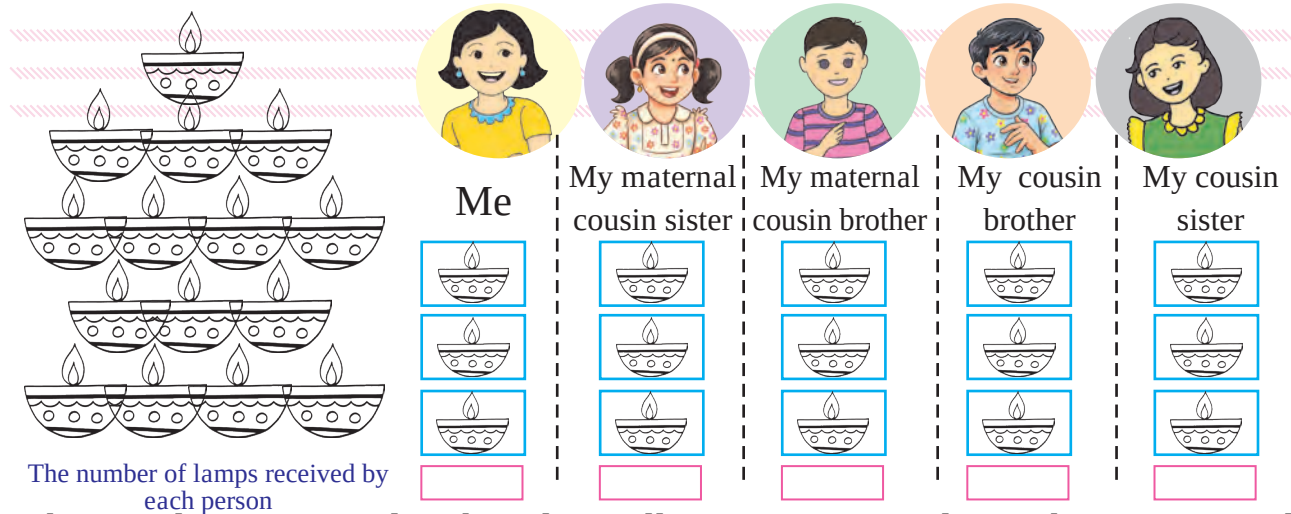


Let's Understand.

I went to my maternal uncle's village with my mother for Diwali. I was overjoyed to meet my grandmother, grandfather, maternal uncle, maternal aunt, maternal cousin-aunt, cousin brothers, and cousin sisters.

Together, everyone prepared *laddus*, *chivda*, *chaklis*, *karanjis*, *shankarpalis*, *jalebis*, and many other festive snacks.

My uncle brought 15 small earthen lamps. He asked the five of our cousins to distribute them equally among ourselves and paint them. We distributed them repeatedly, giving one lamp respectively to each person at a time. This process of distribution continued until all 15 lamps were distributed.



When 15 lamps were distributed equally among 5 people, each one received lamps.

If **15** items are divided into **5** equal groups each group contains **3** items.

This is written as $15 \div 5 = 3$ and is read as "15 divided by 5 equals 3." Here, after distributing the 15 lamps equally among the 5 people, not a single lamp remained.

Mother asked us to distribute 17 karanjis among 5 cousins equally.

Total karanjis	Distributed among how many cousins?	How many each got?	How many over after equal distribution?	How many karanjis were left?
17	5	3	15	2

17 *karanjis* were distributed among 5 people, each person got 3 *karanjis* and 2 *karanjis* were left.

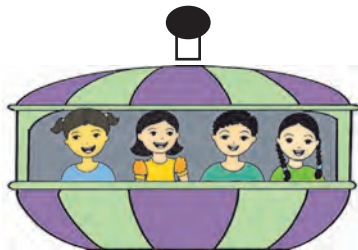
Division by subtracting the same number again and again



Let's Understand.

FOR FREE DISTRIBUTION

In the afternoon, everyone in the village went to funfair. A wheel had been set up in the village. Many boys and girls were standing around it. Only 4 children could sit in 1 carriage at a time. There were 20 children in the line to sit in the Giant wheel. The uncle who was rolling the Giant wheel allowed only 4 children in each carriage



Total number of children = 20

Sat in the first carriage = 4

$$\square - 4 = \square$$

Left in the line = $\square$



Sitting in the second carriage = 4

$$\square - \square = \square$$

Left in the queue = 12



Sitting in the third carriage

$$= 4$$

$$\square - \square = \square$$

Left in the queue = $\square$



Sitting in the fourth

$$\text{carriage} = 4$$

$$\square - \square = \square$$

Left in the queue = $\square$



Sitting in the fifth carriage

$$= 4$$

$$\square - \square = \square$$

Left in the queue = $\square$

$$\begin{array}{r} 20 \\ \text{means, } - 4 \\ \hline 16 \\ - 4 \\ \hline 12 \\ - 4 \\ \hline 8 \\ - 4 \\ \hline 4 \\ - 4 \\ \hline 0 \end{array}$$

If 4 children sat in each cradle, that is, one group of 4, then 20 children sat in 5 carriages. So, there were 5 groups and remainder is 'zero'.

$$20 \div 4 = 5$$



Let's Understand.

Division by equal grouping

- Suman had 24 chikoos. She made groups of 4 each.
Total chikoos = 24, chikoos in each group = 4,
Total groups = 6



- Equal distribution : Complete the table.**

Total number of chikoos	24	24	24	24	24	24	24
How many Chikoos distributed?	2	3	4	5	6	7	10
How many groups?	12	8	6			3	
Remaining chikoos	0	0	0				4

- By forming Equal group : Complete the table.**

Total number of items	35	35	35	35	35	35
Items in each group	7	6	4	10	3	5
How many groups?	5	5	8			
Remaining items	0	5	3			

Different meanings of division



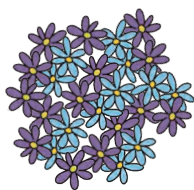
I Understood.

Equal distribution of items or making equal groups of items or subtracting same number of items again and again is division.

Write the dividend, divisor, quotient and remainder by dividing vertically.



Let's Understand.



We had 27 flowers, out of which we made 1 garland of 9 flowers. Like this 3 garlands were made, and there were no flowers left. Let's do the division vertically.



This $\overline{)}$ symbol is used while performing division operations by arranging the division vertically.

- First write 27
- Then draw the division symbol.
- Write 9 to the left of the symbol.
- To do this division, let's write the table of 9.
- Here 27 is the dividend and 9 is the divisor. The quotient is 3. The remainder is 0.

$$\begin{array}{r}
 27 \\
 9 \overline{) 27} \\
 \underline{9 \times 1 = 9} \\
 9 \times 2 = 18 \\
 \underline{9 \times 3 = 27} \\
 00
 \end{array}$$

3 quotient
divisor 9 27 dividend
-27
00 remainder

- 17 *karanjis* are distributed among 5 children. Let's do this division vertically, here we have to do $17 \div 5$.

Here, distributed among 5 children. Hence write the table of 5.

Total = dividend = 17

How many people to distribute =
divisor = 5

How many *karanjis* did each one get =
quotient = 3

How many *karanjis* were left
= remainder = 2

$5 \times 1 = 5$	divisor 5	$\overline{)17}$	3 quotient
$5 \times 2 = 10$			dividend
$5 \times 3 = 15$			-15
$5 \times 4 = 20$			<u>02</u>
But $17 < 20$			remainder

Exercise

Let's Divide

(1) $42 \div 6$ Total = 42 How many people to share? = 6 How many to each? = Remainder =	(4) $63 \div 7$ Total = Group of how many = How many total groups? = Remainder =
(2) $36 \div 4$ Total (dividend) = How many in each group? (divisor) = ... How many groups will there be in total? (quotient) = Remainder =	(5) $72 \div 9$ In total (dividend) = How many items in each group? (divisor) = ... How many groups will there be in total? (quotient) = Remainder =
(3) $45 \div 5$ Total (dividend) = How many items each in group? (divisor) = How many groups will there be in total? (quotient) = Remainder =	(6) $50 \div 10$ Total (dividend) = How many items in a group? (divisor) = Each one will get (quotient) = Remainder =

Grandma has 75 rupees. She wants to divide it equally among 5 people.

So let's see how to divide $75 \div 5$.

$$\begin{array}{r} 5 \overline{)75} \end{array}$$

Here the dividend is 75 and the divisor is 5.

The dividend $75 = 7T + 5U$.

The first 7T will be divided equally among five people.

$$\begin{array}{r} \text{T U} \\ 1 \\ 5 \overline{) 75} \\ - 50 \\ \hline 25 \end{array}$$

Let's say the table of 5. Five ones are five ($5 \times 1 = 5$)
We write, 1 in the tens place in the quotient, 5 tens is 50, write 50 in the tens place of the dividend $7\text{T} - 5\text{T} = 2\text{T}$
2 is in tens place as remainder. Lets write 5 in units place of dividend in front of 2T.

$$\begin{array}{r} \text{T U} \\ 1 \\ 5 \overline{) 75} \\ - 50 \\ \hline 25 \end{array}$$

2 tens and 5 units means 25 units.

Divisor is 5, Therefore take remainder 25 units as dividend. Lets divide it among 5 people, for that lets say table of 5.

$$\begin{array}{r} \text{T U} \\ 15 \\ 5 \overline{) 75} \\ - 50 \\ \hline 25 \\ - 25 \\ \hline 00 \end{array}$$

Five fives are twenty five ($5 \times 5 = 25$), write 5 in the units place of the quotient. Write 25 below 25. ($25 - 25 = 0$). The quotient is 15. That means each of the 5 people will get 15 rupees.
Here Dividend=75, Divisor=5, quotient = 15, remainder = 0
 $75 \div 5 = 15$ and remainder = 0



Let's Think .

How many notes and coins should grandma have of rupees 75, to give 15 rupees to each person? Write down all the possibilities of giving 15 rupees to each person and relate it to the division $75 \div 5$.

Solve

(1) $24 \div 4$

$$\begin{array}{r} 06 \\ 4 \overline{) 24} \\ - 0 \\ \hline 24 \\ - 24 \\ \hline 00 \end{array}$$

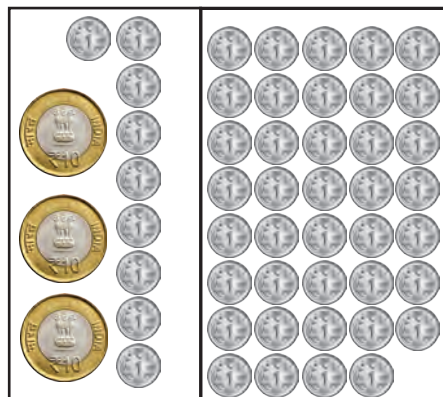
Amount Re. 1 Coins



(2) $39 \div 5$

$$\begin{array}{r} 07 \\ 5 \overline{) 39} \\ - 0 \\ \hline 39 \\ - 35 \\ \hline 04 \end{array}$$

Amount Re. 1 Coins



For teachers : For students to help understand equal sharing, equal grouping, division and repeated subtraction, by giving them sufficient practice and activities. To help students understand the concepts use visual aids and ask to express their ideas verbally. While performing division using the ($\overline{)}$ symbol, make them to write the dividend and identify the divisor/remainder. Practice using examples from daily life.

- Rs. 64 is to be divided equally among 3 people. Do the division vertically.
- 6 tens are to be divided by 3, so say the table of 3. Three twos are 6. Subtracting 6 tens from 6 tens we get 0, write 0 below it. 4 units are divided by 3, write that 1 in units place in quotient.
- $4-3=1$ therefore remainder is 1 and quotient is 21.

$$\begin{array}{r} 2 \\ 3 \overline{) 64} \\ \underline{-6} \\ 04 \end{array}$$

$$\begin{array}{r} 21 \\ 3 \overline{) 64} \\ \underline{-6} \\ 04 \\ \underline{-03} \\ 01 \end{array}$$

Exercise

(1) Write the dividend, divisor, quotient and remainder.

☐ $4 \overline{) 48}$
 ☐ $5 \overline{) 55}$
 ☐ $3 \overline{) 42}$
 ☐ $7 \overline{) 84}$
 ☐ $5 \overline{) 65}$
 ☐ $2 \overline{) 86}$

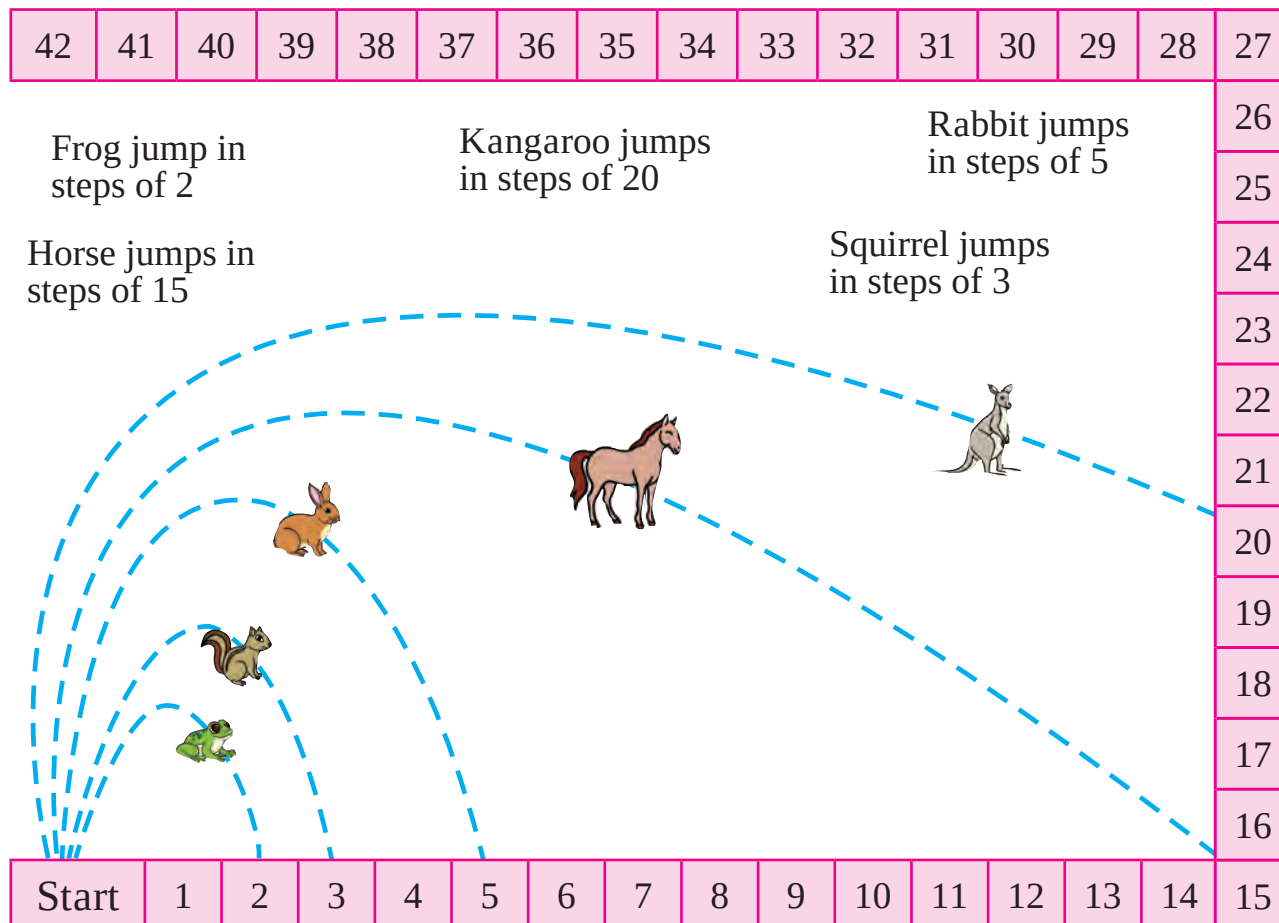
(2) Divide and write the remainder

☐ $31 \div 2$
 ☐ $43 \div 3$
 ☐ $69 \div 6$
 ☐ $98 \div 8$
 ☐ $92 \div 9$
 ☐ $67 \div 3$

(3) Swara decorated 96 lamps to sell during Diwali. If she fills each box with 6 diyas, how many boxes will she need to fill all the diyas? Will there be any diyas left after that?

(4) If 84 kilograms of rice is divided equally among 7 people, how many kilograms of rice will each person get?

Jump in steps.



◎ Who has jumped how many times? Tell and write .

(1) How many jumps will the frog have to reach the number 30?

$$30 \div 2 = \dots\dots$$

(2) How many jumps will squirrel have to make to reach the number 27?

$$27 \div 3 = \dots\dots$$

(3) What number will kangaroo reach in two jumps?

(4) Who all many will come to the number 15?

(5) Will the rabbit hit the number 18?

(6) How many jumps of the rabbit will be equal to one jump of the horse?

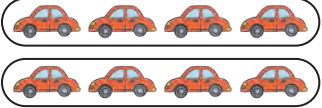
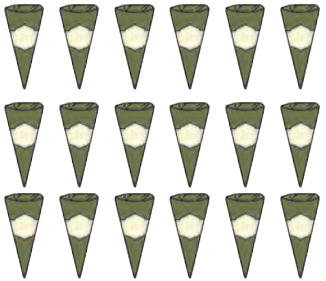
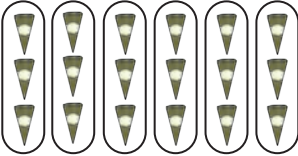
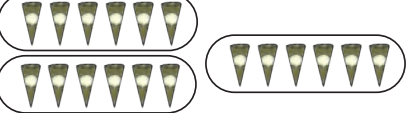
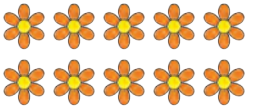
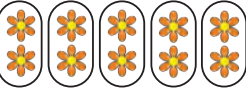
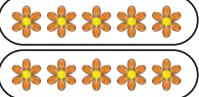
(7) How many jumps will the frog will be equal to one jump of the kangaroo?

(8) If the frog and squirrel start jumping, what is the first smallest number they will jump?

Multiplication-Division Correlation

Multiplication

Division

 $4 \times 2 = 8$	 $8 \div 2 = 4$  $8 \div 4 = 2$
 $6 \times 3 = 18$	 $18 \div 3 = 6$  $18 \div 6 = 3$
 $\square \times \square = \square$	 $\square \div \square = \square$  $\square \div \square = \square$

Exercise

(1) Divide using the table of 2.

⦿ $18 \div 2 = 9$

⦿ $16 \div 2 = \boxed{}$

⦿ $20 \div 2 = 10$

⦿ $9 \times 2 = 18$

⦿ $\boxed{} \times \boxed{} = \boxed{}$

⦿ $\boxed{} \times \boxed{} = \boxed{}$

(2) Divide using the table of 5.

⦿ $10 \div 5 = \boxed{}$

⦿ $\boxed{} \div 5 = 7$

⦿ $40 \div \boxed{} = 8$

⦿ $5 \times 2 = \boxed{}$

⦿ $\boxed{} \times \boxed{} = \boxed{}$

⦿ $\boxed{} \times \boxed{} = \boxed{}$

(3) Using the table of 10, divide into 10 equal groups.

⦿ $20 \div 10 = 2$

⦿ $80 \div \boxed{} = \boxed{}$

⦿ $\boxed{} \div 10 = \boxed{}$

⦿ $10 \times 2 = \boxed{}$

⦿ $\boxed{} \times \boxed{} = \boxed{}$

⦿ $\boxed{} \times \boxed{} = \boxed{}$

(4) How many children can get Rs. 9 each from the total of Rs. 72?

$72 \div 8 = \boxed{}$

- ⦿ I have 72. If this amount is divided equally among 9 children, how much will each child get?

$72 \div 9 = \boxed{}$

- ⦿ If 8 children are given 9 rupees each, how many rupees will be needed?

$9 \times \boxed{} = \boxed{}$

(5) In a garden, 63 flowering plants are planted in 9 rows. If each row containing same number of plants. How many plants are there in each row?

Total trees $\boxed{} \div \boxed{} = \boxed{}$ trees in each row $63 \div \boxed{} = \boxed{}$

..... $7 \times \boxed{9}$ = $\boxed{63}$

(6) The cost of 7 brushes is 84 rupees, so what is the cost of one such brush? $84 \div 7 = \boxed{}$ Rupees

$\boxed{84} \div 12 = \boxed{}$ brushes

$12 \times 7 = \boxed{}$ Rupees

The price of a brush is ₹ $\boxed{}$

(7) Solve. ⦿ $6 \div 6 = \boxed{}$ ⦿ $10 \div 10 = \boxed{}$ ⦿ $15 \div 15 = \boxed{}$

When any number is divided by itself, the quotient is 1.



For Teachers: Explain the relationship between division and multiplication with the various examples.

Dividing a three-digit number by a one digit number



Let's Understand.

Aunty wants to send sweets to her relatives. For this, she called Piyush for the help.

Piyush, we have to fill 108 laddus equally in 9 boxes.



First, let's put 5 laddus in each box. Five nines are forty-five.



$108 - 45 = 63$ laddus
From 63 we will put 5 laddus in every box so that 45 laddus will be placed again.



$63 - 45 = 18$ laddus left. Let's put 2 more laddus in each box, so that all the laddus are distributed equally in each box.



$$\begin{array}{r} 9 \overline{) 108} \\ \underline{- 45} \\ 63 \\ \underline{- 45} \\ 18 \\ \underline{- 18} \\ 00 \end{array} \quad (5 + 5 + 2)$$

Aunty : $5 + 5 + 2 = 12$ laddus in each box

Aunty : Let's do $108 \div 9$ vertically. The table of 9 will help us with that.

$$\begin{array}{r} 0 \\ 9 \overline{) 108} \\ \underline{- 0} \\ 1 \\ 01 \\ 9 \overline{) 108} \\ \underline{- 0} \downarrow \\ 10 \\ \underline{- 9} \downarrow \\ 18 \end{array}$$

- 1 hundred cannot be divided equally among 9 people, so a zero was written in the hundredth place in the quotient.
- 1 hundred is 10 tens.
- After dividing 10 tens equally among 9 persons, each person gets 1 tens, so 1 is written in the tens place in the quotient.
- After subtracting 9 tens from 10 tens, 1 tens remains.
- 1 tens and 8 units make 18 units
- If 18 units are divided equally among 9 persons, each person gets 2 units, so 2 is written in the units place in the quotient.
- $18 - 18 = 0$ is the remainder. The division is 12. We get, $108 \div 9 = 12$.

$$\begin{array}{r} 0 \\ 9 \overline{) 108} \\ \underline{- 0} \downarrow \\ 10 \end{array}$$

$$\begin{array}{r} 012 \\ 9 \overline{) 108} \\ \underline{- 0} \\ 10 \\ \underline{- 9} \\ 18 \\ \underline{- 18} \\ 00 \end{array}$$



For teacher : Explain through various examples that in the division if we get zero in the quotient it is to be written at proper place.

I have 3 notes of 100 rupees, 6 notes of 10 rupees and 4 coins of 1 rupee, lets divide 364 rupees among 4 people

We can solve the above example by division

$$\begin{array}{r} 0 \\ 4 \overline{) 364} \\ - 0 \\ \hline 3 \end{array}$$

- Here 3 notes of 100 rupees cannot be distributed equally among 4 persons. Therefore, 0 is written in the hundredth place in the quotient.

$$\begin{array}{r} 09 \\ 4 \overline{) 364} \\ - 0 \downarrow \\ \hline 36 \\ - 36 \\ \hline 00 \end{array}$$

- 3 hundred = 30 tens (30 notes of Rs.10) Now these 30 notes and the previous 6 notes have become 36 notes of Rs. 10.

- These will be shared among 4 people equally. Each will get 9 notes of 10 rupees. So in the division we wrote 3 in the tens place.

- We have 4 coins of 1 rupee. These 4 rupees are shared equally among 4 people. Everyone will get 1 rupee. So in the division, 1 is written in the units place. The remainder is zero. If 364 rupees is divided equally among 4 people, each person gets 91 rupees.

$$\begin{array}{r} 091 \\ 4 \overline{) 364} \\ - 0 \downarrow \\ \hline 36 \\ - 36 \downarrow \\ \hline 004 \\ - 4 \\ \hline 00 \end{array}$$

Exercise

◉ Solve

(1) $74 \div 4$

(2) $47 \div 6$

(3) $80 \div 7$

(4) $200 \div 8$

(5) $802 \div 9$

(6) $825 \div 5$

(7) $315 \div 3$

(8) $749 \div 7$

◉ Solve

- (1) Ovi has 104 flowers. If she makes a garland of 8 flowers, How many garlands will be there total?
- (2) Akash collected 123 colourful sea shells from the seashore. He filled them equally in 3 jars. So how many sea shells will be in each jar?
- (3) Sheetal has a book of 150 pages, she reads 6 pages every day. So how many day will it take her to read the entire book?
- (4) A shopkeeper has 279 apples, he puts 9 apples in each box. So how many such boxes will he need in total?
- (5) There are 504 books in a library. Each shelf is to hold 6 books. How many such shelves will be required?
- (6) 430 litres of milk were collected at a milk collection centre. If the milk containers are filled with 1 litre of milk each, how many such bags will be made?



For teachers : In division, the largest possible multiple of the divisor is subtracted from the dividend. After subtraction the remainder is always smaller than the divisor. Dividing large numbers and multiplication table upto 10 is not sufficient, then this step by step division method is useful.

Divide by Zero

Let's Understand.



Mom, give laddus to all three of us

Take the laddus from the jar



There is no laddu in the jar.
The jar is empty.

This means that no one
will get a laddu.
Nothing means zero



Zero divided by three is equal to zero! $\rightarrow 0 \div 3 = 0$

$$\begin{array}{r} 0 \\ 6 \overline{) 0} \\ \underline{-0} \\ 0 \end{array} \quad \begin{array}{r} 0 \\ 7 \overline{) 0} \\ \underline{-0} \\ 0 \end{array}$$

Zero divided by 8 is equal to 0 $\rightarrow 0 \div 8 = 0$

Zero divided by 9 is equal to 0 $\rightarrow 0 \div 9 = 0$



I Understood.

If zero is divided by any non-zero number the quotient is always zero.



Let's Understand.

◎ Divide: $90 \div 9$.

$$\begin{array}{r} 1 \\ 9 \overline{) 90} \\ \underline{-9} \\ 0 \end{array}$$

To do this division, first estimate the answer.

(For example, 9 equal parts of 9 tens means that each part will have 1 ten.)

$$\begin{array}{r} 10 \\ 9 \overline{) 90} \\ \underline{-9} \\ 00 \\ \underline{-0} \\ 00 \end{array}$$

First, distribute the tens equally. Everyone will get 1 tens. Now we want to divide 0 units among 9 people. Even if we divide zero by any non-zero number, the quotient is zero, hence write 0 in the units place in the quotient.

So, if 90 items are divided equally among 9 people, each person will get 10 items.



For teachers : Give students the practise of guessing the answer from division examples.

(1) Solve.

- ☐ $70 \div 7$ ☐ $120 \div 8$ ☐ $180 \div 9$ ☐ $600 \div 6$ ☐ $150 \div 5$

(2) From the given information frame word problem on division and solve it.

- ☐ 42 fruits, 6 children ☐ 64 notebooks, 8 children
☐ 120 mangoes, 8 boxes ☐ 75 litre of oil, 5 cans.
☐ 96 rupees, 6 girls ☐ 800 rupees, 5 toys
☐ 238 pages in a book, 7 pages read per day
☐ 525 litres of water in the tank, 5 litres in each bottle



Fun with Maths.

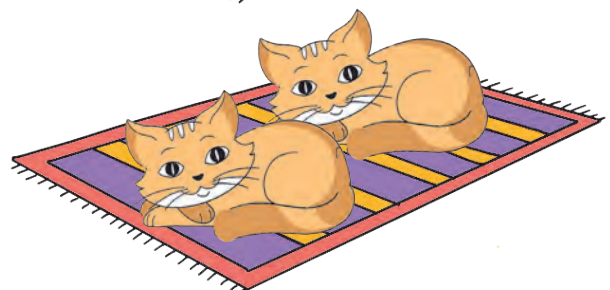
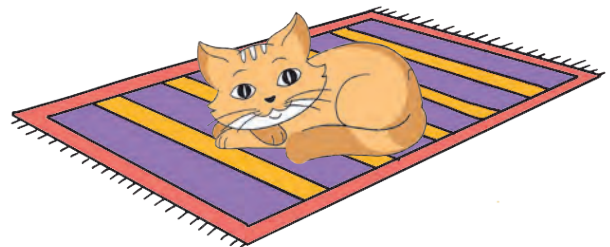
- (1) Yash wants to buy 100 toys for Rs.100. Help him to decide what type and how many bird toys he should buy. The **condition** is that 100 bird toys must be purchased for 100 rupees.

Toy Prices

- 40 sparrows for ₹1
 1 pigeon for ₹3
 1 parrot for ₹5

- (2) Once some cats saw small mattresses.

- If only one cat sits on each mattress, then no mattress is left for one cat.
- If two cats sit on each mattress, then one mattress left with no cats on it. Find how many cats and mattresses are there?





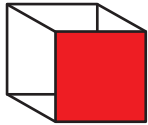
Do The Activity

◎ Colour or paste a paper on one face of each three-dimensional shapes.



I will colour any one face.

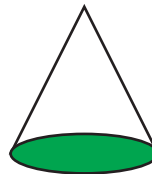
I cut and paste coloured paper of that size on any one face.



.....



rectangle

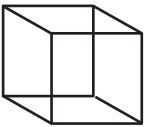
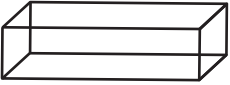
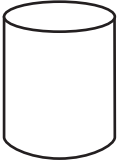
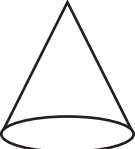
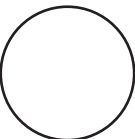


.....



.....

◎ Observe and complete the table below.

Three dimensional shape	Number of flat face	Number of curved face	The name of the three-dimensional shape
			Cube
			
			
			
			

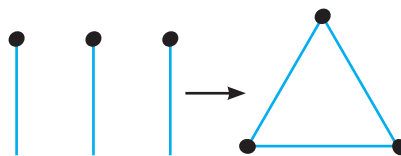


Let's do it

- Priya, Mahesh and Akshita made various shapes using sticks. They discuss the characteristics of their sides, angles and vertices.



Make a closed figure using three sticks.



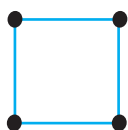
A triangle is formed

A triangle has three angles, three sides, and three vertices.



Let's see what closed shapes are formed by increasing the number of sticks one by one. Let's discuss their angles, vertices, and sides.

Square

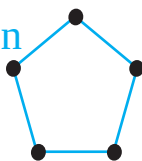


Number of angles = 4

Number of sides =

Number of vertices =

Pentagon

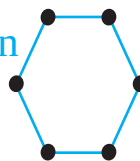


Number of angles =

Number of sides = 5

Number of vertices =

Hexagon

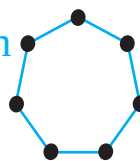


Number of angles =

Number of sides =

Number of vertices = 6

Octagon

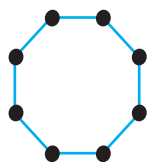


Number of angles =

Number of sides =

Number of vertices = 7

Octagon



Number of angles =

Number of sides =

Number of vertices =

How can you draw a shape with nine sides and nine angles? Think about it and try to draw it in your notebook.

Write the names of the above shapes that have more than four sides.

.....
.....
.....
.....

Identify two-dimensional shapes from three-dimensional shapes

The teacher gave Meena; Shubhangi and Vishal some empty boxes to open. After opening the boxes, they were asked to observe the shapes in the net.

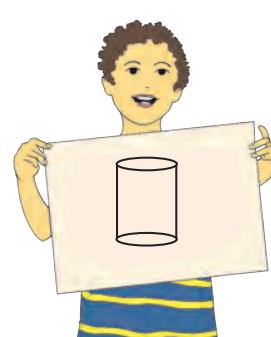
Meena



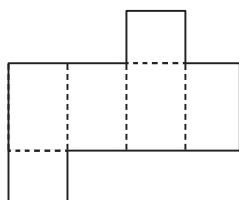
Shubhangi



Vishal

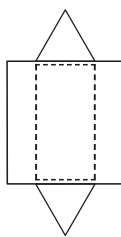


After opening the box, write the names of the two-dimensional shapes from the nets given.



Square

.....



.....



.....

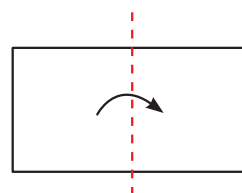
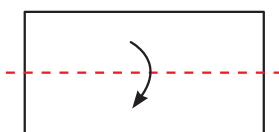
Rectangle

.....

.....

.....

Rectangle



Take a rectangular piece of paper as shown above.

Now fold the paper in half as shown above to connect the opposite sides. This will ensure that the longer side matches up exactly with the opposite side and corners also matches.

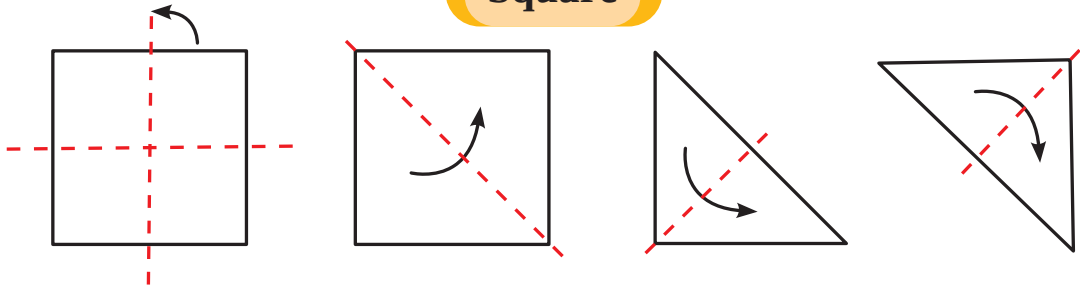
Similarly, the shorter sides also exactly matches with the opposite side.

'Opposite sides of a rectangle are of equal length.'



For teachers : Encourage students to discuss the properties of shapes in the given nets.

Square



Observe the square paper as shown above.

Fold the paper horizontally and vertically and see if the opposite sides are of the same length.

Now fold the paper as shown in the picture to see if the opposite corners match.

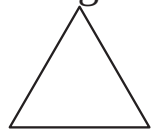
That means all four sides of a square are of equal length.

Triangle

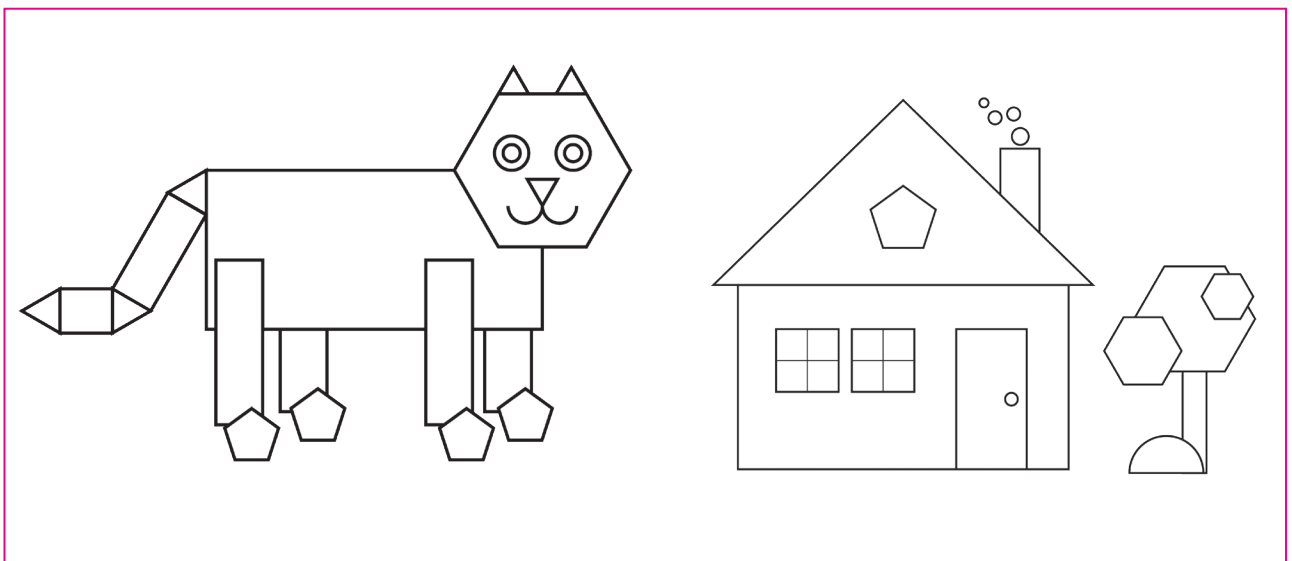
How many sides does a triangle have? How many corners?

Find out where you can find triangular shapes in the your surroundings.

Make squares, rectangles, and triangles using sticks and thin rubber straws.



- Identify the various shapes in the pictures below and make the pictures attractive by filling them with colours according to the shapes.



For teachers : Ask students to use the materials in the math box to create squares, rectangles, and triangles.

Observe if the following shapes are seen in your school or surroundings.



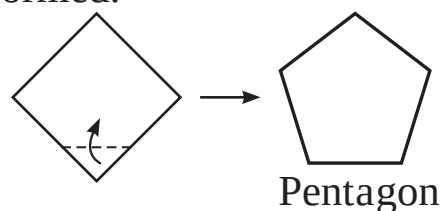
Discuss with your classmates other shapes that are different from the above which you see in your surroundings.



Fun with Maths.

• If one corner of a square is folded then pentagon is formed.

• Take another square piece of paper and fold each corner one by one. Observe which shapes are formed.



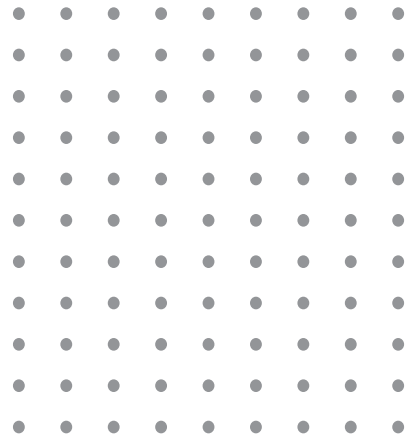
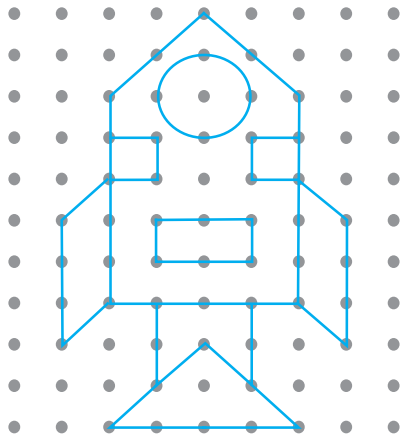
- What shape will be formed if two opposite corners are folded?
- Discuss in class which shapes will be formed, if three and four folds are made.
- Write the names of objects in your surroundings that resemble the following.

			
Circle	Triangle	Square	Rectangle
Bangle			
.....			
.....			
.....			
.....			



Do The Activity

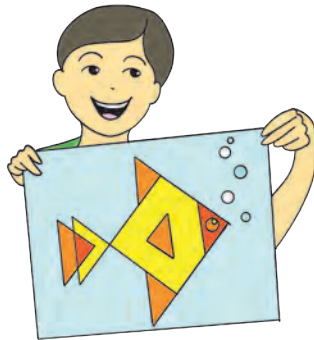
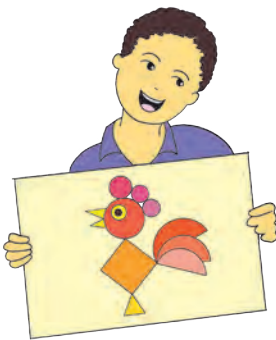
- (1) Make a picture of your choice using various shapes as shown in the picture below.



A picture created with the help of rubber band on a geoboard.

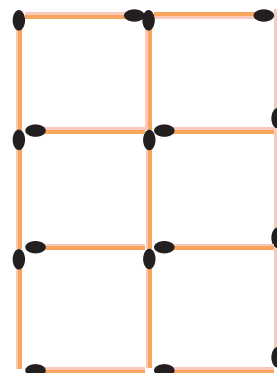
I made my favourite drawing.

- (2) Suraj, Mahesh and Ahmad have drawn some pictures using pages of different sizes. Draw some such pictures in your notebook.



Mathematical puzzle

- (1) Below are some squares made from sticks. Remove 5 sticks so that three squares will be left.



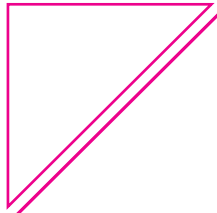


Let's do it

- ◎ Make a rectangle using a square and two triangles.



Square



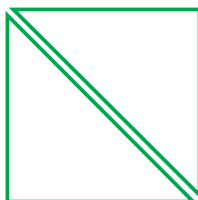
Triangle

Rectangle

- ◎ Construct a hexagon using a rectangle and two triangles.



Rectangle



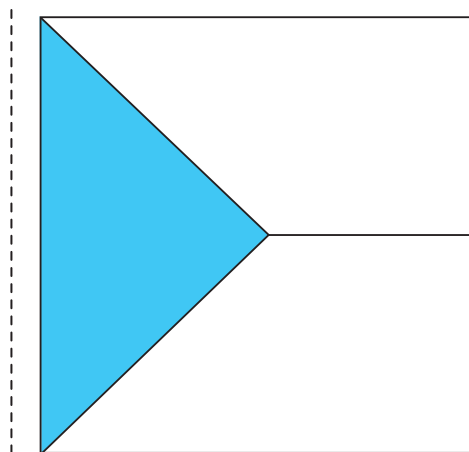
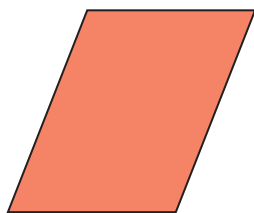
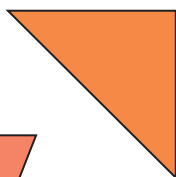
Triangle

Hexagon



Think and answer

- ◎ What shapes are formed by joining various shapes of the tangram given in the mathematical kit?
 - A square and a triangle
 - One square and two triangles
- ◎ Complete the given tangram by using the given pieces.



Tangram of five pieces

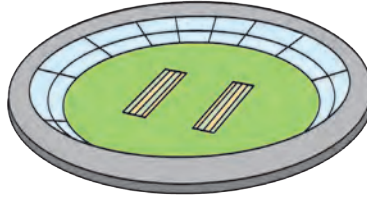
Exercise

(1) Write down the shapes used in the given pictures.

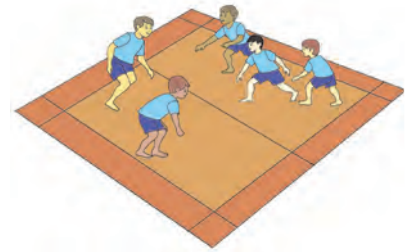
Warli painting



Cricket ground



Kabaddi ground



.....

.....

.....

.....

.....

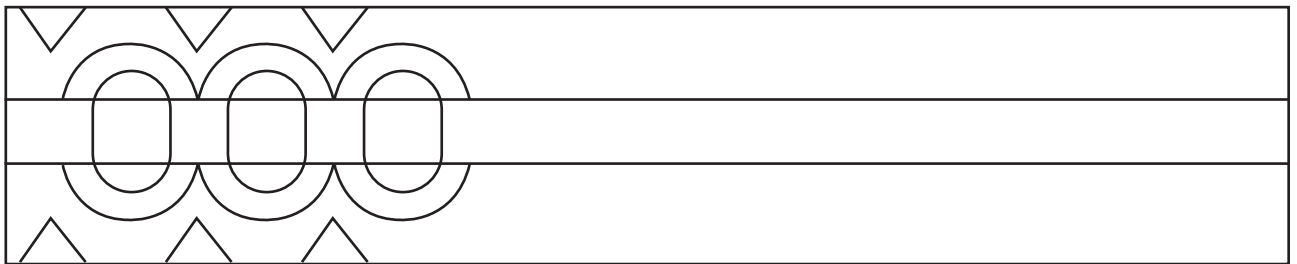
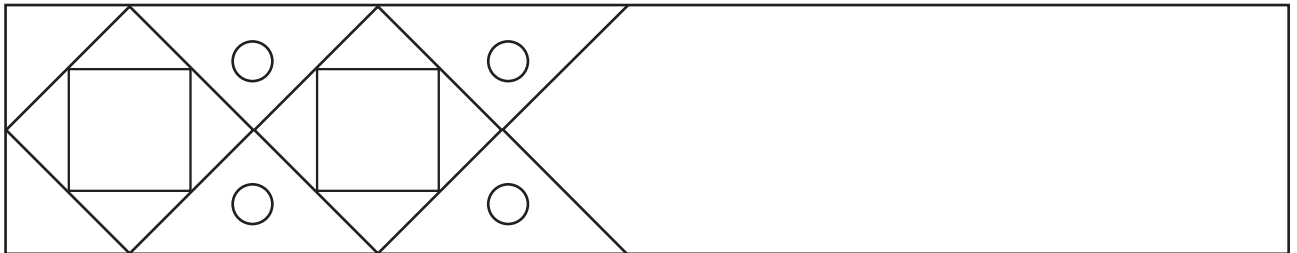
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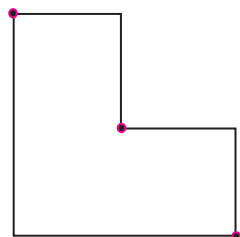
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.....

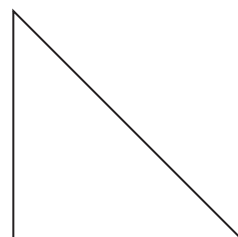
(2) Observe and complete the given design.



(3) Draw a line in the shape below so that three triangles are formed.



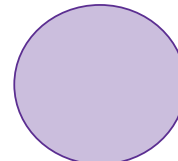
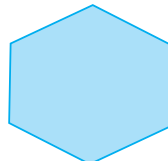
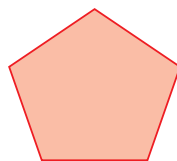
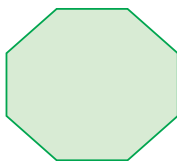
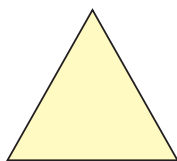
(4) Draw 2 lines in the shape below such that one square and two triangles are formed.





Let's Discuss

- ☉ Select any two of the shapes given below and conduct a class discussion regarding their similarities and differences.



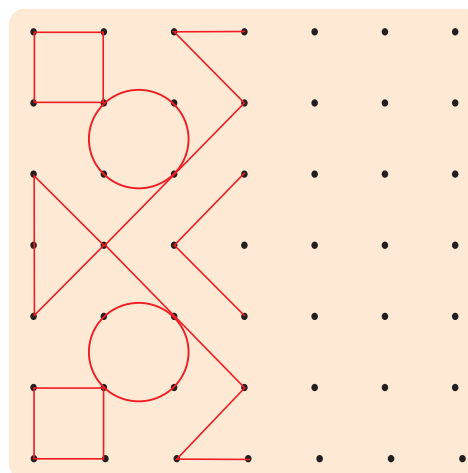
5) Complete the table.

The following letters are formed using vertical, horizontal and diagonal lines, as well as curved lines.

For each letter, place a tick mark (✓) under the specific types of lines used to form it.

Letters	Horizontal Lines	Vertical Lines	Diagonal Lines	Curved Lines
M		✓	✓	
Q				✓
L				
व				
च				
ट				

6) Observe the left side of the Rangoli design and complete the entire Rangoli.



Make this Rangoli beautiful and attractive by colouring the identical shapes within the design with the same colour.



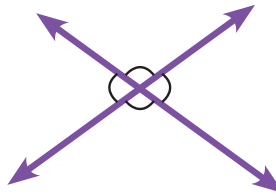
For teachers : Provide students with various two-dimensional and three-dimensional shapes from the Mathematics Kit, and facilitate class discussions regarding their characteristics.

Angle



Let's Understand

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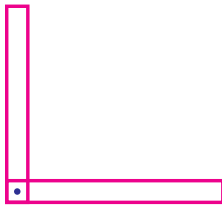


If you observe the above pictures, you will notice that an angle is formed when two sides or two lines meet at a point. Find out where can you see such angles formed in your surrounding or at school.

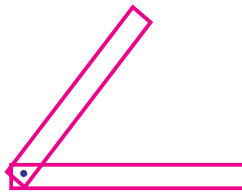


Do The Activity

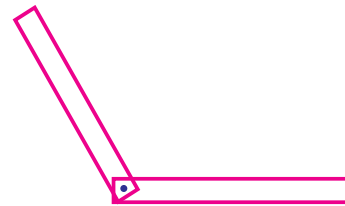
Nagma, Anil and Diksha created rotating corners using paper strips as follows.



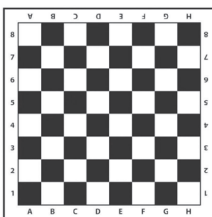
Square Corner
(1)



Smaller than a square corner
(2)



Larger than a square corner
(3)



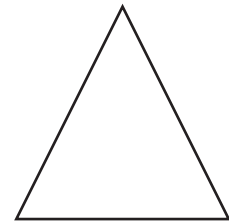
Are all the corners of a chess board square?

.....
.....



Are all the corners of the rectangle 'square corners'?

.....
.....



How many square corners are there in this triangle?

.....
.....

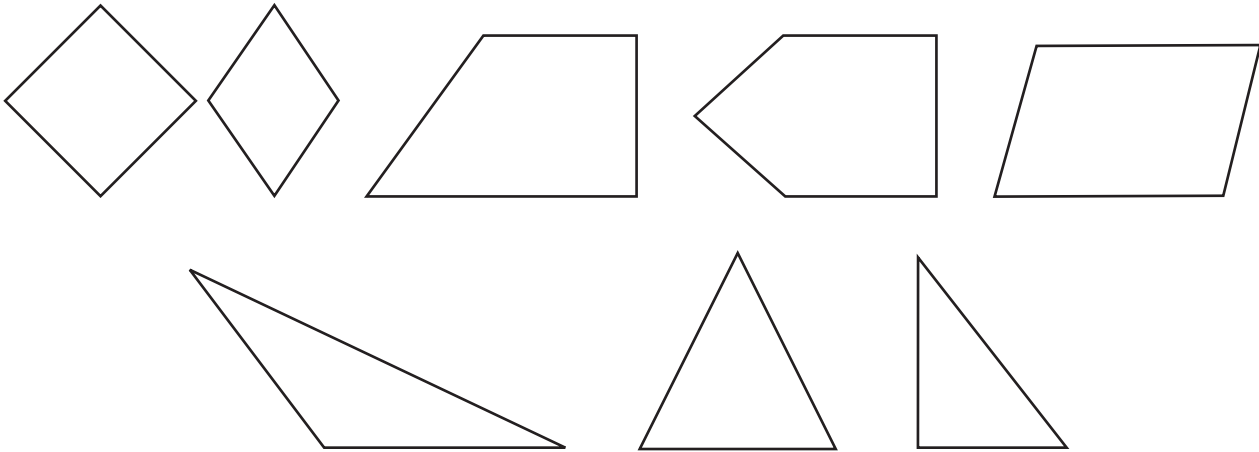


For teachers:

Give a chance to the students to open/close the door in pairs /groups .students should observe the corners of the door and see if the square corner is equal to smaller or bigger than the square corner, Ask them to identify the angles formed by the corners of their hands. This activity can also be done in the context of yogaasan.

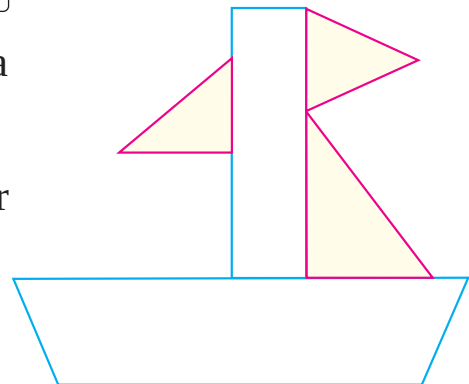
Exercise

- ⊙ Using the square corner, you made from paper strips, mark the corner of the following shapes that have a square corner with a ☒ mark the corners smaller than the square corner with a ☐ mark and mark the corner larger than the square corner with ☐.



- ⊙ Look at the figure below and write the number of corners in it.

- (1) How many square corners are there?
- (2) How many corners are smaller than a square corner?
- (3) How many corners are there that are larger than a square corner?



Do The Activity

Connect the dots on the geoboard with rope or rubber bands to create squares, rectangles, triangles, pentagons, etc.



For teachers : Use geoboards



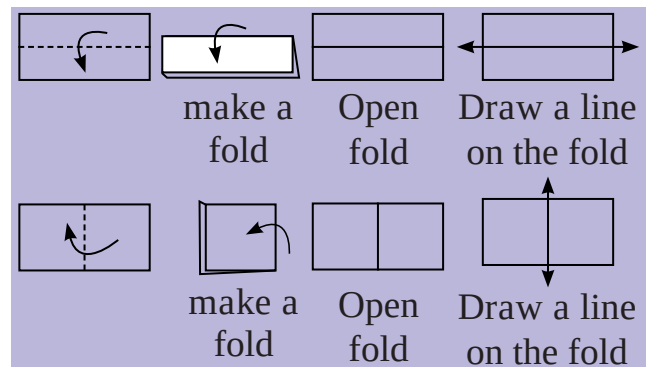
Observe the picture above and discuss which objects on the left and right sides are similar.



Let's Understand

Symmetry in Shape

To make a lantern for Diwali, the children brought coloured paper. They cut it into different shapes. One of the rectangular pieces of paper was folded in such a way that it made two equal parts. When a rectangular piece of paper is folded as shown above, both halves match exactly, means it is symmetrical with respect to the line.

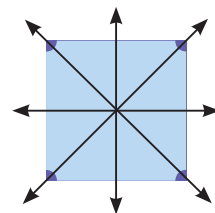


Let's Discuss



If a rectangular piece of paper is folded like this, will the resulting parts match exactly?

A figure that can be divided by a line into two identical halves is a symmetrical figure and that line is the axis of symmetry.



For teachers : Ask students to create symmetrical shapes.



Let's do it

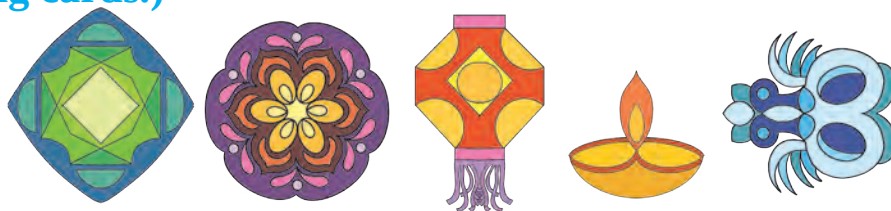
(1) Draw the following shape on paper and cut it along the sides. Fold it and find its symmetry.



(2) Trace the following shapes using trace-paper and then cut it along the sides. Fold the shapes to find the symmetry. After folding separate the shapes which go not give identical halves. Note that these shapes are not symmetrical. Show identical halves.



(3) On the occasion of New Year, we get greeting cards of various shapes. Some shapes on greeting cards are given below. Draw an axis at the right place to make them symmetrical. (Make few such greeting cards.)



(4) Using the property of symmetry, make masks using coloured paper.



(5) Observe the leaves of trees in your surroundings. Find symmetry in them.

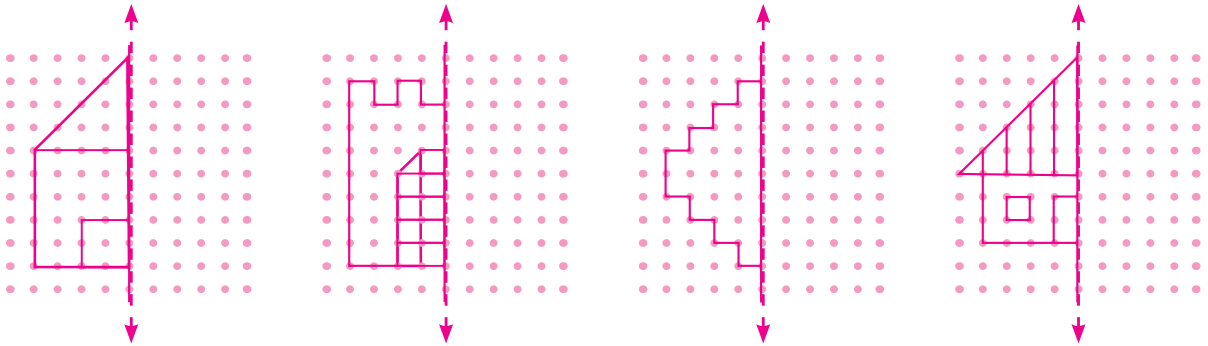
(6) Find the axis of symmetry in the following leaves and trace it with a pen/pencil.



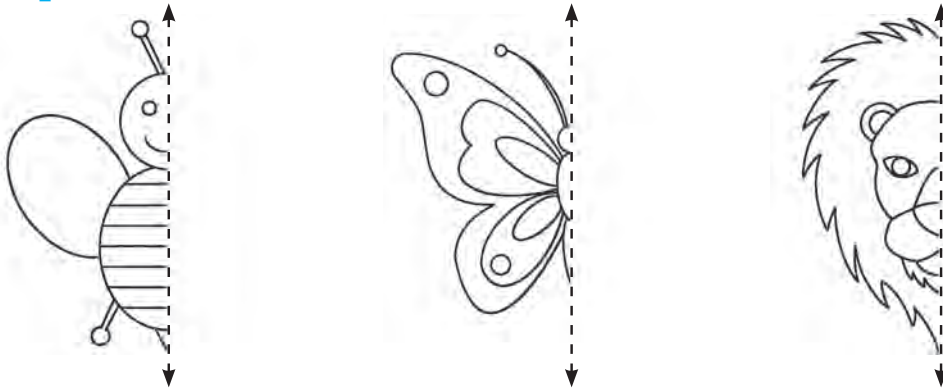
For teachers : Encourage students to experience the difference between symmetrical and asymmetrical shapes around them. Give them the opportunity to explain it, and how symmetry is useful in celebrating various festivals and decorating for various occasions to make it aesthetically appealing. Give students such experiences about this.

◎ Draw symmetrical figures using dot paper.

(1) Complete the diagram so that the given line becomes an axis of symmetry



(2) By completing the image on the other side of the given axis of symmetry, Identify which shape is hidden in the image and complete the picture.

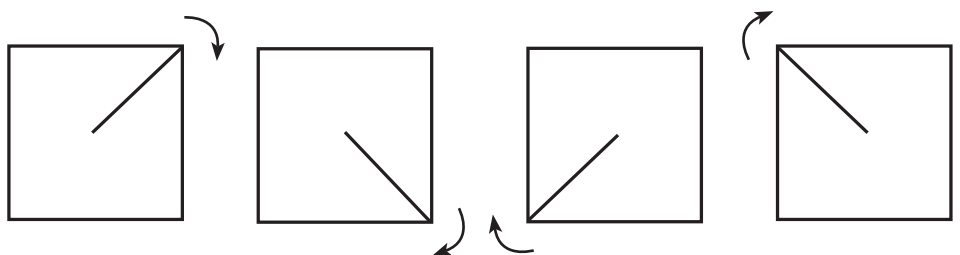


Let's do it

◎ Rotate the symmetrical shape in the clockwise direction and observe.



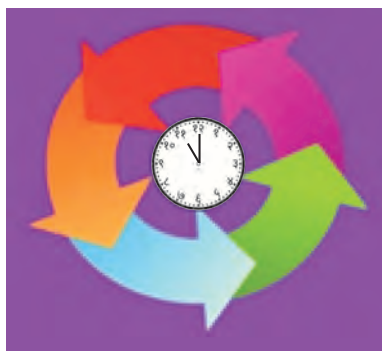
Clockwise direction



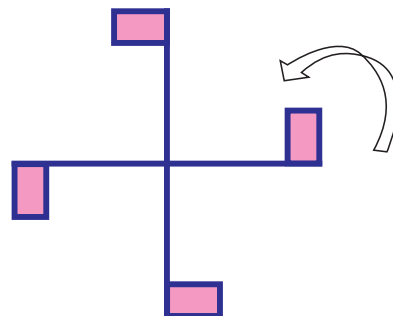
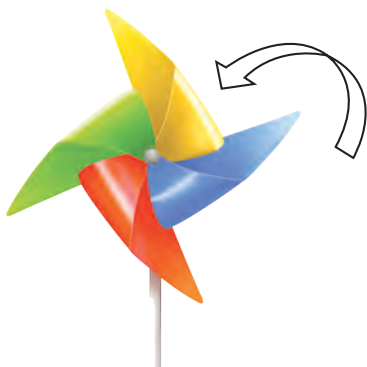
For teachers: In Maharashtra, Rangoli is used for decoration on festivals and occasion In Tamilnadu it is practiced in the form of Kolam, and in Bengal, this art is practiced under the name of Alipana. Explain that it is used to study patterns ,sequences, combinations, and symmetry Make students aware that while preserving our culture we should respect the culture of others. Explain the connection between mathematics and other subjects. In ancient times such artworks were found in many temples, mosques, churches, and monasteries. From this, students should be made aware that this art existed thousands of years ago Amazing symmetry can be seen in the carvings of the Taj Mahal, Bibika Makbara, famous caves, and ancient structures in Maharashtra. Give some examples of such symmetry in your area. Ask them to collect such pictures and photos and stick it in the scrap book.



- ⦿ Rotate the hands of the clock in the opposite direction(anti-clockwise) and observe.



anti-clockwise



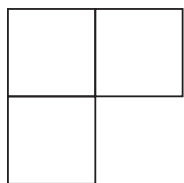
What I Understand

Rotating a symmetrical shape clockwise or anticlockwise does not change its shape.

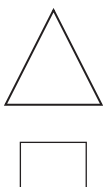
Exercise

- (1) Observe the given figure. Identify the part which makes the given figure symmetrical. From the options choose the part and mark in the box below.

⦿



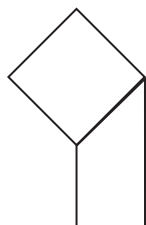
1)



2)



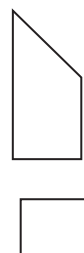
⦿



1)



2)



For teachers: Explain how the position of an object changes when rotated clockwise and anticlockwise, and explain its application. Give examples of this symmetry in daily life. Explain how it is used in various infographics and advertisements.

**Let's Observe**







The above vessels are to be filled completely with water. Make a ✓ mark under the vessels that will hold the most water.

**Let's Discuss**

- (1) Ramesh poured 10 bowls of water of the same size into the vessel, the vessel was completely filled. When Anagha poured 4 pots of water into the same vessel, the vessel was completely filled. Why did this happen?



- The measurement of water to fill the vessel is not same, because different measuring objects were used. The vessel has a capacity of 10 bowls or 4 mugs. To fill the same vessel different utensils (measurements) were used.

**Let's Understand****Capacity**

The capacity of both the pans is 4 copper pots. Because the unit (copper pots) is the same here. From this it can be seen that if anyone uses the same unit to measure the capacity, the measurement will be the same.



Standard units are used to ensure that the volume of a given container remains the same, regardless of who fills it with water using the same objects.

**I Understand**

It is important to use standardized units so that no matter who makes the measurement, it remains the same.



A litre is a standard unit of measurement for liquids.

Volume

Everyday Gauri watches the milkman uncle giving milk using measuring jar.



"I have a one-litre water bottle. I pour the water from it into a mug, I fill the mug to the brim with water."

There was 10 litres of milk in the milk kettle, so the volume of milk in the kettle is 10 litres.

1 litre of it was poured into my bottle, so the volume of milk in my bottle is 1 litre.



Let's Discuss

Discuss, which items are measured in litres? Find out which items have the word "litre" on their sealed packets, cans, bottles, etc. List those items and discuss about it.



Guess

(1) Choose the correct option from the following sentences:

- ⦿ The glass will approximately hold more/less than 1 litre of water.
- ⦿ The bucket will approximately hold more/less than 1 litre of water.
- ⦿ The mug will approximately hold more than/less than/exactly 1 litre of water.



Let's do it

Take two containers of different sizes. For example: a small bowl and a glass. How many small bowls of water do you need to pour into 1 litre bottle? How many glasses of water do you need to pour into 1 litre bottle? First estimate and then actually pour the water and check whether your estimate is correct.

Pots	1 litre bottle	Fill completely with water
	Estimated	Actual
		

List the vessels that can store more than 1 litre of water.



What I Understand

The capacity of any vessel like jug, bucket, drum, pot, pan is the amount of water it holds to fill it completely. Suppose, 2 litres of water is needed to fill the given vessel completely then its capacity is 2 litres.



Tell me



Which of the above vehicles will hold the most diesel in its fuel tank?.....

The vehicle whose tank holds the most diesel has the highest tank capacity.

☉ Observe the pictures below:



One Litre



Half litre



Quarter litre



Three quarter litre

- (1) Here, the first bottle is filled with four parts of kokum syrup. That is, there is one litre of kokum syrup in the bottle.
- (2) In the second bottle, only two parts out of the four parts are filled with kokum syrup. That is, there is half litre of kokum syrup in the bottle
- (3) In the third bottle one part out of four parts is filled with kokum syrup, that is, there is a quarter litre of kokum syrup in the bottle.
- (4) In the fourth bottle, three out of four parts is filled with kokum syrup. That is, there is three quarters of a litre of kokum syrup in the bottle.



Let's do it

If you take a quarter litre of water twice, you get half a litre of water.
A quarter litre and a half litre together makes three quarters of a litre.
A quarter is also equal to three quarter litre.



Do The Activity

- (1) Take a bottle of one litre capacity.
 - (2) With the help of the teacher, make marks like 'Quarter', "Half", "Three quarter" on the bottle.
 - (3) Take an empty vessel.
 - (4) Fill the empty vessel completely with water with the help of the marked bottle. Find the capacity of the vessel. Do the same procedure to find the capacity of different vessels.
- ☉ First estimate and then fill it with water using 1 litre bottle. Mark the correct answer with ✓.
- (1) How much water does a bucket hold? It holds more/less amount of water than one litre.
 - (2) How much water does a mug hold? Holds more/less/ half a litre of water.
 - (3) How much water does one copper pot hold? holds more/less/amount of water than a quarter litre.



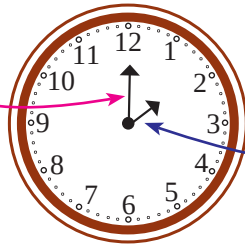
For teachers : Students should be given the opportunity to measure water using different containers and discuss the findings. Also they should practice measuring water using 1 litre and half litre measures and learn the concepts like volume and capacity, also difference between volume and capacity.



Let's Understand

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- I am the big hand.
My name is **minute hand**
- I show minute on the clock.
- It takes me 5 minutes to move from one number to the next number.



- I am the small hand
My name is **hour hand**
- I show hours on the clock
- It takes me 1 hour to move from one number to the next number

The hour hand is at 2 and the minute hand is at 12. That means it is 2 O'clock.

When the minute hand moves from 12 O'clock to 12 O'clock, it makes a total of 60 minutes. That is 1 hour.

There are 60 minutes in an hour. Hours and minutes are the units of time measurement.

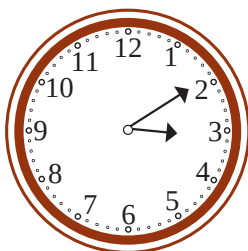
Clock reading



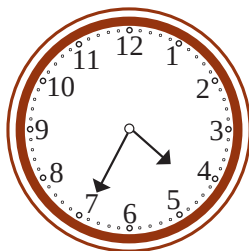
Mom, what time is it?



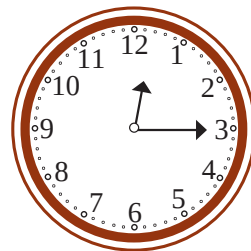
The hour hand on the clock has moved forward from 2, meaning it is past 2 O'clock. There is a gap of 5 minutes between each of the two adjacent numbers, So multiply the number where the minute hand is by table of 5. Here, the minute hand is at 4, so $5 \times 4 = 20$ minutes, thus the clock shows the time 2 hours 20 minutes.



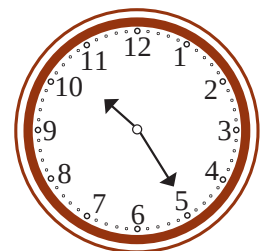
3 hours 10 min.



□ hours 35 min.



12 hours □ min.



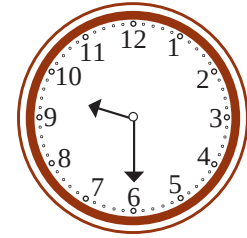
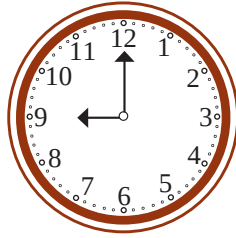
□ hours □ min.



For teachers : Students should be introduced to clocks, explain how to measure time using the clock in the maths kit, and demonstrate how changing the position of the hands on the clock changes the hours and minutes and hence the time.

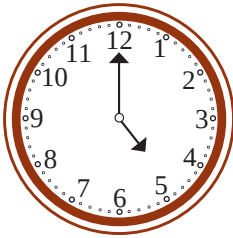
Exercise

1) Shaurya left for school at 9.00 am and reached school at 9.30 am.

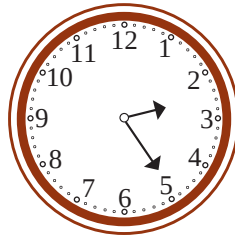


It took Shaurya _____ minutes to reach school. (How did you decide this?)

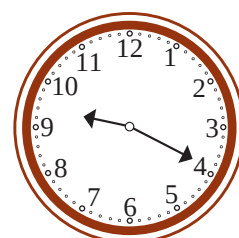
2) Write the time shown on the clocks below in hours and minutes



hours minutes



hours minutes

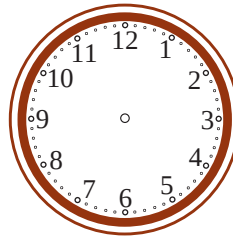
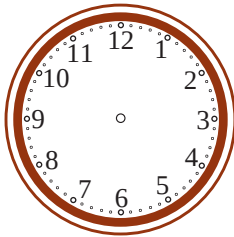


hours minutes

3) Write down your daily routine in the sentences given below and mark the hands on the clock accordingly.

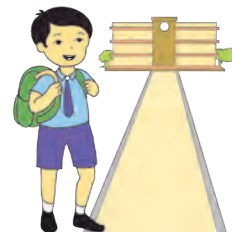
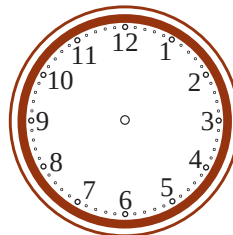
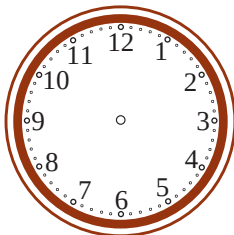
My Routine

(1) I wake up at in the morning. (2) I bathe at in the morning.

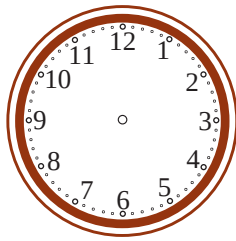


(3) I have breakfast at

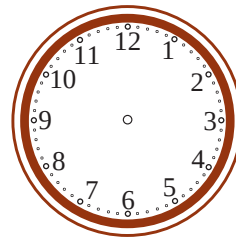
(4) I go to school at o'clock



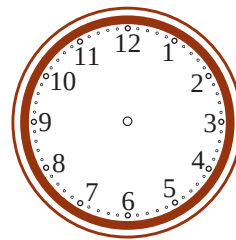
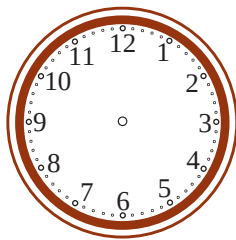
(5) My lunch break is at



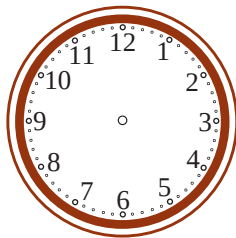
(6) I come home atfrom school.



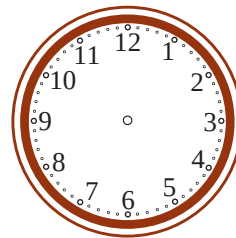
(7) I go on the ground at to play. (8) I sit down at to eat.



(9) I sit down at..... to study.



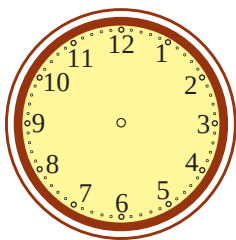
(10) I go at..... to bed at night.



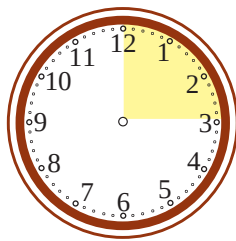
Quarter past, Half, Quarter to



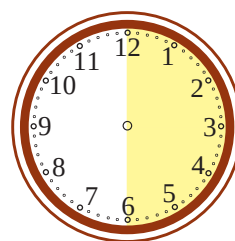
Let's Understand



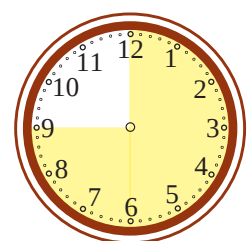
1 hour = 60 minutes



Quarter of an hour
= 15 minutes



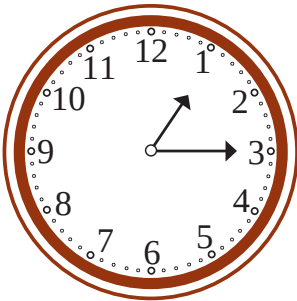
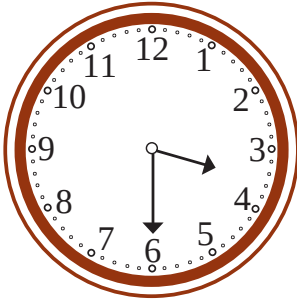
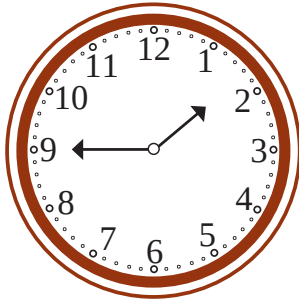
Half an hour = 30
minutes



3 quarters of an
hour = 45 minute

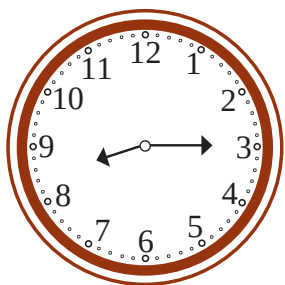


For Teachers : Discuss with students the relationship between the hands on a clock and time, as well as how to tell time in hours and minutes.

Quarter Past	Half Past	Quarter To
		
The hour hand is between 1 and 2, and the minute hand is at 3. It is 15 minutes past 1, which is called as ' quarter past one '.	The hour hand is between 3 and 4, and the minute hand is on 6. So, the time is 3:30, which is called as ' half past three '.	The hour hand is between 1 and 2 and the minute hand is on 9, which means it is 45 minutes past 1. It is 15 minutes before 2. That is, it is called as ' quarter to two '.
15 minutes past 1 = Quarter past one 15 minutes past 2 = Quarter past two 15 minutes past 4 = 15 minutes past 10 =	1:30 = Half past one 2:30 = Half past two 5:30 = Half past five 11:30 = 7:30 =	3:45 = Quarter to four 6:45 = Quarter to seven 9:45 = 12:45 =

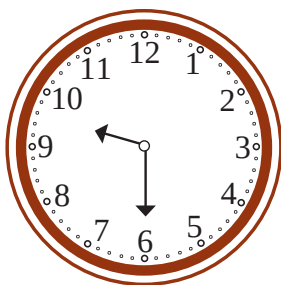
Exercise

(1) Write the time shown on the clock.

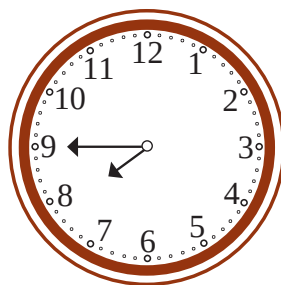


For example:
8 hours 15 minutes

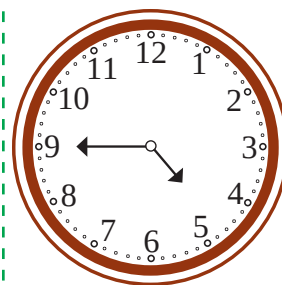
8.15



Half past nine



7 hours 45 minutes



Quarter to five

For teachers : Discuss with student the concepts of half past, half and quarter to. Give them practice of telling time using quarter to, quarter past, of hands of clock.

(2) Write down the events of your daily life according to the time period given below.

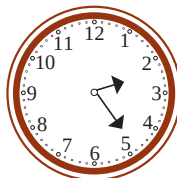
Sr. No	Work to be completed in few weeks	Work to be completed in few days	Work that can be completed in few hours	Work to be completed in few minutes
1.	Completing the lesson in class	Knit a sweater	Sleep at night	To dine
2.				
3.				
4.				

(3) Write a list of tasks to be done in the given time in the table below.

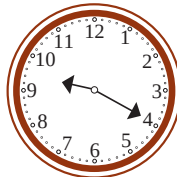
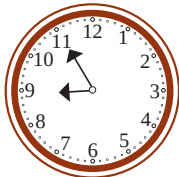
Sr.No	Time	For what work?
1.	5 minutes	,,
2.	15 minutes	,, To come home from school
3.	30 minutes	,,
4.	60 minutes	,,

(4) Write down the number of minutes taken by observing the position of the hour and minute hands.

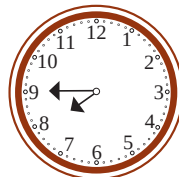
Starting Time	Ending Time	Time taken
---------------	-------------	------------



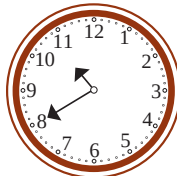
.....20..... minutes



..... minutes



..... minutes



..... minutes

Calendar

- ◎ Prepare a calendar for the month of the current year in which your birthday falls. Look at the calendar and answer the questions given below.

Month	Year					
Sun						
Mon						
Tue						
Wed						
Thu						
Fri						
Sat						

- (1) How many Fridays are there in this month?
- (2) What dates are on Tuesday?
- (3) Write down which festival is coming up this month.
- (4) Which day is on the 25th of this month?
- (5) What is the day on your birthday?
- (6) What day is on the 15th day of the month?
Which other dates falls on the same day?

Fun with Age

- ◎ Discuss with your parents and write the exact age in the box below.



My mother's, present age is years.



My father's present age years.



My grandmother's present age is years. My grandmother's age at the time of my birth was years.



My grandfather is years old. My grandfather's age at the time of my birth was years.

For teachers:

Students should be motivated to look at the calendar by giving them information about the days of the month and the days of the week. Learners should be advised to take help from friends and parents where necessary. Teachers should get students to prepare a 12-month calendar for the current year. Learners should complete a 1-year table of national festivals, Indian festivals with dates and time.





3 years ago my age
was years.



My age today
is years.



My age after 7 years
will be years.

◎ Complete the table by writing the dates in the box below.

My date of birth	Mother's date of birth	Father's date of birth	My friend's date of birth

Exercise

(1) How many months are there in a Gregorian year? Write their names in order.

(2) How many months are there in a year with 30 days? Write their names.

(3) How many months are there in a year with 31 days? Write their names.

(4) Prepare the following table to enlist the National and Indian festivals that will be celebrated in this academic year in your notebook.

Sr.No	Festivals	Month	Date	Day



For teachers: Seek the help of parents to facilitate discussions about other relationships with respect to their current and past ages.

(5) Observe the total number of days in February 2024 and 2025. Write down in the given boxes.

February 2024							February 2025						
Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3							1
4	5	6	7	8	9	10	2	3	4	5	6	7	8
11	12	13	14	15	16	17	9	10	11	12	13	14	15
18	19	20	21	22	23	24	16	17	18	19	20	21	22
25	26	27	28	29			23	24	25	26	27	28	

Number of days in February 2024: Number of days in February 2025:

(6) Write the names and dates of the days in the box below.

Day Before	← Yesterday	Today →	Tomorrow →	Day after
.....				
.....				
...../...../.....	/...../.....	/...../.....	/...../.....	/...../.....



Activity

- Read the printed information on the packed items such as food, drinks and medicines that we use in our daily life. Complete the table below based on the information.

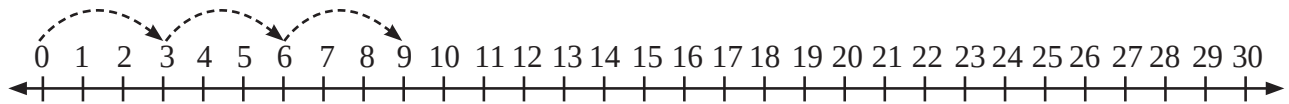
Sr. No	Name of the item	Manufacturing date	Expiry date	Price	Weight/Capacity



For teachers: Information about festivals, celebrations, eclipses, sunset, moonrise, lunar, days, constellations etc. of religious events, various religions should be given to the students by using the calendar. While purchasing an item, we write the manufacturing date of that item and the date up to which that item can be used. This should be discussed in the class.



Do The Activity



(1) Write the next numbers that Paresh will jump to.

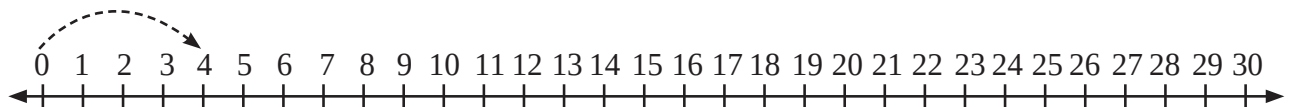
0, 3, 6, 9,

(2) How many steps did he jump each time?

(3) Write the steps of 3 from the place of the number 5.

5	8								
---	---	--	--	--	--	--	--	--	--

(4) If Paresh jumps from zero in steps of 4, then complete the steps of his jumps



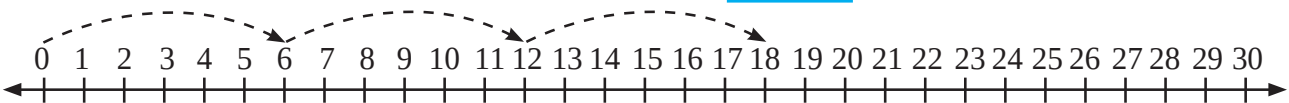
(5) If Paresh jumps further with steps of 5 from the given number, write his jumps in numerical form.

5, 10, —, —, —, —, —, —, —, —, —, —, —, —.

(6) Anil and Kavita are currently at position 31, Anil jumps forward by 2 steps. Kavita jumps back by 2 steps. Write the numbers they will jump to.

—, —, —, —, —, —, 31, —, —, —, —, —, —, —.

(7) Alka is jumping forward by a step of 6 each time. So, after the number 18, to which number will Alka jump?



(8) If Ketki is jumping back by steps of 6 from the number 60, then complete the following table.

					36	42	48	54	60
--	--	--	--	--	----	----	----	----	----



For teachers: Explain number patterns with the help of various examples and explain the diversity in patterns by giving examples.



Let's do it

(1) Look at the following arrangement. Identify the patterns that resemble ABC, ABB, AAB. AB and write its style in the box using the letters ABC.

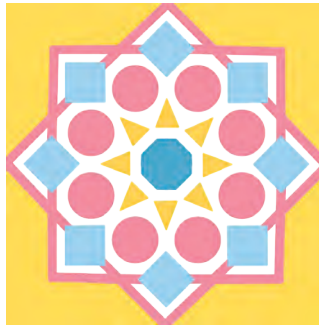
- AB
- ABC
-
-

(2) Using geometric figures ($\square, \Delta, O$), create a pattern of the types AABB, AA, AABC, ABCC, ABBC in your notebook.

•



Akash, look at the picture on the tile. What shapes do you see on the tile?



Shyamal, I see shapes like triangles, squares, and circles on the tile.



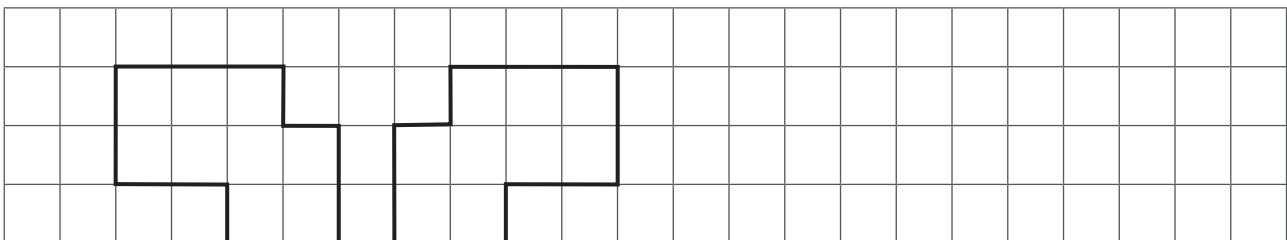
•



The embroidery on my mother's saree.

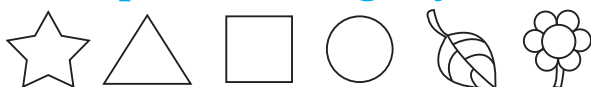


• Complete the pattern below.



For teachers : Students should practice various examples of patterns with the help of friends, parents and teachers as needed

- Make patterns using any three of the following shapes.



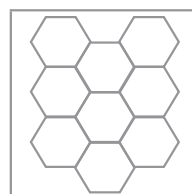
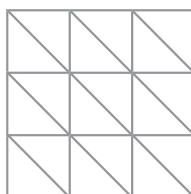
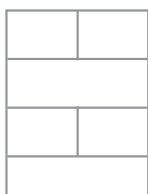
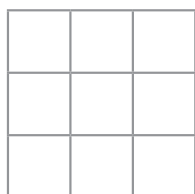
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- Fill in the tiling below with your favourite colours and create a design.

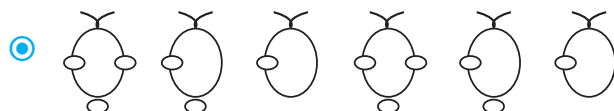


Activity

Take rectangular strips of paper. Fold them. Draw designs on them. Cut out some parts of it. After cutting, unfold the paper strip. Observe the pattern so formed. Use the designs formed for classroom decoration.

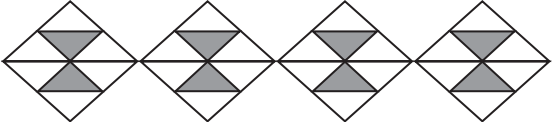
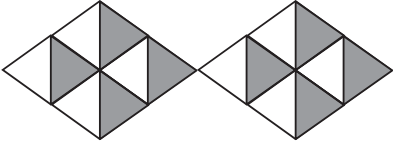
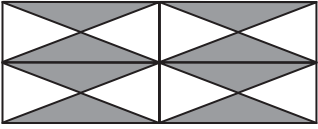
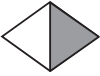
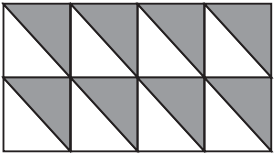
Exercise

(1) Identify and complete the patterns.



For teachers: Ask students to observe the pattern created with lines, discuss it, and practice completing shapes with the help of different lines. Practice making shapes by folding and cutting paper. Practice making different shapes using the different geometric shapes given in the maths kit.

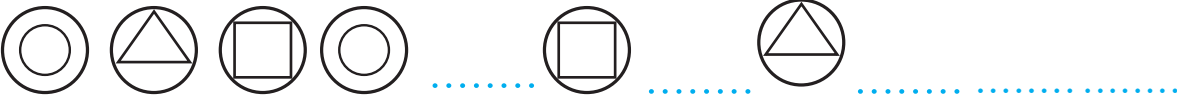
(2) Match the pairs of the given shape and the patterns formed from it.

Given shape	Patterns formed
☐ 	
☐ 	
☐ 	
☐ 	

(3) Complete the given arrangement by observing the patterns of the following numbers.

- ☐ 2, 5, 10, 10, 5, 2, 2, 5, 10, , ,
- ☐ 6, 8, 10, 12, 14, , , -
- ☐ 21, 24, 27, 30, 33, , ,
- ☐ $1 \times 1 = 1$, $2 \times 2 = 4$, $3 \times 3 = 9$, $4 \times 4 = 16$, , ,

(4) Complete the patterns of the following shapes.

- ☐ 
- ☐ * * * # * * * # 



For teachers : Student should be encouraged to observe the use of tiling on their floors at home and in school premises .Examples of such patterns and tiling in daily life should be discussed.



Let's do it

(1) Write the table of 5

Table of 5,,,,,,,,,

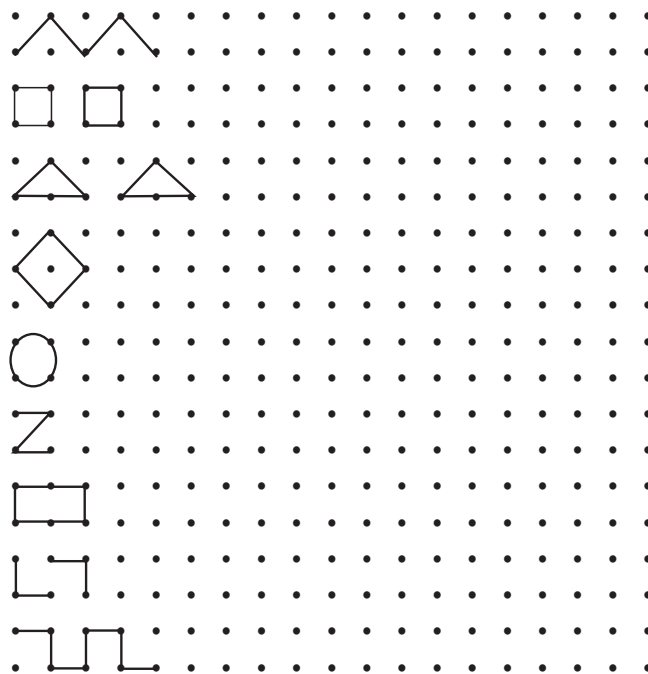
Complete the arrangement by identifying the patterns of digits in the units place from the table of 5.

5 0
.....,.....,.....,.....,.....,.....,.....,.....,.....,.....

(2) A pattern is formed by adding the numbers horizontally, vertically and diagonally, giving the answer 33. (5, 11, 17), (4, 11, 18), (10, 11, 12). Like this find a pattern of groups of three numbers that have the same sum like this.

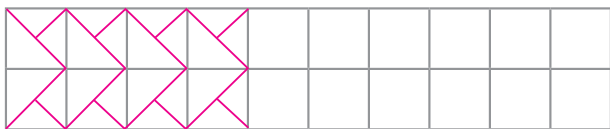
Sun		6	13	20	27
Mon		7	14	21	28
Tue	1	8	15	22	29
Wed	2	9	16	23	30
Thu	3	10	17	24	31
Fri	4	11	18	25	
Sat	5	12	19	26	

(3) Complete the diagram using the dots below.



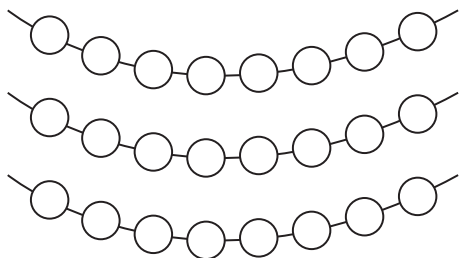
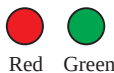
For teachers : Form groups of students and ask them to write the steps and tables and ask them to describe the patterns in them. Using dots and calendars encourage them to make patterns with the help of friends and parents and ask them to create various patterns.

(4) Complete the following design. Colour it with the colours of your choice.

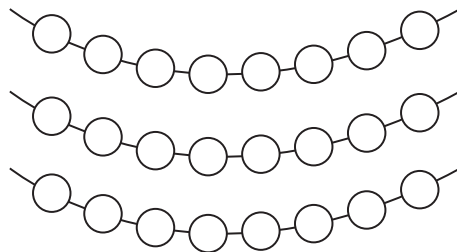
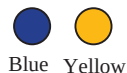


(5) Create different patterns using the given colours

Suyash



Mayuri



Let's Observe

- Ramesh was very hungry when he came back home after playing. There were three fruits in the basket - guava, apple and chikoo. His mother asked Ramesh to choose any two fruits. Write down the pairs of fruits that Ramesh could choose.

Apple	Guava	Apple	Chikoo
	Guava	Chikoo	



Let's Understand.

Combination

- We have seen how to obtain various patterns using shapes. Now let's see how to create patterns, determining their order or according to given conditions.

Selection for any two games out of three



For teachers : Through all the activities mentioned above, explain to the students that here, the 'selection' matters and not the 'order'. This concept is known as 'Combination'.

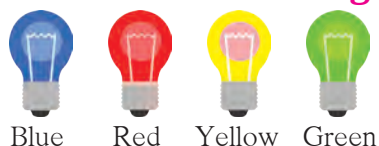
- Which two games can student choose from the three given options?
The pairs of these games are as follows:

.....		
-------	-------	-------

If Kabaddi had been included among the sports above, in how many ways can two games be selected from those four? Write down the pairs in your notebook

Exercise

- (1) Four coloured lights are given below. If you wanted to install any two of these coloured lights in your room. In how many different ways you can make selection of lights. Write down all the pairs.

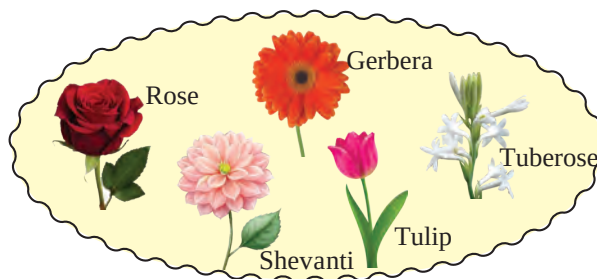


.....,

.....,

.....,

- (2) From the given five flowers , select any two flowers for flower pot. Write in your notebook how many different types of selection can be done for the same.



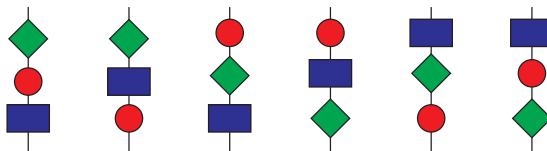
Let's Understand

Permutation

- We have observed how pairs are formed by selecting from given objects in various ways. Now, let us explore how these objects can be arranged in a specific order and according to given conditions.



Let's Discuss



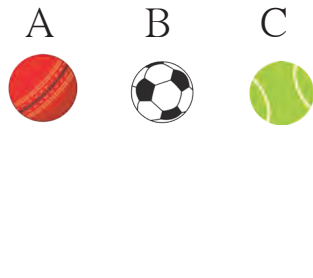
- How many different types of shapes are used to make each of the decorative *toran* shown above?
- Is it possible to make designs different from the ones above by using three distinct shapes? Think about it and discuss with your friends.



For teachers : Through all the activities mentioned above, explain to the students that here 'order' is important. If the sequence in which the objects are arranged is altered, a distinct arrangement or configuration is formed. This concept is known as Permutation.

Exercise

- (1) Arrange the following balls in a row within the cupboard in various different sequences. Use letters given below for representing the balls.



- (2) Meena bought three balloons—red, yellow, and green from a fair. In how many different ways can she arrange these balloons in a line? (Use the letters R for the red balloon, Y for the yellow balloon, and G for the green balloon to represent the arrangements.)

.....

.....

.....

- (3) A magician wants to create a password to lock his box and keep the items inside safe, he intends to do this by using three specific digits from his box only once. Write down how many different types of passwords he can create.



Hello Children!

If you feel uneasy about someone touching you inappropriately, you should not keep quiet. You must :

1. Not blame yourself
2. Tell someone whom you trust.
3. You can also inform National Commission for Protection of Child Rights through the **POCSO e-box**.

When you get an unsafe touch, you may feel bad, confused and helpless.

You need not feel “bad” because it’s not your fault.



Press This Button

POCSO e-box available at ncpcr@gov.in

If you are below 18 years of age, and are troubled or confused or abused or in distress or know some other child who is...

*Call 1098...because some numbers are good!
They change lives!!!*



CHILDLINE 1098 - a national 24 hours toll free emergency phone service for children in distress is an initiative of CHILDLINE India Foundation supported by Ministry of Women & Child Development.



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Towards a new dawn



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